



Calix Engagement Cloud Help

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Chapter

About Calix Engagement Cloud Help

Welcome to Calix Engagement Cloud Help! Visit Help to get current information on using the next-generation Calix Engagement Cloud.

Calix Engagement Cloud Help provides instructions to help you accomplish application tasks, which are conveniently grouped in the following sections:

- **Calix Engagement Cloud Overview:** This section provides an application overview and detailed information on the origin of Engagement Cloud data, data ingestion resources, and FAQs/Key Concepts/Terminology.
- **Getting Started:** This section includes help topics for simple, everyday tasks such as launching Calix Cloud applications and logging out.
- **Using Calix Engagement Cloud for Subscriber Retention and Upsell:** This chapter contains topics related to using features common to both Engagement Cloud and Engagement Cloud Plus.
 - **Gather Insights to Manage Your Subscribers:** This topic highlights elements of the Home dashboard.
 - **Explore Residential Subscriber Insights Data:** This section includes help topics covering how to use Dashboards to better understand and manage your subscribers.
 - **Work with Subscriber Segments:** This section includes help topics covering how to view subscribers geographically, define and manage target segmentation, and export subscriber data by region.
 - **Work with Campaigns:** This section includes help topics covering how to view, schedule, and manage campaigns/mobile notifications sent to subscribers per engagement channels.
 - **Configure Calix Engagement Cloud Settings:** This chapter contains topics covering how to improve billing data mapping, upload subscriber and prospect lists, and upload ZIP lists.
- **Using Calix Engagement Cloud Plus for Subscriber Acquisition:** This chapter contains topics that discuss how to use demographic data for residential and business subscriber acquisition.
 - **Understanding Demographic Data:** This topic introduces demographic insights data and segments in Engagement Cloud Plus.

- **Explore Subscriber Demographic Data:** This section includes topics related to filtering data and viewing insights.
- **Explore Prospect Demographic Data:** This section contains topics covering how to view subscribers geographically, define and manage target segmentation, and export subscriber data by region.
- **Work with Demographic Segments:** This section includes information on recommended segments based on demographic data and instructions for using demographic data to create segments.
- **Administering Calix Cloud Applications:** This section provides instructions for administering Calix Cloud applications.
- **Reference Information:** The Reference Information section provides additional information on topics such as how ProtectIQ identifies threats and how subscriber matching to demographic data works in Engagement Cloud Plus.

Intended Audience

This document is intended for marketing professionals and decision makers in your organization.

Chapter 1

Calix Engagement Cloud Overview

Calix Engagement Cloud empowers service providers (SPs) with targeted, on-demand analytics and intelligence based on subscriber segmentation. Engagement Cloud enables SPs to identify Marketing Qualified Leads (MQLs) to drive marketing campaigns that may immediately increase ARPU, increase retention, and deliver an unmatched subscriber experience, all while delivering a compelling return on investment on their marketing dollars. Further, the upsell features help BSPs to identify the most strategic geographies or demographics in which to invest their marketing dollars (i.e., gamers or streaming video subscribers in a specific community).

Calix Engagement Cloud is the only marketing-focused analytics solution that is “out of the box” ready to help service providers engage their subscribers. With the Engagement Cloud, service providers may experience the following benefits:

- **Discover actionable insights on subscribers:** Engagement Cloud provides data analytics to empower you with actionable insights on demand, enabling you to target subscribers in a timely manner and engage with them successfully with on-demand data. These insights may help you to:
 - Review metrics that drive KPIs.
 - Target marketing spend where it makes sense.
 - Identify the need for new services or right-size existing offers.
- **Simplify subscriber data analytics to reveal subscriber preferences:** Engagement Cloud provides a variety of off-the-shelf dashboards to segment your base and identify high-opportunity targets based on subscriber usage and line of business data.
- **Elevate your business with targeted data for campaigns:** Engagement Cloud provides you with the data you need to run highly successful targeted marketing campaigns to increase ARPU and improve the subscriber experience and maximize the return on your marketing spend.
- **Create custom audiences, drive more efficient campaigns, and reach more subscribers:** Engagement Cloud allows you to upload addresses of subscribers who can get your service,

allowing you to gather insights and deploy campaigns for specific subscribers within your serviceable geographic territory.

- **Expand service offerings to new areas:** Engagement Cloud allows you to define service areas by ZIP/ZIP+4 code to acquire new subscribers, plan service expansions, and drive a higher ROI.

Using Engagement Cloud, some service providers have achieved:

- 40%+ close rates on targeted marketing campaigns
- 20%+ increases in ARPU on targeted subscribers
- Measurable reduction in churn
- Exceptional returns on their marketing investment dollars

The Calix Engagement Cloud Plus entitlement extends Engagement Cloud's capabilities with the addition of third-party demographic data (all 50 U.S. states, no U.S. territories) for households and businesses, enabling new subscriber acquisition, subscriber and prospect insights, service expansion planning, and data enrichment. Using Engagement Cloud with demographic data, service providers may experience the following benefits:

- **Enhance the subscriber experience:** Entice subscribers with targeted service offerings based on rich demographic and experiential insights.
- **Simplify subscriber acquisition:** Identify new market opportunities with prospect recommendations based on the data usage habits of existing subscribers in your service area.
- **Discover actionable insights on non-subscribers:** Craft targeted campaigns for specific audiences with rich demographic data that provide insight into the lifestyle, shopping, financial, and technology usage habits of non-subscribers in your service area.
- **Create custom audiences, drive more efficient campaigns, and reach more prospects:** Upload addresses of subscribers and prospects who can get your service to gather insights and deploy campaigns for specific subscribers and prospects within your serviceable geographic territory.
- **Expand business service offerings:** Identify SmartBiz upsell opportunities with insights data for business subscribers.

For more information, see Using Demographic Insights Data.

Calix Engagement Cloud Features and Benefits

The next-generation Calix Engagement Cloud provides the following features and benefits:

Feature Description	Actionable Insights and What You Can Accomplish
The Home page presents a quick, holistic overview of the subscriber base and recent fluctuations.	Provides quick access to Engagement Cloud dashboards and presents key subscriber data at-a-glance, including: • New Subscribers • Churned Subscribers • Total Subscribers • Active Subscribers • Internet Service ARPU • Service Tier & Technology • Reach by Campaign Type • Recent Campaign Details
Dashboards presents insights data for both residential and business subscribers and prospects ² .	<i>Business Insights</i> provides an overview of subscriber trends including new and churned subscribers, service adoption, and campaign effectiveness.
	<i>Subscribers</i> presents insights data for your subscribers, allowing marketers to quickly and easily monitor subscriber growth and churn trends, track managed services adoption rates, and understand subscriber data usage habits. The Calix Engagement Cloud Plus entitlement adds demographic data for both residential and business subscribers.
	<i>Residential Prospects</i> ² provides demographic data for residential non-subscribers. From the Residential Prospects dashboard, marketers can filter non-subscriber data to create segments for acquisition campaigns.
	<i>Business Prospects</i> ² provides demographic data for business non-subscribers.

Feature Description	Actionable Insights and What You Can Accomplish
Segments provides details for custom and recommended segments and allows you to create new segments.	• View details for custom segments including the number of campaigns associated with the segment, the date the segment was created, and the date the segment was last modified. • View details for pre-built segments including the number of segment members and average revenue per user (ARPU). • Create new subscriber, residential prospect², or business prospect² segments.
Campaigns lists existing campaigns, allows you create new campaigns, and allows you to configure engagement channels.	Enables marketers to perform the following tasks: • View existing triggered and scheduled campaigns • Create/edit triggered and scheduled campaigns • Configure engagement channels: - Mailchimp[®] - HubSpot[®] - Constant Contact[®] - Postalytics[®] - Facebook[®] - NISC - Mobile Notification
Configuration provides access to data configuration and file upload settings.	Enables marketers to complete the following tasks from within Engagement Cloud: • View billing data ingestion (pipeline or API) details • Upload ZIP code filter lists • Upload residential subscriber and residential prospect² lists

Feature Description	Actionable Insights and What You Can Accomplish
Calix Cloud Administration is easily accessible from the top right menu for Organization Administrators to quickly administer applications.	Enables the Organization Administrator to complete the following tasks for Calix Cloud applications: • Set up roles and associate users with roles • Deactivate/reactivate users • View and configure ACS Settings • View Organization info • Set up the Flow Configuration • Configure the Access Control List • Configure Single Sign-on (SSO) • Customize Call Outcome¹ • Customize blocked page templates • Add SmartTown™ communities and community users • Update USOC data • Customize mobile app branding • View API usage • View billing integration status • Configure the SmartBiz™ Customer Portal banner
¹ Feature available only with Calix Service Cloud entitlement. ² Feature available only with Calix Engagement Cloud Plus entitlement.	

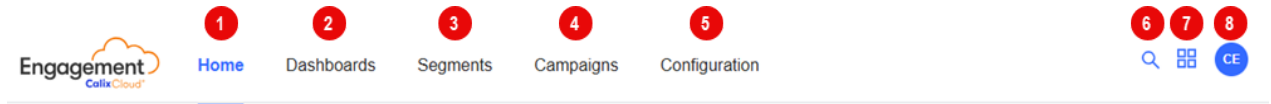
Chapter 2

Getting Started

This topic describes how to navigate Calix Engagement Cloud and presents simple steps for getting started using the application.

Navigating Calix Engagement Cloud

The top menu bar provides access to the following pages and functions within Engagement Cloud:



Menu Bar Item	Description
1. Home	Presents a quick, holistic overview of the subscriber base.
2. Dashboards	Presents subscriber data at-a-glance, allowing marketers to quickly and easily monitor trends in subscriber growth and churn, track managed services adoption rates, and understand subscriber data usage habits. The Calix Engagement Cloud Plus entitlement adds third-party residential demographic data and business data for both subscribers and non-subscribers (prospects).
3. Segments	View pre-built and custom segments, view segment insights, and create and modify segments.
4. Campaigns	View existing campaigns, create new campaigns, view campaign results, and configure channel integrations.
5. Configuration	Monitor billing data quality, upload subscriber and prospect contact lists, and upload ZIP code filter lists.
6. Search	Click to quickly search by name for subscribers and campaigns. As you type, the search list returns subscribers and campaigns that match your input.
7. Cloud apps menu	Click to switch between Calix Cloud applications.
8. Profile menu	Click to log out of Calix Cloud, change the display language, configure accessibility options, and configure Administration settings (OrgAdmin role required).

Getting Started in Engagement Cloud

Engagement Cloud provides all the tools and data you need to turn insights into action. Use the following getting started flow to guide your journey through Engagement Cloud:

1. View Home page insights to see what or who may need attention.
2. Visit Dashboards to determine why.
3. Go to Segments to target affected subscribers.
4. Go to Campaigns to deploy a campaign to the subscribers.
5. Track results to monitor campaign effectiveness.

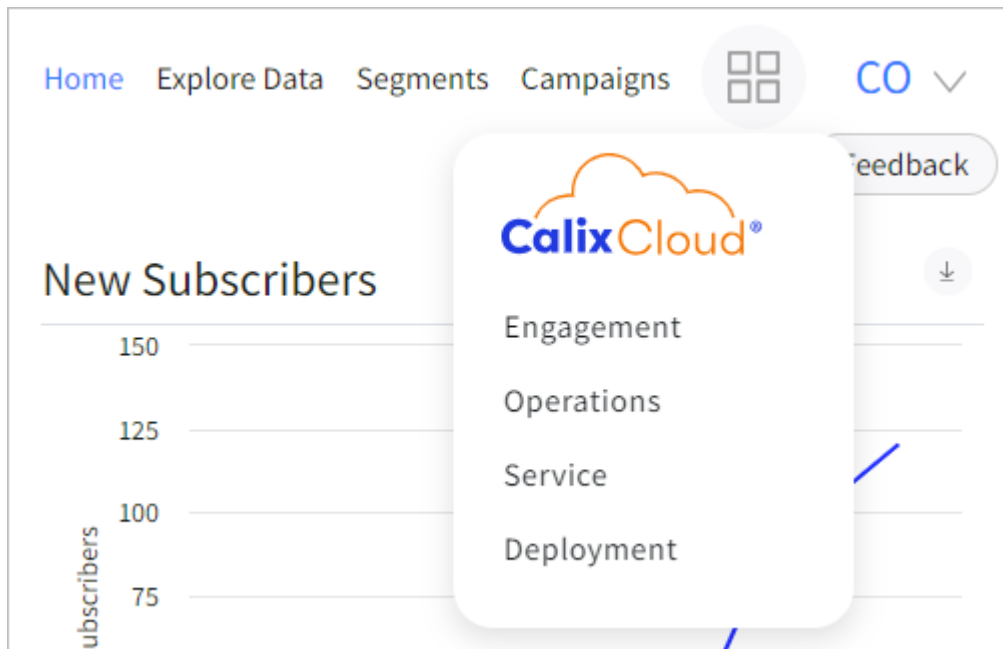
Considerations

- Upon logging in to Calix Engagement Cloud for the first time, users must review and accept the Calix Cloud Terms of Service (ToS).
- To improve data response times and promote a better user experience, Calix Engagement Cloud caches users' frequently visited pages. This caching behavior *may make it seem* that infrequently viewed pages load at a slower rate by comparison, but they actually load at a normal rate.
- For complete access to all Engagement Cloud features and functions, ensure users have the *MarketingAllAccess* role assigned to their user. See *Set Up Roles for Calix Engagement Cloud* (on page [177](#)) for more.
- For subscribers to show in Engagement Cloud, the following billing fields must contain data:
 - Technology
 - USOC
 - Service Tier
 - Customer Type
- Viewing the Insights content on the Engagement Cloud Home page may require an update to your firewall. To ensure delivery of content, please note the following:
 - For CalixCloud.calix.com users, you must permit traffic from host *clouddashboards-us.calix.com*, port 4000.
 - For CalixCloud-ca.calix.com users, you must permit traffic from host *clouddashboards-ca.calix.com*, port 4000.
- Home page charts and KPIs update daily and display insights from the previous 30 days.
- Key performance indicators (KPIs) and charts that do not rely on subscriber data usage—such as Total Subscribers, Residential, and Business—include all unchurned subscribers (both active and inactive) in the billing file. Insights based on data usage—such as Gaming and Work from Home—include only active subscribers.

Launch Cloud Applications

This topic describes how to launch Calix Cloud applications.

If you are logged in to the Calix Cloud environment, you can easily switch to other Cloud applications for which your organization maintains subscriptions/entitlements. To launch Calix Cloud applications, click the bento menu to select available applications.



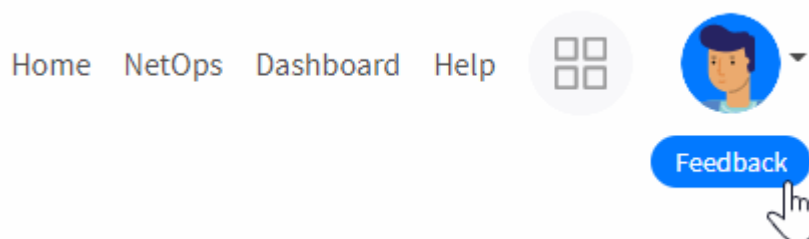
Provide Feedback

This topic describes how to provide feedback on Calix Cloud applications.

Calix would love your feedback on Calix Cloud applications. Your suggestions can help us improve our offerings in the future.

To share your feedback, follow these steps:

1. Click **Feedback**.



2. Type your feedback and suggestions.

Share your feedback

Thanks for using the new Calix Support Cloud. We would love to know what you think of it! Please share your feedback and suggestions.

NOTE: If a feature is not working please open a case through the Service Station in My Calix.

I love the new look of Calix Cloud, but I would like to see...

Submit Feedback

3. Click **Submit Feedback**.

Note: For technical issues, please submit a case through the **Service Station** in My Calix.

Change the Language

This topic describes how to change the display language in Calix Cloud.

Calix Engagement Cloud is presented in the English language by default, but also supports the following languages:

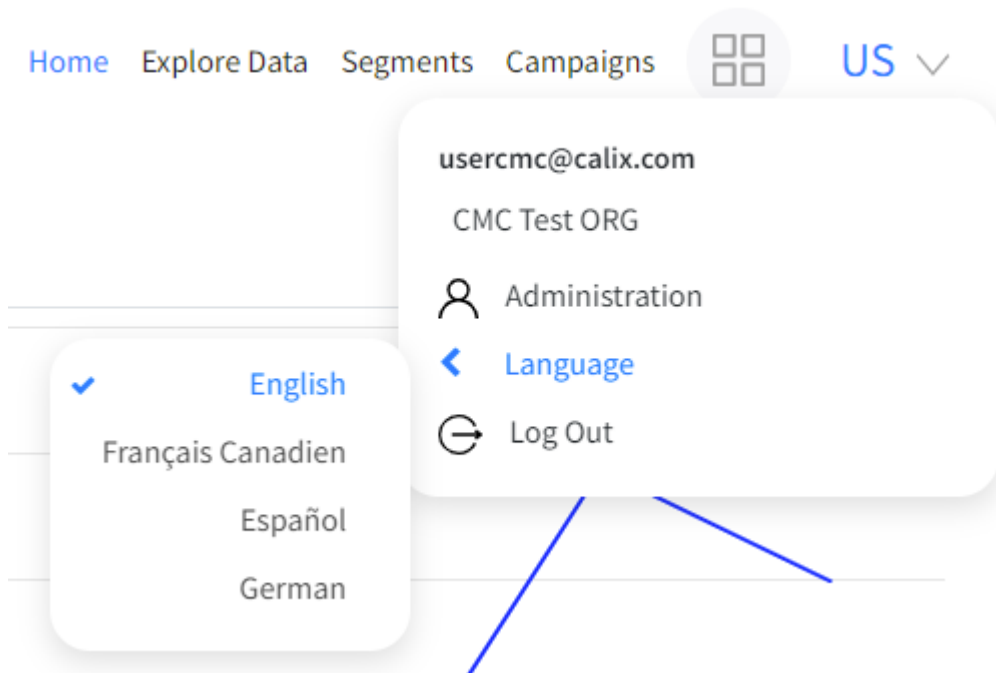
- French Canadian
- Spanish
- German

You can change the language to your desired option.

To change the Marketing Cloud Plus display language

1. Click the down arrow in the top right corner and select **Language**.
2. Select the desired language. The display text changes to the selected language.

Note: All Calix Marketing Cloud product documentation, including export files, is available in English only.



Use the Help Center

This topic describes how to use the Calix Cloud Help Center. The Help Center, available throughout Calix Cloud, provides quick access to a variety of help resources. From the Help Center, you can view the release notes and Help Documentation, pose a question to the Calix Community, and provide product feedback. To open the Help Center, click on the question mark icon in the lower-right corner of the screen.

Use Accessibility Options

This topic describes how to use accessibility options in Calix Cloud.

Using Accessibility Options

The Accessibility Help Desk powered by AudioEye® provides accessibility options to help you more easily read, locate, and interact with page content in Calix Cloud applications. From the Accessibility Help Desk, you can perform the following actions:

- View the page structure, including a list of headings and links
- Enlarge the cursor
- Highlight page headings, links, and/or buttons
- Change page color contrast and color shift
- Change text size, spacing, and font type
- Highlight your cursor's position on the page

Accessibility settings apply per user and persist across Calix Cloud applications. Calix Cloud retains your preferences on log out and applies them on subsequent logins. To open the Accessibility Help

Desk, click on the  icon in the lower-right corner of the browser window. See the AudioEye Help for more information. If you have an existing AudioEye account, you can log in to and customize the Accessibility Help Desk.

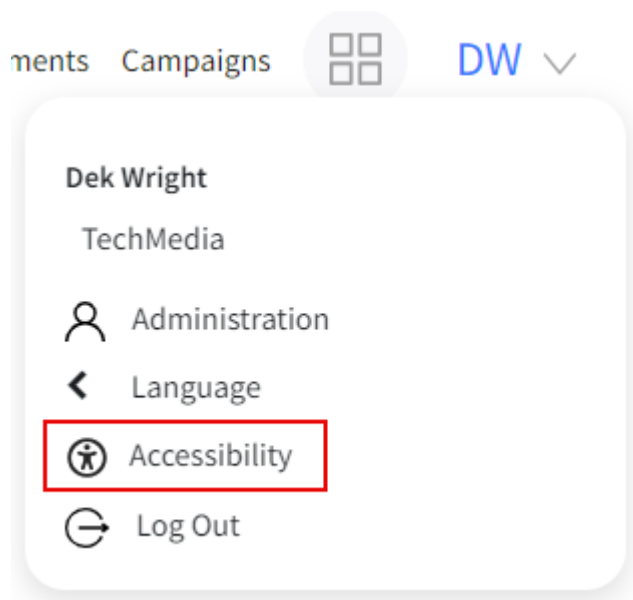
Considerations

- Ad blockers may prevent the Accessibility Help Desk from appearing in your web browser. If you do not see the Accessibility Help Desk, check your ad blocker settings.
- Text enlargement does not affect content in iframes (for example, charts and graphs).

Deactivate/activate the Accessibility Help Desk

If desired, you can hide the Visual Toolkit.

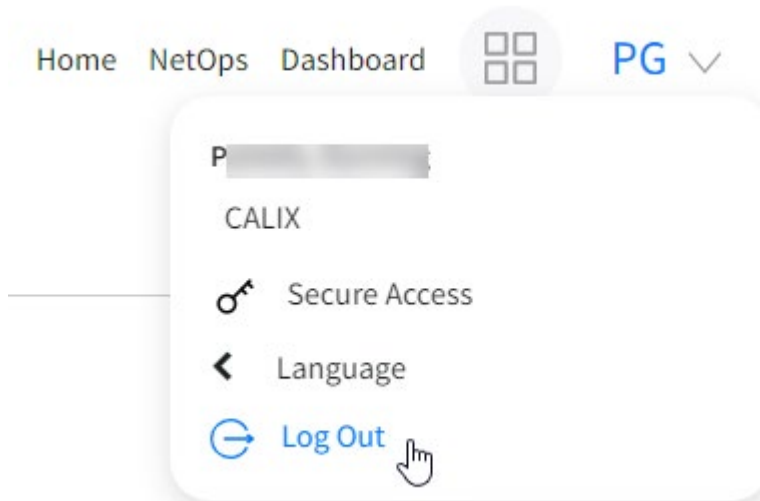
1. Click the down arrow in the top right corner and select **Accessibility**.



2. Click **Deactivate Accessibility Help Desk**.
To restore, click **Activate Accessibility Help Desk**.

Log Out of Calix Cloud

This topic describes how to log out of Calix Cloud. To log out of Calix Cloud, click the down arrow in the top right corner and select **Log Out**.



View the Landing Page

The Calix Cloud landing page is a combination of critical key performance indicators (KPIs) and trend charts from the Service Cloud, Operations Cloud, and Engagement Cloud applications. For users assigned to one of the following roles, the landing page displays upon logging into the Calix Cloud environment:

- Service BusinessInsights
- Operation BusinessInsights
- Engagement BusinessInsights

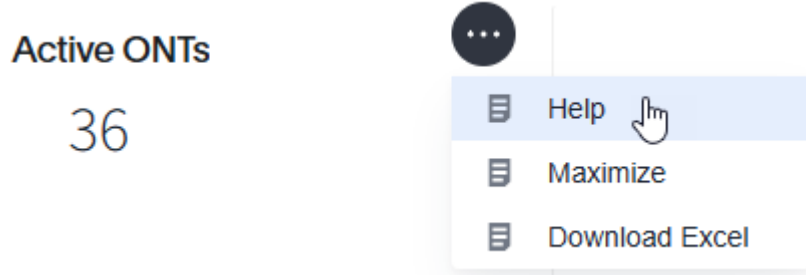
To assign a role to a user, see Define ACLs and Roles for Calix Cloud Users.

Note that a user does not need to have a Service Cloud, Operations Cloud, or Engagement Cloud role to view the landing page. However, for a user assigned to a landing page role without the corresponding Calix Cloud entitlement, the related KPI and trend charts display as blank with a "restricted" label.

Navigation and viewing options

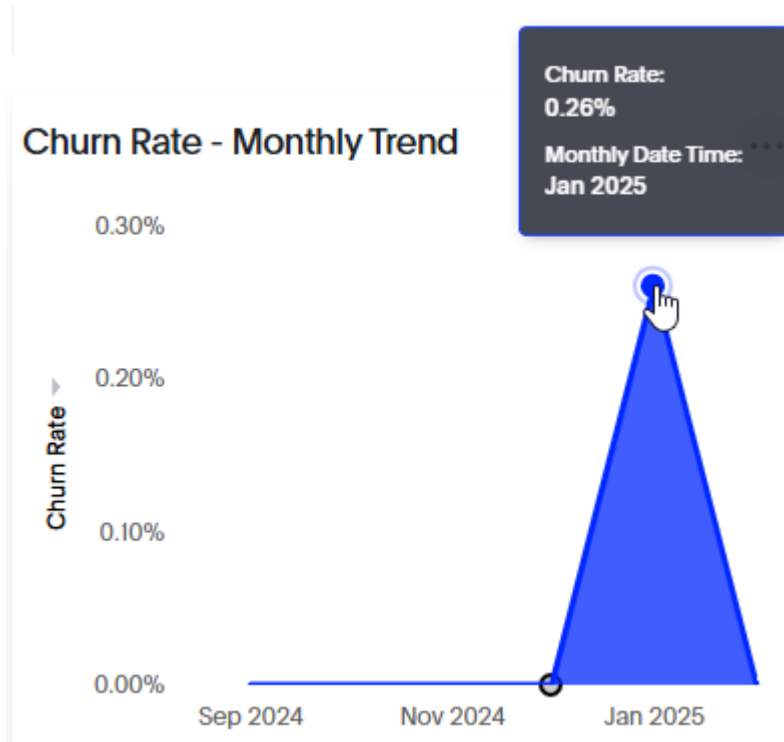
The landing page supports the following navigation and viewing options:

- To view the definition of a KPI or chart in the UI, click the corresponding ellipsis and select **Help**.



- To expand a chart, click the corresponding ellipsis and select **Maximize**.
- To export a chart as a CSV file, click the corresponding ellipsis and select **Download Excel**.

- To view detailed chart information, hover over an axis point:



Definitions

The landing page displays the following KPIs:

KPI	Description
Active Subscribers	Total number of active subscribers with flow data or as defined by billing status over the last 30 days.
New Subscribers	Total number of new subscribers over the past 30 days.
Churn Rate	Total number of subscribers lost in the past 30 days divided by the current total subscribers.
ARPU (Average Revenue Per User)	The total revenue divided by the total number of subscribers.
Active ONTs	Total number of active ONTs in the last 30 days.
Active RGs	Total number of Active RGs in the last 30 days. "Active" implies RGs that have checked in the last 24 hours over the last 30 days. This includes the count of systems that are RGs (not mesh units).
RGs with 4+ Reboots	Total number of RGs having 4+ Reboots in the last 30 days.
Blocked Threats Count	Blocked Threats Count in the last 30 days.

KPI	Description
Active Subscribers Variance %	Percentage change between the total number of Active Subscribers in the last 30 days compared to the previous 30 days.
New Subscribers Variance %	Percentage change between the total number of New Subscribers in the last 30 days compared to the previous 30 days.
Churn Rate Variance %	Percentage change between the total churn rate in the last 30 days compared to the previous 30 days.
ARPU Variance %	Percentage change between the total ARPU in the last 30 days compared to the previous 30 days.
Active ONTs Variance %	Percentage change between the total number of Active ONTs in the last 30 days compared to the previous 30 days.
Active RGs Variance %	Percentage change between the total number of Active RGs in the last 30 days compared to the previous 30 days.
RGs with 4+ Reboots Variance %	Percentage change between the total number of RGs having 4+ Reboots in the last 30 days compared to the previous 30 days.

The landing page displays the following trend charts:

Trend Chart	Description
Active Subscribers - Monthly Trend	Shows the trend of active subscribers for the last six months.
New Subscribers - Monthly Trend	Shows the trend of new subscribers for the last six months.
Churn Rate - Monthly Trend	Shows the churn rate trend for the last six months.
ARPU - Monthly Trend	Shows the ARPU trend for last six months.
Active ONTs - Daily Trend	Shows the daily trend of active ONTs over a period of 180 days.
Active RGs - Daily Trend	Shows the daily trend of active RGs over a period of 180 days.
RGs with 4+ Reboots - Daily Trend	Shows the daily trend of RGs having 4+ Reboots over a period of 60 days.
Subscriber Data Usage	Shows the distribution of subscribers by consumed bandwidth for the last 30 days.
Blocked Threats Insights	Shows the distribution of blocked threats by categories for the last 30 days.
CommandWorx - Daily Trend	Shows the number of subscribers using CommandWorx for the last 30 days.

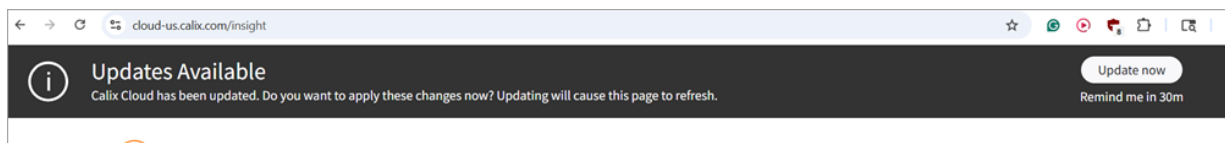
Trend Chart	Description
CommandIQ - Daily Trend	Shows the number of subscribers using CommandIQ for the last 30 days.

Landing page limitations

- ADA text re-sizing does not affect landing page charts or KPI boxes.
- Downloaded CSV files may not sort columns as expected.
- Individual KPIs—including blank placeholder KPIs—display download and maximize options.
- The sort and time bucket filters applied to charts do not take effect or carry over to downloaded CSV files.
- If no data is available, the following default error message displays:
No data found for this query. Try changing your filters.

Apply Calix Cloud Updates

To improve the user experience, Calix periodically releases user interface (UI) updates for Calix Cloud applications. When updates are available, a banner displays above the application homepage upon login. Click the **Update now** button on the banner to refresh the page and apply the updates, or click **Remind me in 30m** to clear the banner for 30 minutes.



Chapter 3

Using Calix Engagement Cloud for Subscriber Retention and Upsell

Topics covered

See the topics in this chapter for information on using Engagement Cloud dashboards:

- Gather insights on the Home dashboard
- View subscriber insights and create segments
- View recommended and custom segments
- Create and deploy campaigns
- Configure channel integrations
- View billing data quality metrics and upload ZIP code filter and subscriber/prospect lists

For topics relevant to working with demographic, SMB, and prospect data, see the *Using Calix Engagement Cloud for Subscriber Acquisition* (on page [143, http://www.](#)) chapter.

Gather Insights to Manage Your Subscribers

This topic describes insights available on the Calix Engagement Cloud Home page.

The **Home** dashboard in Engagement Cloud presents a quick, holistic overview of the subscriber base. Use Home page insights to monitor overall subscriber health and campaign performance.

Home KPIs/Charts

The Home page displays the following charts and KPIs.

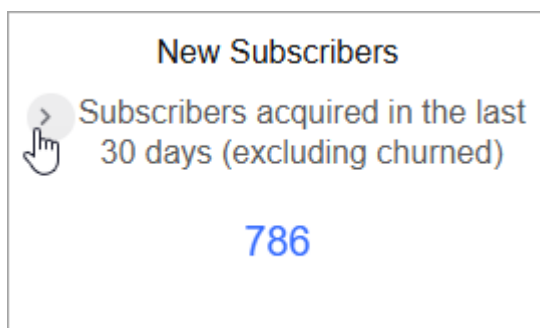
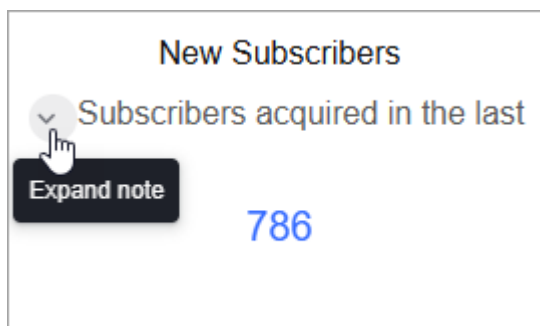
KPI/Chart	Description
KPIs	
New Subscribers	Shows the number of subscribers acquired (added to billing) within the last 30 days.
Churned Subscribers	Shows the number of subscribers removed from billing within the last 30 days.
Total Subscribers	Shows the total number of subscribers in the billing file and the percentage change (increase or decrease) over the last 30 days.
Active Subscribers	Shows the number of subscribers with internet usage (flow data) within the last 30 days.
Internet Service ARPU	Shows the average revenue per user (ARPU) for the current day. This value updates daily. Note: To view ARPU, users must have the 'Revenue' permission assigned to their user role. For users without the Revenue permission, this KPI is hidden. See <i>Set Up Roles for Engagement Cloud</i> (on page 177) for more.
Charts	
Service Tier & Technology	Displays the number of subscribers by access technology within each service tier. Each colored bar segment represents a different access technology, and the size of a bar segment indicates the number of subscribers with that technology within the service tier. Axes: - Horizontal axis: Service tier - Vertical axis: Number of subscribers

Reach by Campaign Type	Displays the total number of contacts—as well as the number of subscribers, residential prospects, and business prospects—who received at least one campaign within the last 90 days.
Recent Campaign	Displays data for the most recently deployed campaign: • Campaign name • Campaign start and end dates • Campaign deployment status and channel failures • Number of subscribers who upgraded their service tier during the campaign Click View campaign to open the campaign Results page.

Interact with KPI/Chart Data

To expand KPI description text

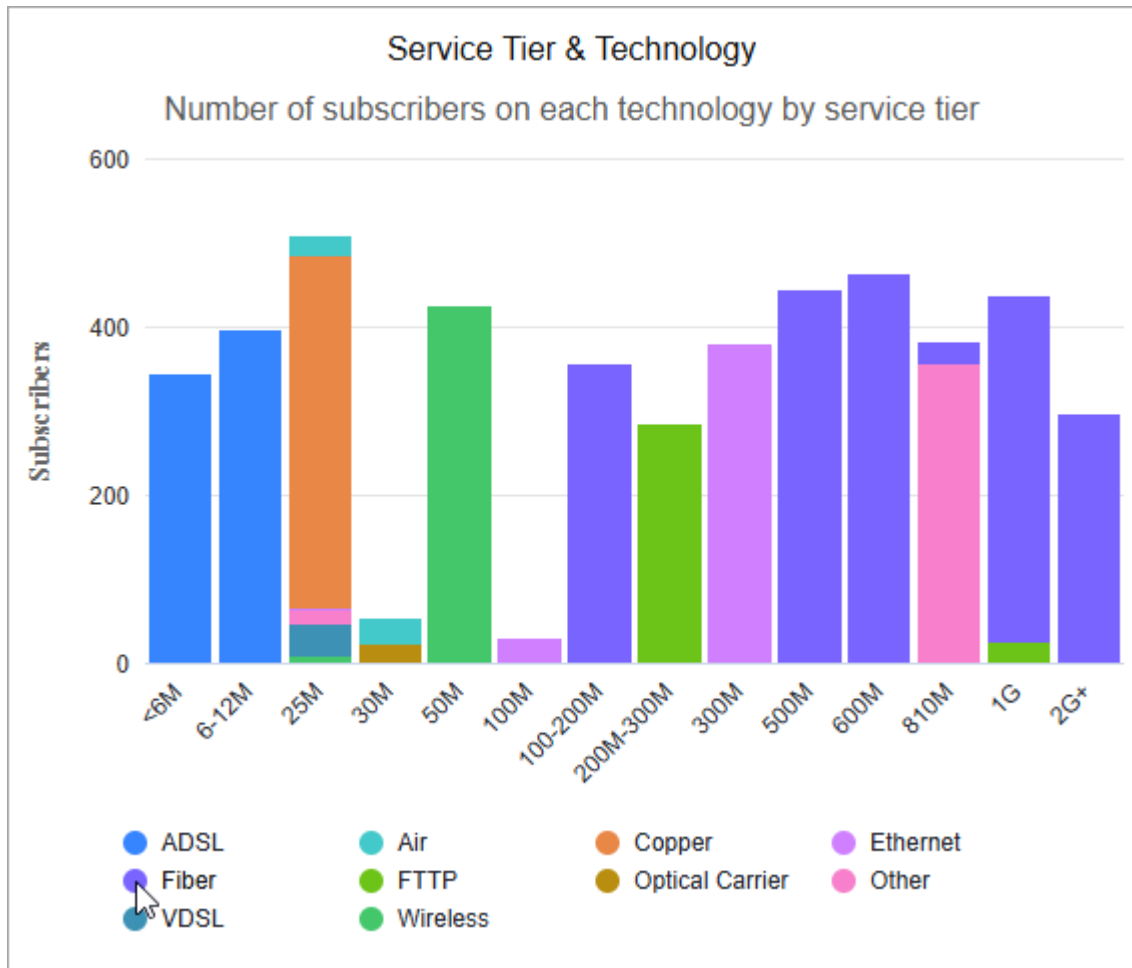
Some KPI description text may be hidden from view. To reveal, click the *down-facing arrow*.



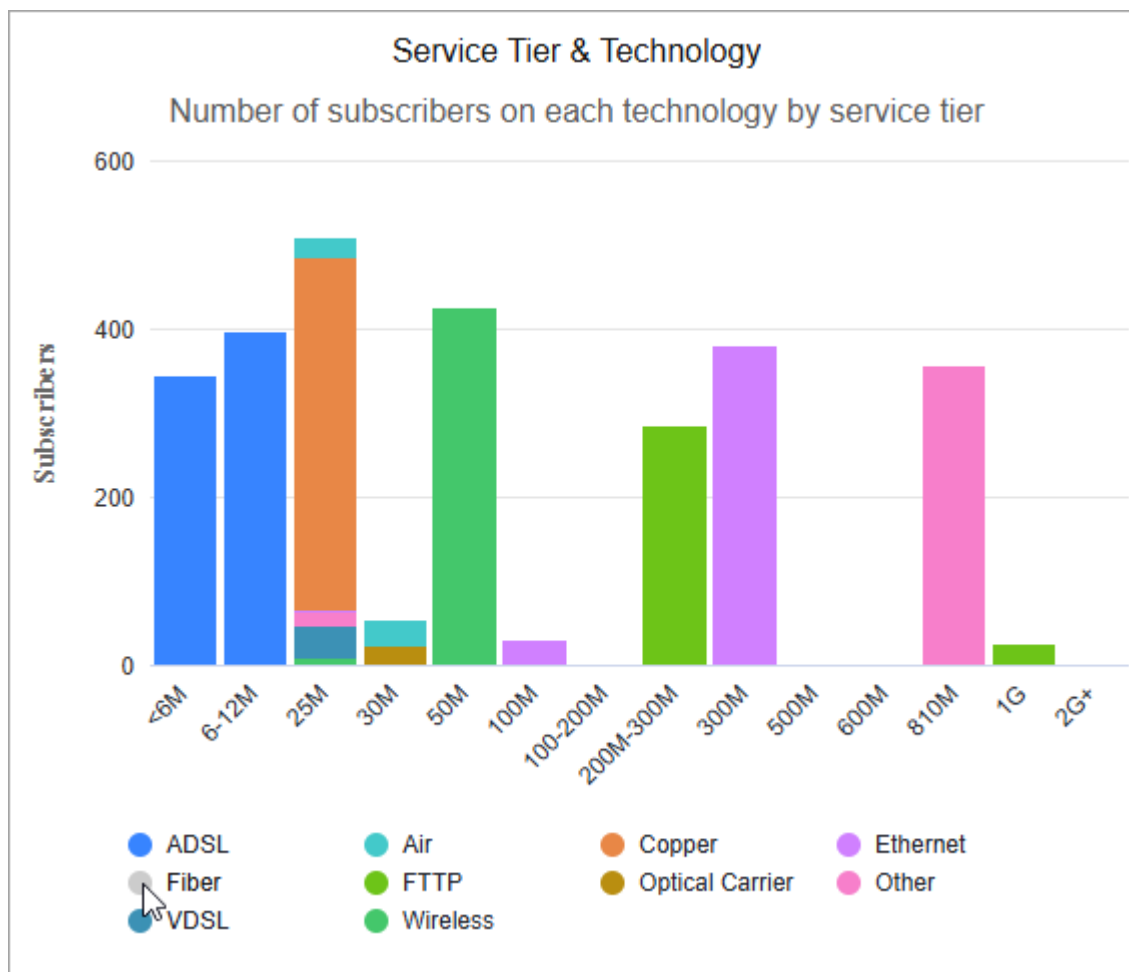
To show or hide chart layers

You can hide chart data to focus on specific data. Click on a chart legend key to show or hide that layer.

Example bar chart with layers visible

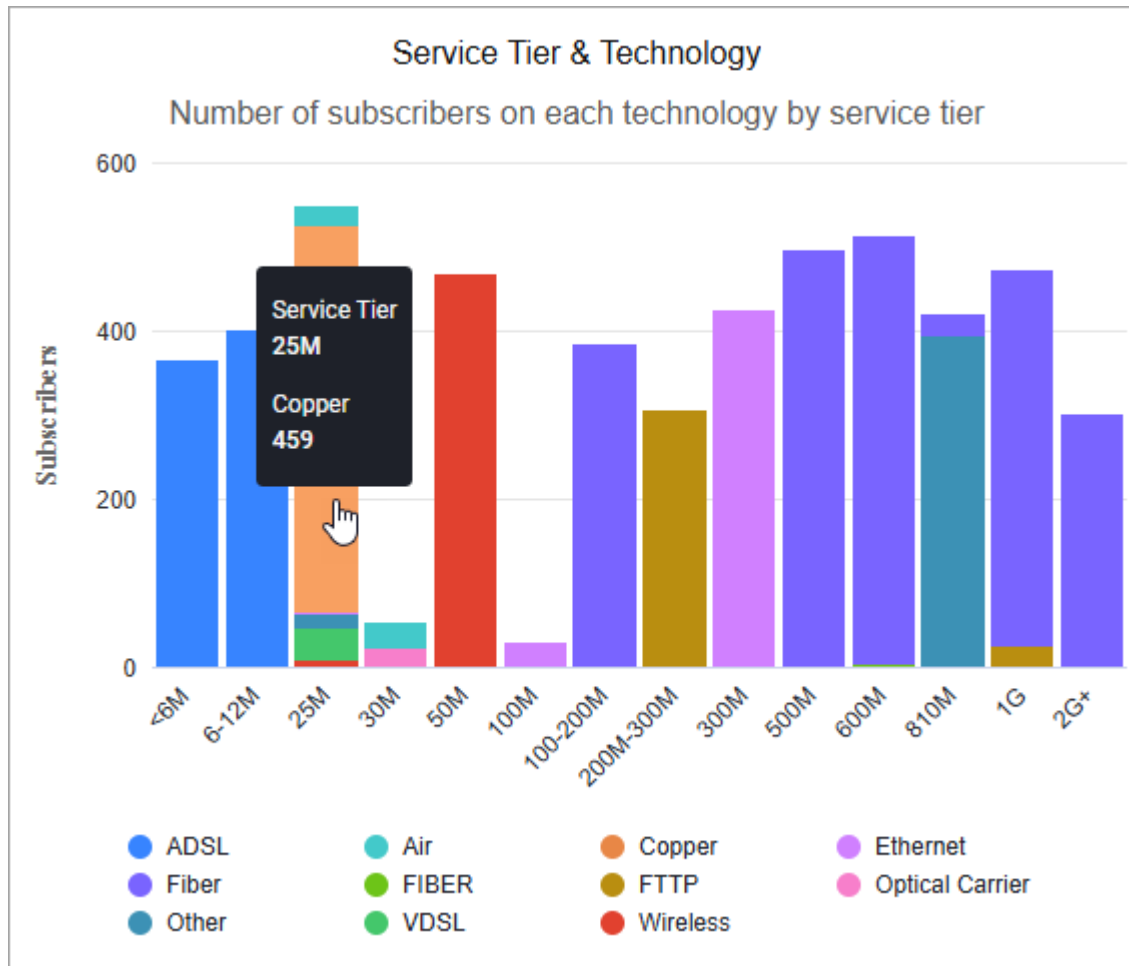


Example bar chart with a layer hidden



To view data for a chart data marker

Hover over a data marker (for example: a bar segment, a pie chart slice) to view data for that segment.



Explore Insights Dashboards

This topic provides an overview of the **Dashboards** page in Calix Engagement Cloud.

The Dashboards page presents subscriber data at-a-glance, allowing marketers to quickly and easily monitor trends in subscriber growth and churn, track managed services adoption rates, and understand subscriber data usage habits.

The Dashboards page has two tabs for viewing subscriber data:

- **Business Performance** presents overall trends in subscriber growth, churn, and service adoption.
- **Subscribers** provides an overview of subscriber composition and tracks subscriber data and application usage trends for both residential and business subscribers.

The Calix Engagement Cloud Plus entitlement adds two dashboards for prospect data. See *Explore Residential Prospect Demographic Data* and *Explore SMB Demographic Data* (on page [153](#)) for more.

Limitations

- Dashboards charts do not support sorting data along the axes.

Filter, View, and Export Dashboard Data

This topic describes how to filter, view, and export Dashboards data in Calix Engagement Cloud.

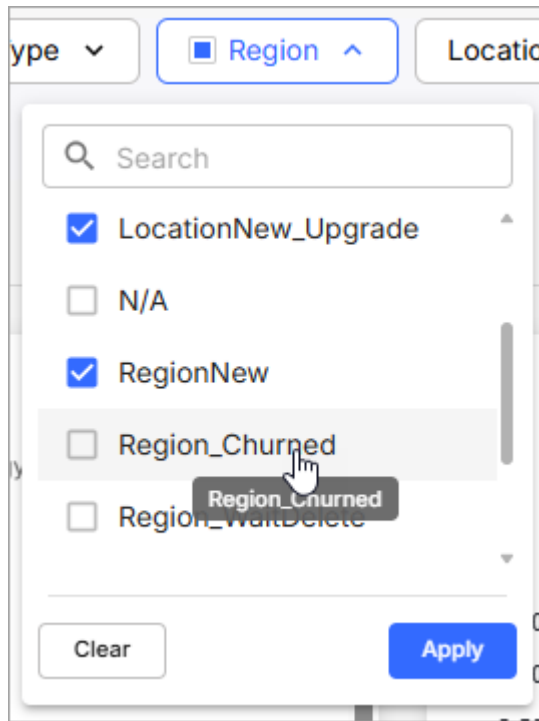
Note for orgs migrated to the next-generation Calix Cloud platform: In traditional Engagement Cloud, segmentation was performed on the Explore Data page. In next-generation Engagement Cloud, perform all advanced filtering and segmentation on the Segments page. See *Work with Subscriber Segments* (on page [58](#)) for more.

Filter Dashboards Data

To apply filters

1. Click on a filter button to expand its options menu.

2. Select the desired option(s), and then click **Apply**. Additionally, you can search the filter list by typing into the Search box within the drop-down menu.



A blue square on a filter button indicates the filter has options applied.

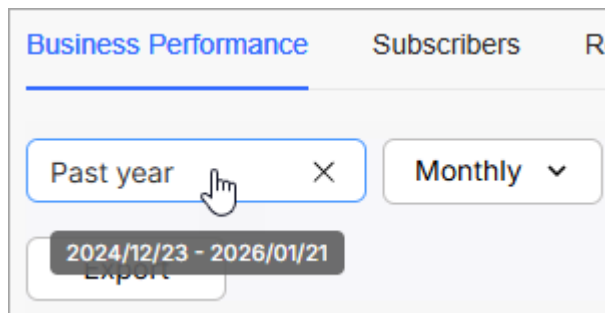


To clear a filter, expand the filter drop-down menu and click **Clear**.

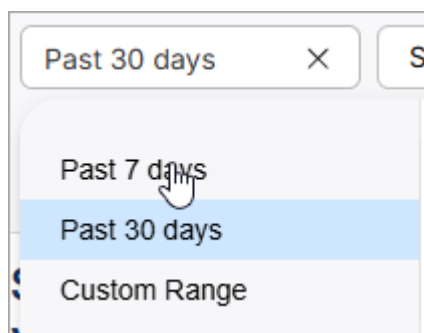
To filter Business Performance dashboards by date

Use the date filter to view historical Business Performance data for up to the previous five years.

1. Click on the **Date** filter.



2. Select the desired time period.



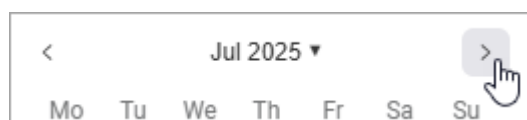
3. To configure a custom time range:

- a. Click **Custom Range**.

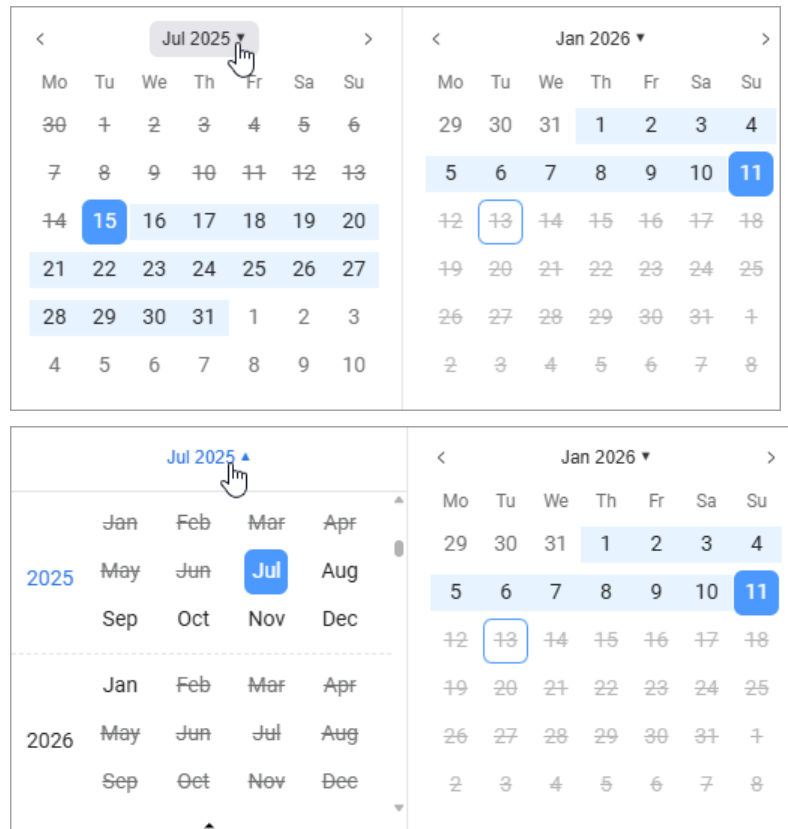
The calendar menu opens.

- b. To select the start date and end date:

- Click on the left and right-facing arrows to scroll to the desired month:

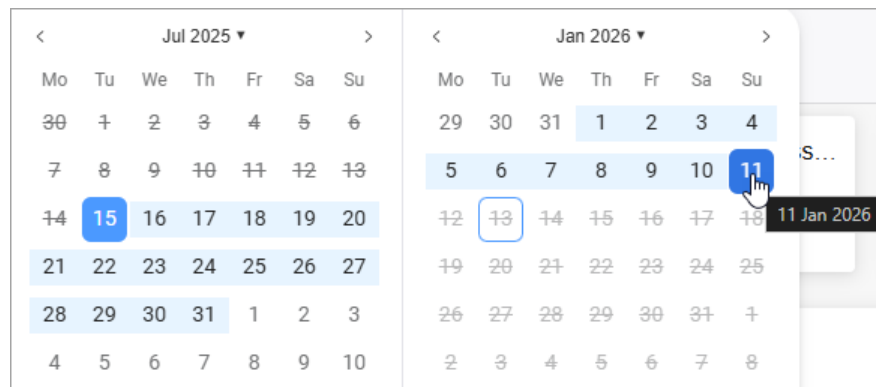


- Alternately, click the *down-facing arrow* to the right of the year, and then select the desired month within the year:



- Click dates on the calendar to select the start date and end date.

The calendar menu shows two months. If the campaign spans multiple months, choose the start date on the left and the end date on the right.



- Click the **Ok** to apply the custom date range.

Export Subscriber Data

From the Business Performance and Subscribers dashboards, you can export data for either individual charts or an entire dashboard.

To export data from Business Performance dashboards

You can export data from Business Performance dashboards to an Excel (.xlsx) file.

1. Go to **Dashboards > Business Performance**.
2. Select either the **Health** or **Adoption** tab.
3. Apply the desired filters.
4. Click **Export**.
5. Choose how to group the data:
 - Customer Type
 - Region
 - None
6. Click **Export**.

To export data from Subscribers dashboards

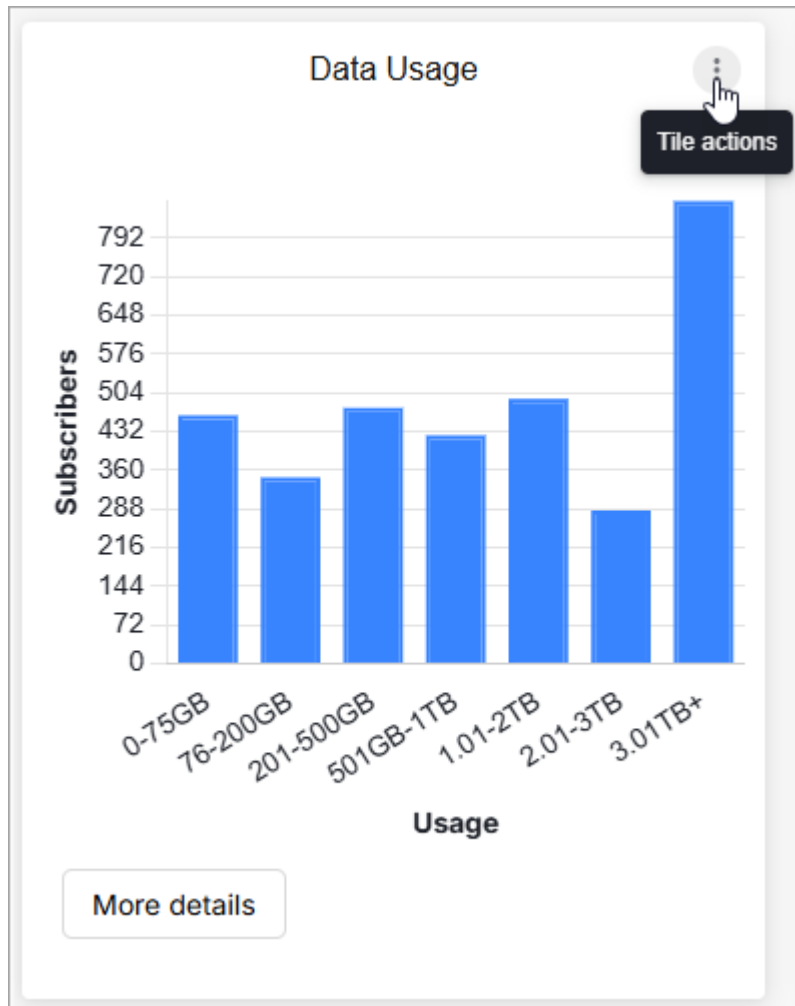
You can export data from Subscribers dashboards to a CSV file.

1. Go to **Dashboards > Subscribers**.
2. Select either the **Overview**, **Experience**, **Demographic**, or **Business** tab.
3. Apply the desired filters.
4. Click **Export**.
5. Select the checkbox for each data column to include in the file.
6. Click **Export**.

To export data from a chart

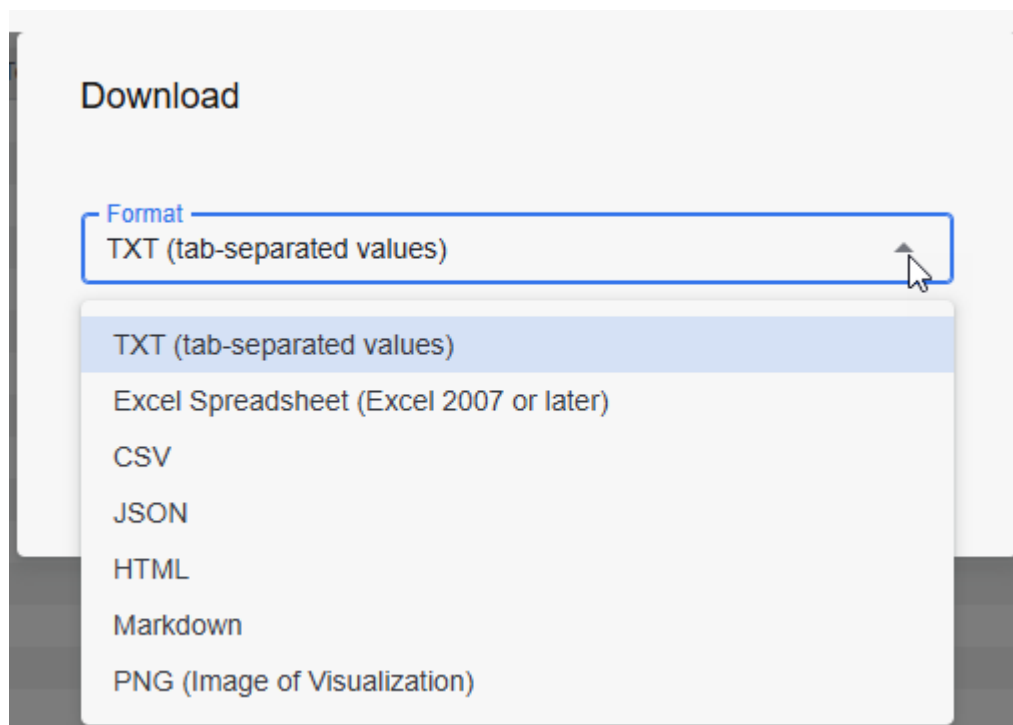
From Dashboards, you can export data from individual charts.

1. Click on the *Tile actions* icon in the upper right corner of the chart tile and select **Download data**.



2. Select your preferred file type.

Note: The PNG option captures only the chart portion visible on screen and excludes portions that require scrolling to see.

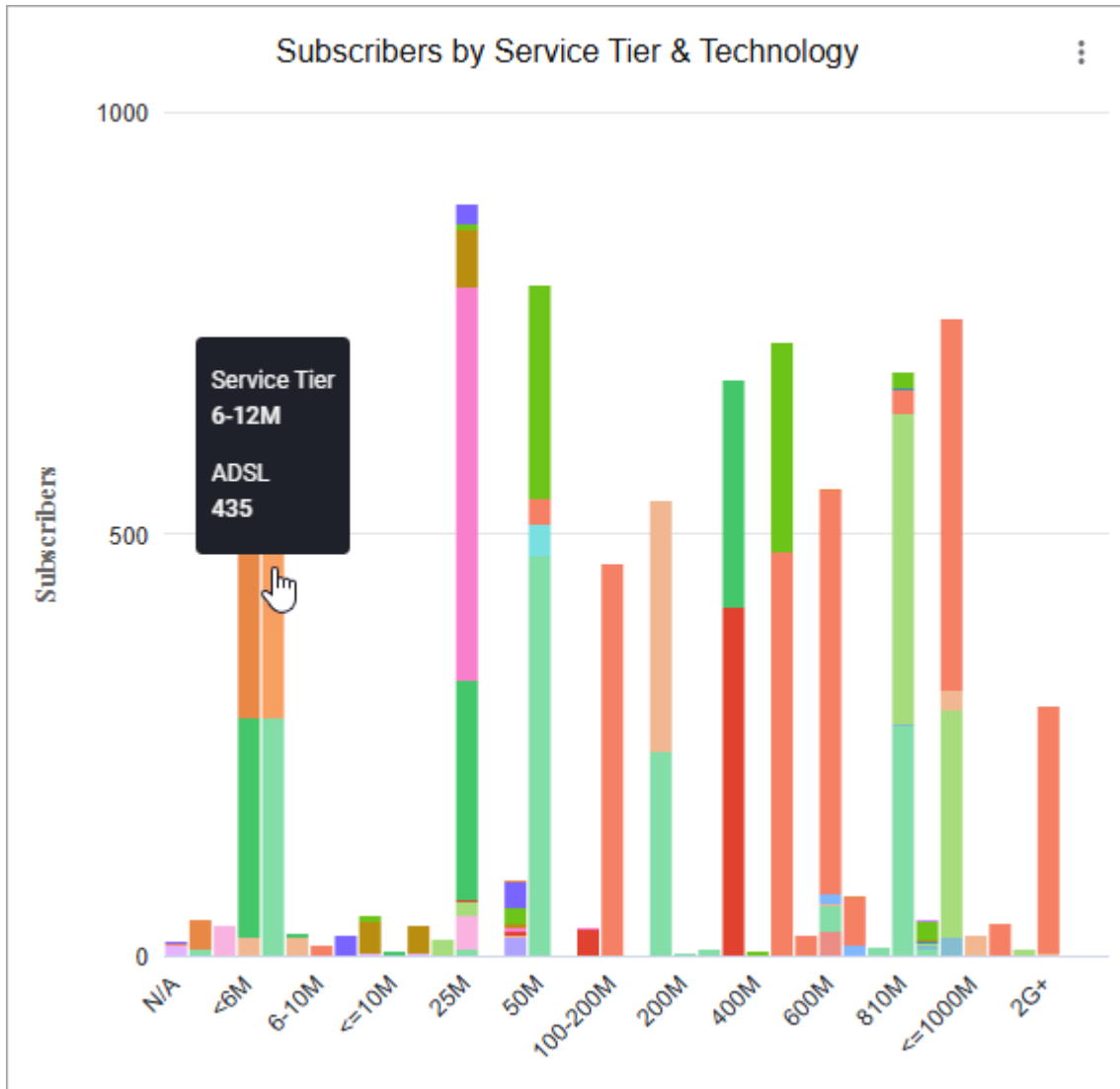


3. (Optional) Click **Advanced data options** to configure additional download settings:
 - Results: Choose to download results with or without data visualizations (for example: re-labeled columns, hidden values) applied.
 - Data values: Choose either *Formatted* or *Unformatted*.
 - Number of rows to include: Choose whether to include the current results table, all results, or a specified number of rows.
4. (Optional) Click **Open in Browser** to view the table in your chosen format in a new browser window.
5. Click **Download**.

View Chart Data

To view additional data for a data marker

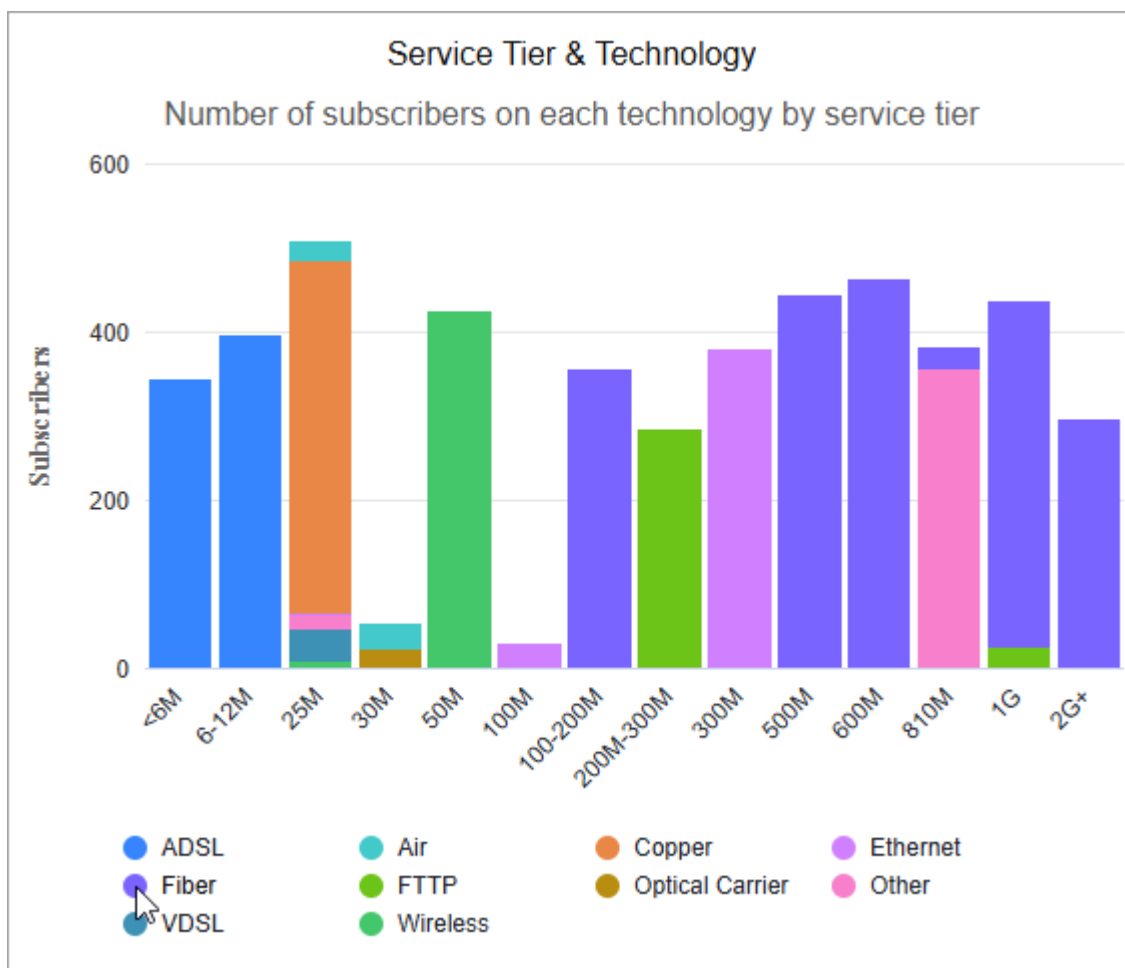
Hover over a data marker (for example: a line, a bar segment, or a pie chart slice) to view additional data.



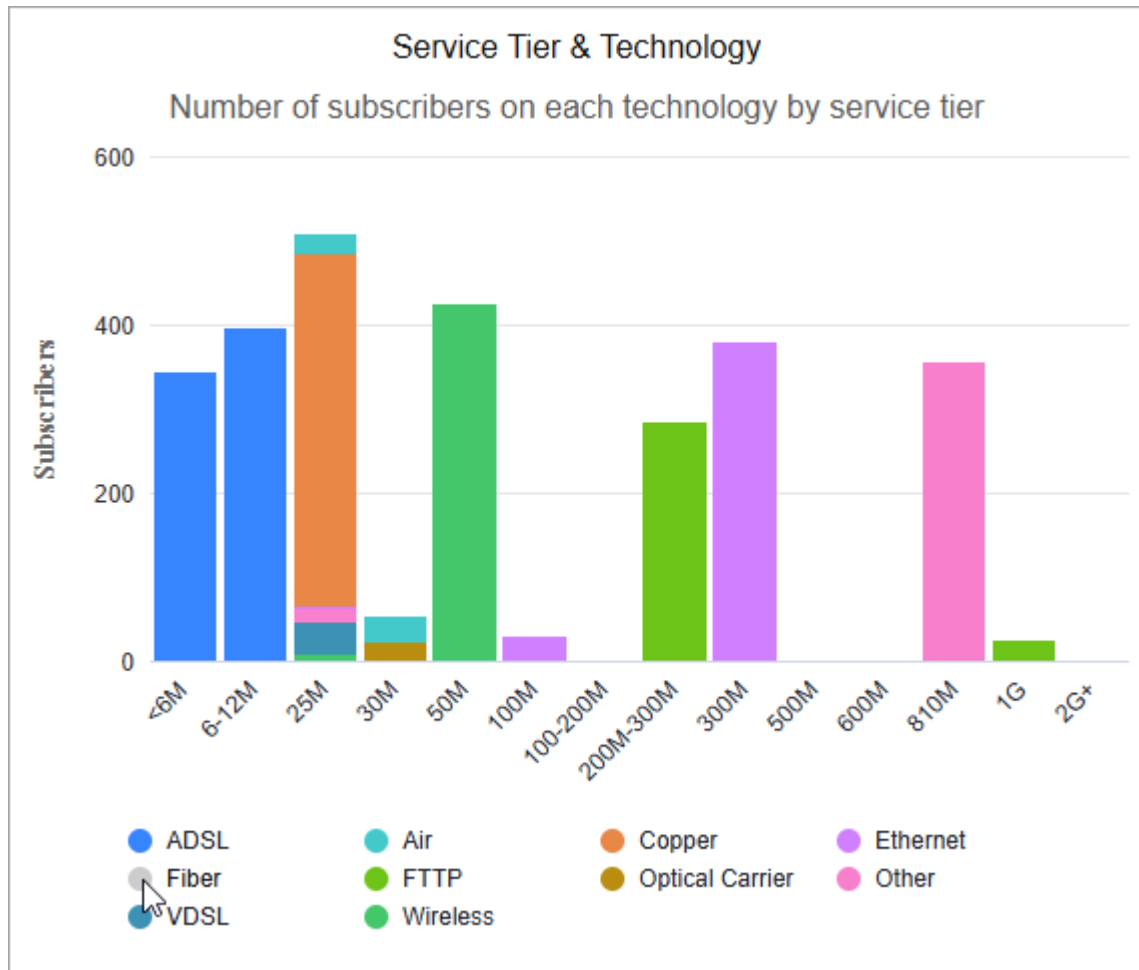
To show or hide chart layers

Hide chart layers to focus on specific data. Click on a legend key to hide that layer.

Example chart with layers visible



Example chart with a layer hidden

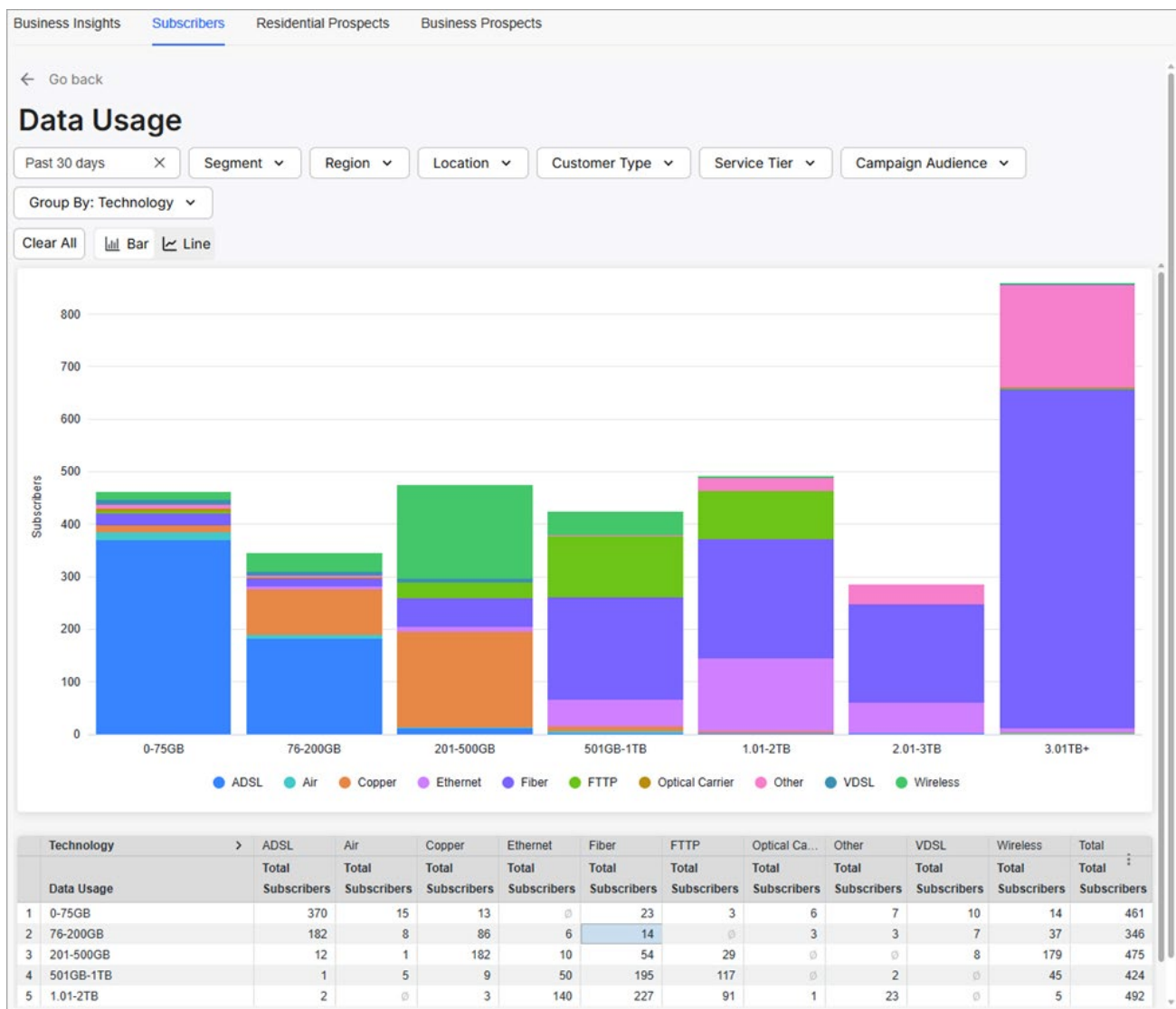


Use the Chart Drilldown

Some charts on the Business Insights and Subscribers dashboards allow you to explore data on a deeper level. To open the drilldown page, click the **More details** button on a chart.

The drilldown view includes data filters, a chart view, and a pivot table. To download chart data, click on the *three dots* icon in the upper-right of the chart and select **Download data**.

Sample chart drilldown

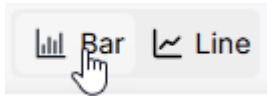


Access the drilldown for the following charts:

- Business Insights > Health tab:
 - New Subscribers
 - Churned Subscribers
 - Service Tier Changes
- Subscribers > Experience tab:
 - Data Usage
 - Blocked Threats
 - Average Devices

To change the drilldown chart type

View the chart as either a bar chart or a line chart.

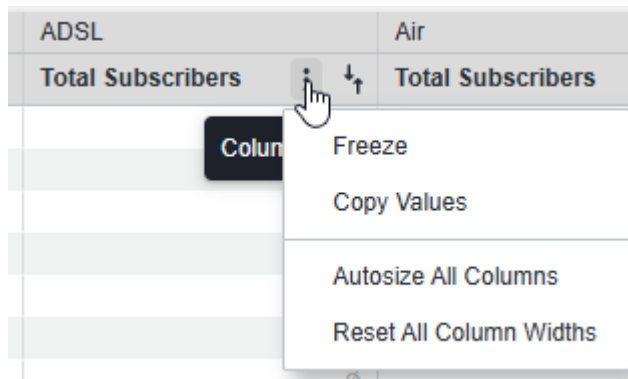


To use the drilldown pivot table

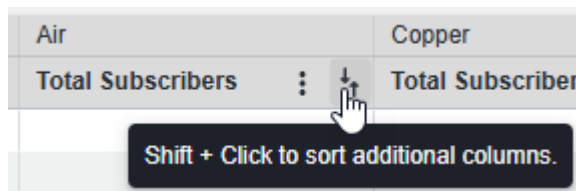
Manipulate pivot table columns

Hover over a column header to access additional column options:

- Click the *three dots* icon to access the column options menu.

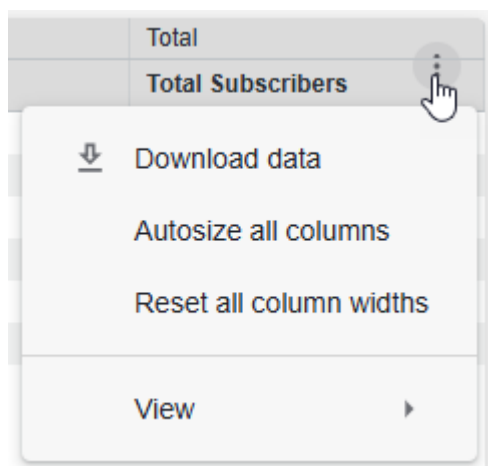


- Freeze:** Pins the column to the left side of the table and keeps the column visible (stationary) during horizontal scrolling. To unfreeze a column, click the three dots for the column and select **Unfreeze**.
- Copy Values:** Copies all values in the column to the clipboard.
- Autosize All Columns:** Shrinks column widths to fit the data in each column.
- Reset All Column Widths:** Expands columns evenly to fit the entire table width.
- Click the *opposing arrows* icon to sort column values in ascending or descending order.



Manipulate the pivot table

Click the *three dots* icon in the upper right corner of the table to access the table options menu:



- **Download data:** Exports table data to a file type of your choosing. See the "To export chart data" procedure above for more.
- **Autosize all columns:** Shrinks column widths to fit the data in each column.
- **Reset all column widths:** Expands columns evenly to fit the entire table width.
- **View:** Opens the table in a popup window.

View Business Performance Insights

This topic describes the **Dashboards > Business Performance** page in Calix Engagement Cloud.

Business Performance provides operational and marketing insights for your organization, helping you take immediate, proactive action in response to subscriber growth and adoption trends.

Business Performance presents data on two dashboards:

- **Health** displays trends in subscriber growth, churn, and average revenue per user (ARPU)
- **Adoption** displays trends in service tier changes, GigaSpire[®] adoption, and SmartHome[™] Managed Services adoption

Business Performance Dashboard Quick Filters

Refine Business Performance data with the following filters:

- Date range (default: Past year)
- Time increment (granularity)
 - Choose time increments for the chart display axes (for example: Daily, Weekly, Monthly, Quarterly). Options vary depending on the selected date range.
- Customer Type (Residential or Business)
- Region

- Location
- Service Tier
- Campaign Audience

The Customer Type, Region, Location, and Service Tier filters populate data from billing. For more on using filters, see Filter, View, and Export Dashboard Data.

Health Tab Charts and KPIs

The following table describes each chart and KPI on the Business Performance > Health tab. These data update daily with data from billing.

KPI/Chart	Description
Total Subscribers KPI	Shows the number of subscribers in billing with service tier and technology information.
Active Subscribers KPI	Shows the number of billing subscribers with internet (data) usage within the selected time frame.
Total Subscribers versus Active Subscribers chart	Compares the number of total subscribers to the number of active subscribers within the selected time frame. Axes: • Horizontal axis: Time increments • Vertical axis: Number of subscribers
New Subscribers chart	KPI: Shows the number of subscribers added to billing within the selected time frame. Chart: Tracks new subscriber trends within the selected time frame. Axes: • Horizontal axis: Time increments • Vertical axis: Number of subscribers
Churned Subscribers KPI and chart	KPI: Displays the number of subscribers removed from billing within the selected time frame. Chart: Tracks trends in subscriber churn within the selected time frame. Axes: • Horizontal axis: Time increments • Vertical axis: Number of subscribers

KPI/Chart	Description
Internet Service ARPU KPI and chart	KPI: Shows the average revenue per user (ARPU). ARPU is calculated as Total Revenue / Total Subscribers. Chart: Tracks monthly trends in ARPU within the selected time frame. Axes: - Horizontal axis: Time increments - Vertical axis: ARPU (dollars)
Service Tier Changes Sankey chart	Shows how many subscribers either upgraded or downgraded their service tier within the selected time frame. - Chart left: Service tier at the beginning of the selected time frame. - Chart right: Service tier at the end of the selected time frame. Note: An <i>upgrade</i> occurs when a subscriber switches to a service tier with a higher download speed. A <i>downgrade</i> happens when a subscriber switches to a service tier with a lower download speed.

Adoption Tab Charts and KPIs

The following table describes each chart and KPI on the Business Performance > Adoption tab. These data update daily with data from billing.

KPI/Chart	Description
Subscribers by Service Tier & Technology chart	Displays the number of wireless, cable, fiber, and DSL subscribers within each service tier. Each colored bar segment represents a different technology, and the size of a segment indicates the number of subscribers with that technology within that service tier. Axes: - Horizontal axis: Service tier - Vertical axis: Number of subscribers
Subscribers by Service Tier chart	Displays the number of subscribers within each service tier over the selected time frame. Axes: - Horizontal axis: Service tier - Vertical axis: Number of subscribers

KPI/Chart	Description
Subscribers with GigaSpires chart	Displays the number of subscribers with a Calix GigaSpire system over the selected time frame. Axes: - Horizontal axis: Time increments - Vertical axis: Number of subscribers
SmartHome Subscribers Overview KPIs and chart	Shows how many subscribers have CommandIQ[®], ExperienceIQ[®], ProtectIQ[®], Arlo[®], and Bark[®]. Axes: - Horizontal axis: Managed service - Vertical axis: Number of subscribers
Subscribers with CommandIQ chart	Displays two trendlines: - Has CommandIQ: Tracks subscribers who have created a CommandIQ account and successfully onboarded a system in the app. - Has used CommandIQ: Tracks subscribers who have logged in to the app. Axes: - Horizontal axis: Time increments - Vertical axis: Number of subscribers
Subscribers by SmartHome Managed Services chart	Tracks ExperienceIQ, ProtectIQ, Arlo, and Bark adoption rates over the selected time frame. Axes: - Horizontal axis: Managed services - Vertical axis: Number of subscribers
Business Subscriber Overview KPIs	Shows business subscriber metrics: - Business subscribers: Number of subscribers with a customer type of "business" in billing. - SmartBiz subscribers: Number of subscribers with the SmartBiz[™] service enabled in Calix Cloud. - CommandWorx subscribers: Number of subscribers who created a CommandWorx[®] account and successfully onboarded a system in the app.

KPI/Chart	Description
Subscribers with SmartBiz chart	Shows how many subscribers purchased SmartBiz over the selected time frame. Axes: - Horizontal axis: Time increments - Vertical axis: Number of subscribers
Subscribers with CommandWorx chart	Displays two trendlines: - Has CommandWorx: Tracks business subscribers who created a CommandWorx account and successfully onboarded a system in the app. - Has used CommandWorx: Tracks business subscribers who have logged in to CommandWorx. Axes: - Horizontal axis: Time increments - Vertical axis: Number of subscribers
SmartTown Users chart	Tracks the number of subscribers with a SmartTown™ subscription and the number of hotspot connections. Axes: - Horizontal axis: Time increments - Vertical axis: Number of connections

View Residential Subscriber Insights

This topic describes each chart and KPI on the **Dashboards > Subscribers** page in Calix Engagement Cloud.

The Subscribers page provides an overview of your subscriber base and presents internet experience and data usage trends for both residential and business subscribers.

The Subscribers dashboard has two sub-tabs:

- **Overview** displays subscriber growth and service adoption trends
- **Experience** provides insight into subscriber data usage and internet experience trends

Subscribers Dashboard Quick Filters

Refine subscriber data with the following filters:

- Segment (includes both pre-built and custom segments)

- Region
- Location
- Customer Type (Residential or Business)
- Service Tier
- Campaign Audience

The Customer Type, Region, Location, and Service Tier filters populate data from billing. For more on using filters, see Filter, View, and Export Dashboard Data.

Overview Tab Charts and KPIs

See the following table for a description of each Subscribers > Overview dashboard chart and KPI. Charts and KPIs show data for the previous 30 days.

Chart/KPI	Description
Total Subscribers KPI	Shows the total number of subscribers with service and technology information in billing. This count includes subscribers not matched with an endpoint.
Active Subscribers KPI	Shows the number of subscribers in billing with data (internet) usage.
Subscribers Data Match Rate KPIs	Provides an overview of matches rates for subscribers: • Endpoint Match: Percentage of total subscribers matched to an endpoint. • Demographic Match*: Percentage of total subscribers matched to a residential demographic record. • Business Data Match*: Percentage of total subscribers matched to an SMB data record.
Subscriber Type KPIs	Shows the number of residential and business subscribers in billing.
Internet Service ARPU KPIs	Shows the daily average revenue per user (ARPU) for both residential subscribers and business subscribers. To calculate ARPU, Engagement Cloud sums the monthly fee value for each subscriber in billing and divides that sum by the total number of subscribers in billing.
Subscribers with GigaSpire KPI	Shows the number of subscribers (based on total subscribers) with a Calix GigaSpire [®] system.
SmartHome Subscriber Overview KPIs	Shows how many subscribers (based on total subscribers) have a CommandIQ [®] , ProtectIQ [®] , ExperienceIQ [®] , Arlo [®] , and Bark [®] SmartHome managed services.

Chart/KPI	Description
SmartBiz Subscriber Overview KPIs	Provides counts of subscribers with SmartBiz and CommandWorx: • SmartBiz: Number of subscribers with a SmartBiz™ subscription. • CommandWorx: Number of subscribers who have onboarded a GigaSpire system in CommandWorx™.
Subscribers by Service Tier & Technology chart	Shows the distribution of access technologies (for example: Fiber, DSL, Ethernet) across the service tiers and the number of subscribers with each technology within each service tier. Axes: • Horizontal axis: Service tier • Vertical axis: Number of subscribers
Subscriber by System Family chart	Shows subscriber distribution by system family (for example: GigaSpire, GigaPro, Third-Party). Axes: • Horizontal axis: System family • Vertical axis: Number of subscribers

* Feature available with Calix Engagement Cloud Plus entitlement.

Experience Tab Charts and KPIs

See the following table for a description of each Subscribers > Experience dashboard chart and KPI.

Chart/KPI	Description
Data Usage chart	Displays the distribution of subscribers by the total amount of data used (both sent and received) within the selected time frame. Axes: • Horizontal axis: Data increment • Vertical axis: Number of subscribers Click More details to view detailed insights data in the chart <i>drilldown</i> (on page 37).

Chart/KPI	Description
Average Quality of Experience (QoE) chart	Displays the distribution of subscribers by average QoE score. QoE presents a score between 1 and 5 that represents a subscriber's internet quality over a given time frame. QoE uses metrics such as Wi-Fi signal strength, internet speed, and service interruptions to evaluate a subscriber's overall service experience. This metric applies only to Calix GigaSpire systems. Axes: - Horizontal axis: QoE score range - Vertical axis: Number of subscribers
Downstream Service Limit Hits chart	Shows the distribution of subscribers by the number of times they exceeded 80% of their provisioned downstream bandwidth within the selected time frame. Axes: - Horizontal axis: Number of service limit hits - Vertical axis: Number of subscribers
Upstream Service Limit Hits chart	Shows the distribution of subscribers by the number of times they exceeded 80% of their provisioned upstream bandwidth within the selected time frame. Axes: - Horizontal axis: Number of service limit hits - Vertical axis: Number of subscribers
Speed Tests chart	Shows the distribution of subscribers by the number of minutes spent on speed testing websites. Axes: - Horizontal axis: Time increments (in minutes) - Vertical axis: Number of subscribers
Competitor Site Visits chart	Shows the distribution of subscribers by number of minutes spent on competitor websites. Axes: - Horizontal axis: Time increments (in minutes) - Vertical axis: Number of subscribers

Chart/KPI	Description
Blocked Threats chart	Shows blocked intrusions, web threats, and viruses as percentages of the total number of threats blocked by ProtectIQ. Click More details to view detailed insights data in the chart <i>drilldown</i> (on page 37).
Average Devices chart	Displays the distribution of subscribers by the average number of devices (both wired and wireless) connected to the subscriber's system. Axes: • Horizontal axis: Number of devices • Vertical axis: Number of subscribers Click More details to view detailed insights data in the chart <i>drilldown</i> (on page 37).
Device Types	Shows the distribution of subscribers by device type (for example: Phone, Computer, Gaming Console).
Smart Camera Ownership	Shows the distribution of subscribers by camera brand: • Arlo • Nest • Ring • Other

Work with Subscriber Segments

This topic provides information on subscriber segments in Calix Engagement Cloud.

What is a segment?

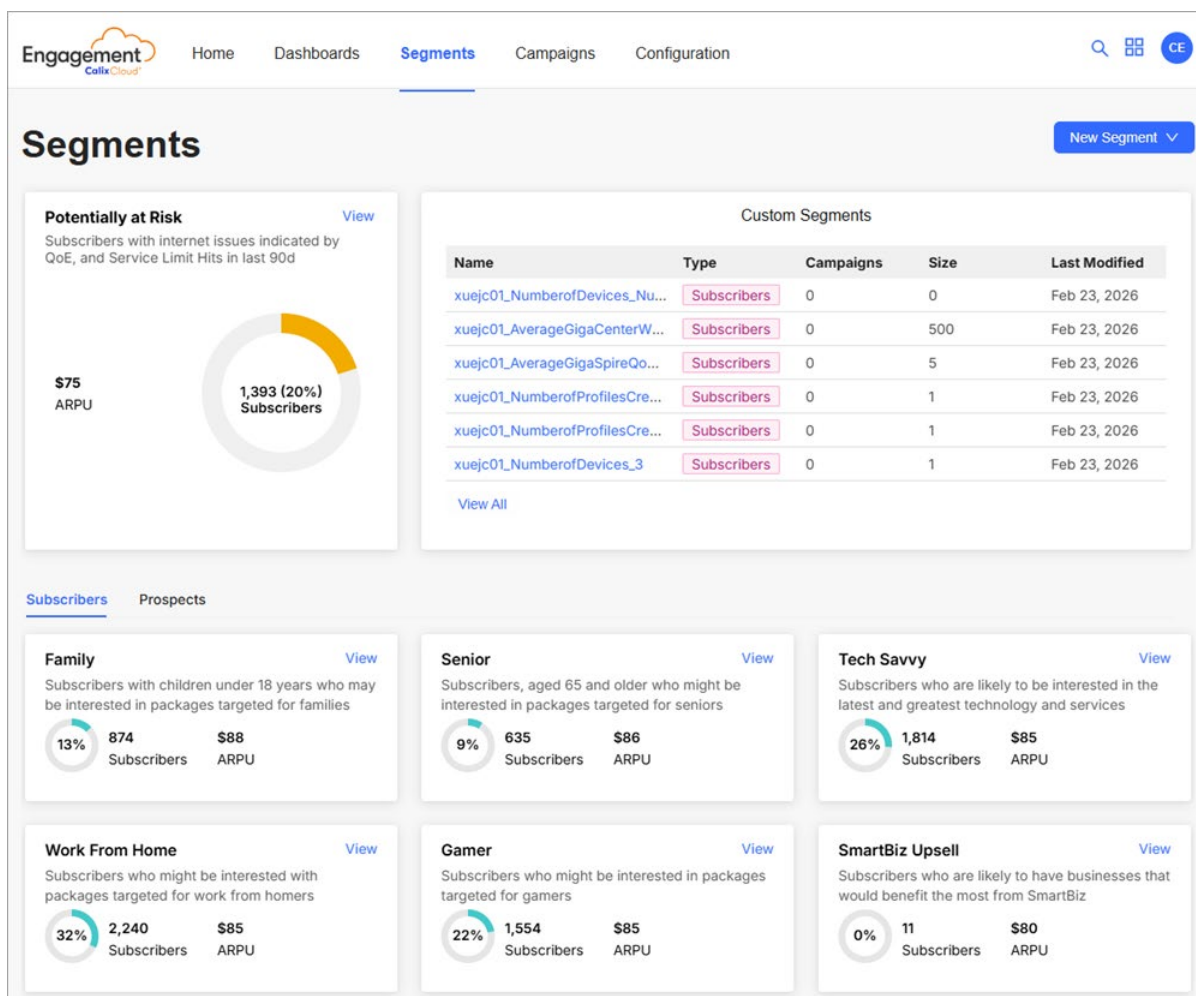
A *segment* is a subset of subscribers that shares common characteristics, needs, or behaviors. Segments help you tailor products, services, and marketing efforts to better meet subscriber preferences. View detailed segment insights from the Dashboards page.

Engagement Cloud provides pre-built segments for specific personas (for example: gamers) and allows for custom segmentation by filtering subscriber data. Use pre-built segments for quickly gaining insights or as a starting point for further data refinement. Pre-built segments update daily with data from billing.

Viewing Segments

The **Segments** page lists custom segments, displays insights for pre-built segments, and allows for advanced segmentation. View details for each custom segment, including the number of campaigns associated with the segment, the date the segment was created, and the date the segment was last modified (custom segments only).

To quickly locate a specific custom or pre-built segment, filter the segments list by type or search for the segment by name. You can also delete segments not used in campaigns. Click on a segment to view detailed insights in **Dashboards > Subscribers**.



The Segments page includes the following sections:

Section	Description
New Segment button	Click to create a subscriber, residential prospect*, or business prospect* segment.
Potentially at Risk KPI	Shows the percentage of subscribers with a poor Wi-Fi experience and the average revenue per user (ARPU).
Custom Segments table	Displays a list of custom segments along with segment details.
Subscribers tab	Provides insights into each pre-built subscriber segment, including segment size and ARPU.

Section	Description
Prospects tab*	Provides insights into each pre-built prospect segment, including segment size and ARPU.

* Calix Engagement Cloud Plus entitlement required.

View Pre-Built Segment Details

From Segments, you can view high-level insights for pre-built subscriber segments, including:

- Segment size
- Internet service ARPU
- Number of GigaSpires deployed within the segment
- Service tier and technology
- Data usage

To view insights, click **View**.

Additionally, you can take the following actions on the segment:

- Create Campaign: Click to use the segment in either a triggered campaign or a scheduled campaign.
- View in Dashboard: Click to explore additional segment insights on the Dashboards page.

View Custom Segment Details

Click on a custom segment to view high-level insights for the segment, including:

- Total number of subscribers in the segment
- Number of active subscribers in the segment
- Internet service ARPU
- Number of GigaSpires deployed within the segment
- Number of subscribers with CommandIQ and CommandWorx
- Number of subscribers with each access technology within each service tier
- Campaigns using the segment

Additionally, you can take the following actions on the segment:

- Edit, copy, or delete the segment
- Use the segment in a scheduled or triggered campaign
- Explore detailed segment insights in Dashboards

Pre-Built Segments

Engagement Cloud provides the following pre-built subscriber segments:

Note: For information on pre-built demographic segments, see the *Work with Demographic Segments* (on page [156](#)) chapter.

Segment	Description	Metrics
Potentially at Risk	Identifies subscribers who meet at least one of the following criteria: - Over 15 service limit hits (downstream + upstream) within the last 90 days - Average QoE score < 3 over the last 90 days These subscribers may be at high risk of churning and may require proactive support.	Click on the segment tile to view the following metrics: - Segment Size KPI: Total number of unhappy subscribers - Internet Service Revenue Impact KPI: Total annual revenue gained from this segment (calculated as the subscriber's monthly fee * 12) - Internet Service ARPU KPI: Percentage of subscribers who receive the highest service levels - GigaSpires KPI: Number of GigaSpire[®] systems deployed within this segment - Service Tier Breakdown chart: Percentage of potentially at risk subscribers in each service tier - Potentially at Risk Subscribers chart: Percentage of potentially at risk subscribers over the last 90 days - Subscriber details table - Used in Campaigns table: Lists the campaigns using the segment
Gamers	Identifies subscribers with gaming consoles and high data usage from gaming applications (for example: Twitch). Subscribers in this segment want internet that delivers a lag-free gaming experience.	Click on the segment tile to view the following metrics: - Number of subscribers in the segment - Average revenue per user - Number of GigaSpires deployed within the segment - Number of subscribers in the segment with CommandIQ[®] - Number of subscribers in each service tier with fiber, wireless, cable, or DSL - Distribution of subscribers by tenure - Distribution of subscribers by data usage - Distribution of subscribers by number of devices - Most popular gaming devices by number of subscribers - Most popular gaming applications by number of subscribers - List of campaigns using the Gamers segment

Segment	Description	Metrics
Work from Home	Identifies subscribers with high data usage from work-related applications (for example: Microsoft Outlook). Subscribers in this segment expect a stable, reliable, and secure internet connection.	Click on the segment tile to view the following metrics: • Number of subscribers in the segment • Average revenue per user • Number of GigaSpires deployed within the segment • Number of subscribers in the segment with CommandIQ • Number of subscribers in each service tier with fiber, wireless, cable, or DSL • Distribution of subscribers by tenure • Distribution of subscribers by data usage • Distribution of subscribers by number of devices • Most popular business applications by number of subscribers • List of campaigns using the Work from Home segment

Create Subscriber Segments

This topic describes how to use the segment builder and how to create a subscriber segment in Calix Engagement Cloud. To begin, go to the **Segments** page.

Using the Segment Builder

The Segment Builder organizes all settings onto a single page for efficient segmentation:

Segments / Custom Segments / New Segment

New Segment

Subscribers

Cancel Save Segment

1

Name

Description (Optional)

Enter a name

Enter a description

2

Attribute Filters

Add attributes to narrow your segment

Clear Manage Filters

3

Map Filter

Draw a boundary on the map to further narrow your segment

Search address or latitude...

Layers

Select Boundaries



4

Preview

Size : 0

Download CSV

Search

Account No.	Name
No Data	

The following table describes each section of the Segment Builder.

Section	Description
1 Name and Description fields	Enter a unique, helpful name and description for the segment.
2 Attribute Filters	Displays currently applied filters and allows you to apply/remove filters. For a full list of filters and descriptions, see <i>Segment Attribute Filters</i> (on page 68).
3 Map Filter	Enables filtering by geographic area. See the 'Using the Map Filter' section below for more.
4 Segment preview	Lists details—including name and email address—for subscribers in the segment. Click Download CSV to export table data to a CSV file.

Using the Map Filter

The map enables filtering by geographic area for precise subscriber targeting. Map boundaries become filters that either include or exclude subscribers within the selected area. Use the map filter to define and visualize the geographic reach of both segments and campaigns.

By default, the map shows all your subscribers, with each subscriber represented by a blue dot. For prospect segments, each blue dot represents a prospect household within your base area.

Considerations

- Map views that display more than 200,000 prospects may affect browser performance and result in an "Unresponsive" error.
- The map filter supports up to one inclusion boundary (free-form or circle) and one exclusion boundary (free-form or circle).

To navigate the map

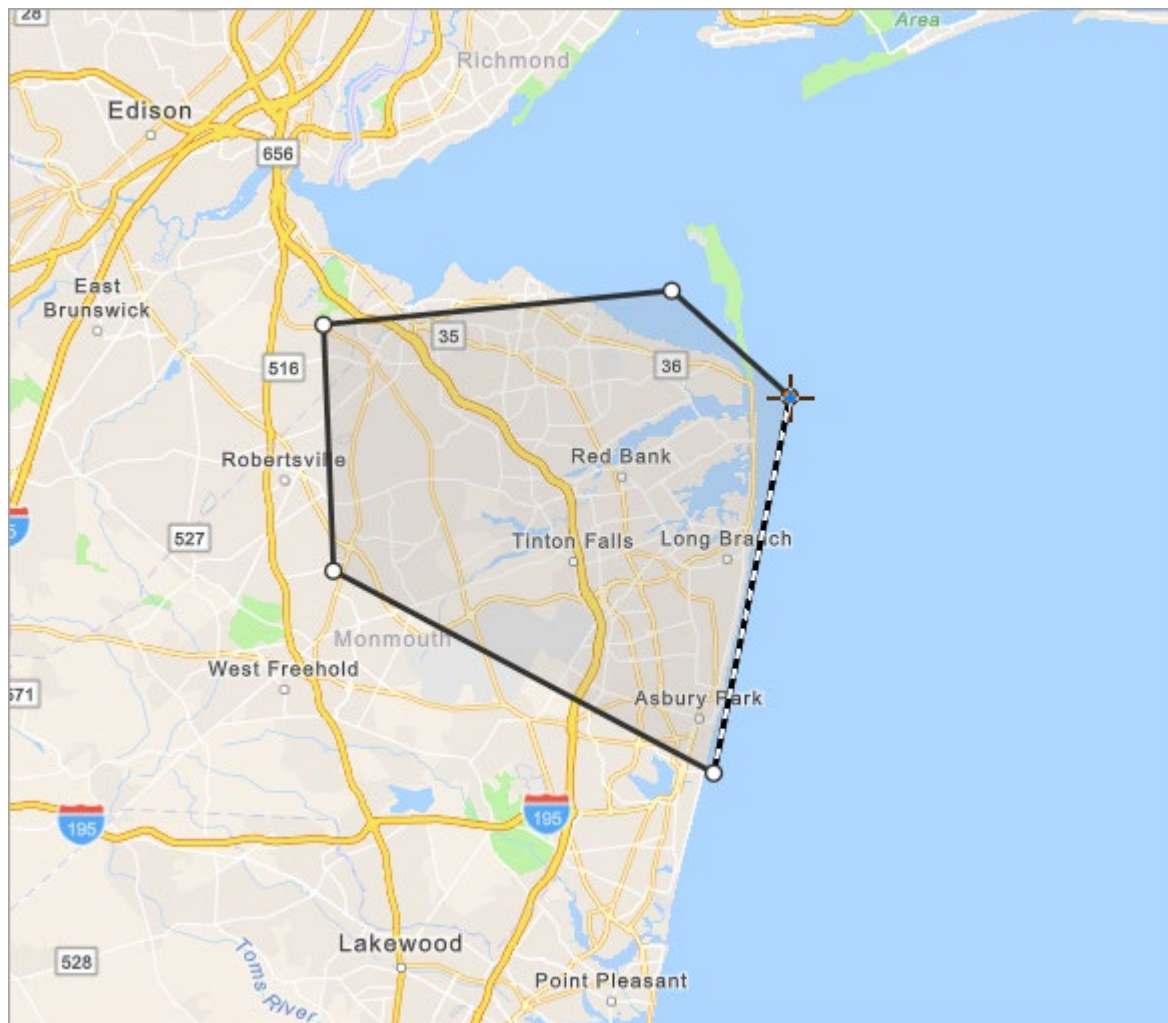
- **Adjust position:** Click on and drag the map to the desired location. Alternately, type an address or latitude and longitude coordinates into the Search field. When searching by address, enter the entire address, including street, city, and state (for example: 123 Main Street, Anywhere, KS).
- **Zoom in/out:** Zoom in or out with the scroll wheel on your mouse. Alternately, click on the + or the - in the bottom right corner of the map.

To select an area

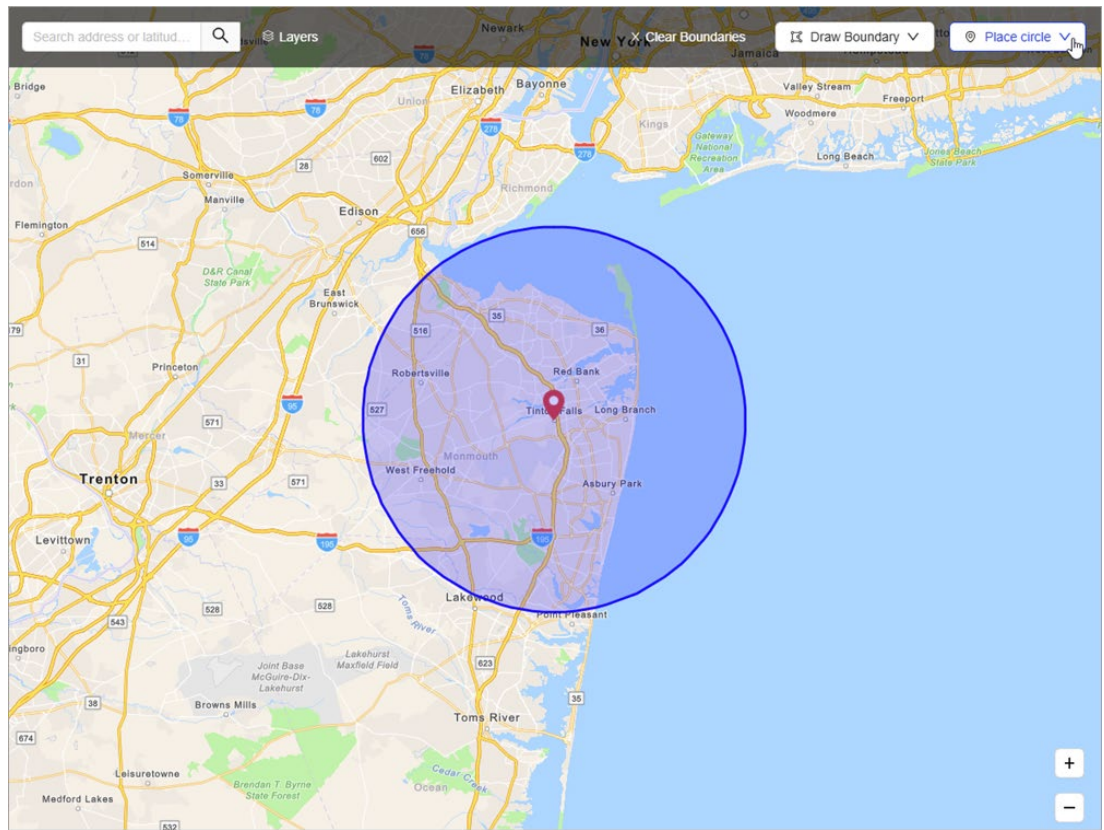
Use the following methods to define areas of inclusion or exclusion:

- **Draw Boundary:** Use this option to create a free-form, polygonal shape on the map. Click on the map to set points and outline the desired area. A dashed boundary line indicates the shape is

still open (see following image). To close it, either double-click or place the last point on the first point. Once closed, the map automatically centers on the selected area.



- **Place Circle:** Select this option to add a circle with a defined radius (representing a tower). Click on the map to set the center point of the circle, then enter the desired radius in the popup dialog. Note that the dialog may take a few seconds to appear after clicking on the map.



To clear or cancel a boundary selection

To remove a boundary, click **X Clear Boundaries**.

To cancel a selection while placing a boundary, click **Cancel**.

To use map layers

Map layers provide a visual aid for plotting the geographic reach of a segment or a campaign. Layers are not filters; use them to inform and guide boundary placement.

To create layers, Organization Administrators can upload GeoMap files via Administration > GeoMap Overlays. See "Create GeoMap Overlays (Layers)" for more.

Create a Subscriber Segment

To create a subscriber segment

1. Go to **Segments**.
2. Select **New Segment > Subscribers**.
3. Enter a name for the segment.
4. (Optional) Enter a description of the segment.
5. Apply the desired attribute filters:
 - a. Click **Manage Filters**.
 - b. Click **+ Add Filters**. See *Segment Attribute Filters* (on page [68](#)) for filter descriptions.
 - c. Select a filter from the drop-down menu.
 - d. Configure the filter options:
 - Select an operator. Options vary depending on the selected filter.
 - Select or enter the filter criteria.
 - e. Repeat steps b through d for each desired filter.
 - f. Click **Save**.
6. Select a region on the map using one of the following methods:

Note: The map supports up to one inclusion boundary (free-form or circle) and one exclusion boundary (free-form or circle).

- Draw a free-form boundary:
 - a. Click **Select Boundaries**.
 - b. Select either **Inclusion Boundary** or **Exclusion Boundary**.
 - c. Click on the map to set the initial point.
 - d. Move the cursor to a different spot on the map and click again to set the next point.
 - e. Continue until you enclose the desired area.
 - f. To close the shape, double-click or place the last point on the first point. Once the shape is closed, all boundary lines turn solid.
- Place a circular boundary:
 - a. Click **Place circle**.
 - b. Select either **Inclusion circle** or **Exclusion circle**.
 - c. Click on the map to set the center point of the circle.
 - d. Enter the desired radius (in miles) in the popup dialog, and then click **Save**.

Note: The dialog may take a few seconds to appear.

7. Click **Apply Selection**. To reset the boundary, click **Clear Boundaries**.

The Preview table lists subscribers in the segment who meet the filter criteria. Click **Download CSV** to export the segment to a CSV file.

8. Click **Save Segment**.

Next Steps

- View segment insights in *Dashboards* (on page [37](#))
- Use the segment in a *campaign* (on page [80](#))

Segment Attribute Filters

This topic describes how to apply segment attribute filters and provides descriptions for attribute filters in Calix Engagement Cloud.

Attribute filters enable precise segmentation. Choose from a robust set of filters that allows you to segment based on usage data, billing data, or demographic* data.

* Available with Calix Engagement Cloud Plus entitlement.

To add an attribute filter

1. Go to **Segments**.
2. Click **New Segment** and select the segment type.
3. Click **Manage Filters**.
4. Click **+ Add Filter**.
5. Select the desired filter from the drop-down menu.

Alternately, type into the **Select a filter** field to search for a filter. The search populates results as you type.

6. Select a logic operator. Available operators depend on the selected filter.

Logic operators allow you to include (*Equals, In*) or exclude (*Not Equals, Not In*) subscribers who meet more (*Greater Than, Greater Than or Equal*) or less (*Less Than, Less Than or Equal*) than the filter criteria. The *In* and *Not In* operators use "OR" logic (subscribers can meet any one of the selected options) and allow you to multi-select options from the filter options list. *Equals* and *Not Equals* use "AND" logic and allow only one selection.

7. Configure the filter criteria. Some filters—like Service Limit Hits—require you to manually input a number, while others—like ZIP Code List—allow you to select options from a drop-down menu.

To remove a filter, click the **X** on the right of the filter row. To remove all filters, click **Clear**.

Segment Attribute Filters for Residential Subscribers

Engagement Cloud provides the following attribute filters for creating segments. For a description of each demographic and business attribute filter available with Calix Engagement Cloud Plus, see *Work with Demographic Segments* (on page [156](#)).

Filter Category	Filter	Description	How to Apply the Filter
Pre-built Subscriber Segment	Family	Includes subscribers in the Family pre-built segment.	Select an operator, and then select either <i>false</i> or <i>true</i> .
	Tech Savvy	Includes subscribers in the Tech Savvy pre-built segment.	
	Work from Home	Includes subscribers in the Work from Home pre-built segment.	
	Gamer	Includes subscribers in the Gamer pre-built segment.	
	SmartBiz Upsell	Includes subscribers in the SmartBiz Upsell pre-built segment.	
	Potentially At Risk	Includes subscribers in the Potentially At Risk pre-built subscriber segment.	
Upload List	Subscriber Lists	Includes subscribers in files uploaded via Configuration > Subscriber/Prospect Lists.	Select an operator, and then select the desired list(s).
Account Information	Account Number	Filters subscribers by account number. You can enter up to 20 account numbers separated by a comma.	Select an operator, and then enter the account number(s).
	Customer Type	Filters subscribers by the customer type set in billing (<i>Residential, Business</i>).	Select an operator, and then select the customer type.
	Field1 - Field4	Filters subscribers by billing rider fields 1-4.	Select an operator, and then enter the desired value.
	Location	Filters subscribers by location.	Select an operator, and then select the desired location(s).
	Mobile App	Identifies subscribers with or without a mobile app: - Does not have Mobile App - Has CommandIQ - Has CommandWorx	Select an operator, and then select an option.

Filter Category	Filter	Description	How to Apply the Filter
	Monthly Data Fee	Filters subscribers by the billing monthly data fee value from billing.	Select an operator, and then select an option.
	Opt In/Opt Out Preference	Identifies subscribers who have either opted in to or opted out from marketing communications.	Select an operator, and then select either <i>Y</i> or <i>N</i> .
	Region	Filters subscribers by region.	Select an operator, and then select the desired region(s).
	Technology	Filters subscribers by access technology (for example: fiber, DSL, copper).	Select an operator, and then select the desired option(s).
	ZIP Code List	Filters subscribers by ZIP code. This filter populates with ZIP code lists uploaded from Configuration > ZIP Code File Upload. See <i>Manage ZIP Code Filter Lists</i> (on page 135) for more.	Select an operator, and then select the desired ZIP code(s) from the list: - To select an entire list, select the checkbox next to the list name. - To select ZIP codes within a list, click on the expand arrow for the list, and then select the checkbox for each ZIP code.
	ZIP/ZIP+4	Filters subscribers by billing or prospect record ZIP/ZIP+4 code.	Select an operator, and then select the desired value(s).
Campaigns	Campaign Audience	Filters by groups of subscribers previously used in a campaign.	Select an operator, and then select the desired audience(s).
Engagement and Behavior	Business Application Usage (GB)	Identify Work from Home subscribers by the amount of data used (both send and received) by business applications within the selected time frame. See <i>VPN and Business Applications</i> (on page 214) in the Reference Information section for a complete list of business applications. Tip! This filter replaces the "WFH" filter in traditional Engagement Cloud.	- Select an operator. - Enter the desired threshold (in gigabytes). - Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.

Filter Category	Filter	Description	How to Apply the Filter
	Competitor Site Visits (min)	Filters subscribers by the amount of time (in minutes) they spent on competitor websites within the selected time frame. Competitor websites are defined during the Engagement Cloud onboarding process.	1. Select an operator. 2. Enter the desired threshold (in minutes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Gaming Usage (GB)	Identify gamers by the amount of data used (both sent and received) by gaming applications and consoles within the selected time frame. Tip! This filter replaces the "Gaming" filter in traditional Engagement Cloud.	1. Select an operator. 2. Enter the desired threshold (in gigabytes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Last Connected to SmartTown	Filters subscribers by the most recent date one of their devices connected to a SmartTown hotspot.	Select a time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.

Filter Category	Filter	Description	How to Apply the Filter
	Maximum Number of Devices Connected to SmartTown	Filters subscribers by the number of devices connected to their SmartTown hotspot.	1. Select an operator. 2. Enter the desired threshold. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Number of Profiles Created in App	Filters subscribers by the number of Person profiles created in CommandIQ.	Select an operator, and then select the desired value(s).
	Social Usage (GB)	Filters subscribers by the amount of data used (both sent and received) by social applications within the selected time frame. Tip! This filter replaces the "Social" filter in traditional Engagement Cloud.	1. Select an operator. 2. Enter the desired threshold (in gigabytes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Streaming Usage (GB)	Filters subscribers by the amount of data used (both sent and received) by streaming applications within the selected time frame. Tip! This filter replaces the "Streaming" filter in traditional Engagement Cloud.	1. Select an operator. 2. Enter the desired threshold (in gigabytes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.

Filter Category	Filter	Description	How to Apply the Filter
	Time Spent Running Speed Tests (min)	Filters subscribers by the amount of time (in minutes) they spent running speed tests within the selected time frame.	1. Select an operator. 2. Enter the desired threshold (in minutes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
Internet Experience	Average GigaSpire QoE	Filters subscribers by average quality of experience (QoE) score. This score is the average of QoE scores over the selected time frame. QoE uses metrics such as Wi-Fi signal strength, internet speed, and service interruptions to evaluate a subscriber's overall service experience. This metric applies only to Calix GigaSpire systems. For more on QoE scores, see <i>View Quality of Experience (QoE) Data</i> in <i>Calix Service Cloud Help</i>.	1. Select an operator. 2. Enter the desired value. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.

Filter Category	Filter	Description	How to Apply the Filter
	Average GigaSpire Wi-Fi Experience	Filters subscribers by average Wi-Fi Experience score. Wi-Fi Experience presents a score between 1 and 5 that represents a subscriber's Wi-Fi quality over a given time frame. Wi-Fi Experience uses different metrics such as Wi-Fi signal strength, packet rates, packet errors, and airtime analytics to evaluate a subscriber's wireless experience. This metric applies only to Calix GigaSpire systems. For more on Wi-Fi Experience, see <i>Understand Wi-Fi Experience, Whole Home Efficiency, and Continuity Tests in Calix Service Cloud Help</i>.	1. Select an operator. 2. Enter the desired value. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Average Wi-Fi Score	Filters subscribers by average Wi-Fi score. This metric applies only to Calix GigaCenter systems. Wi-Fi score is a simplified indicator of Wi-Fi performance for GigaCenter systems measured on a scale of 0 (worst) to 5 (best). The Wi-Fi score considers factors such as signal strength, interference, and data transmission rates for each device connected to the wireless access point (WAP).	1. Select an operator. 2. Enter the desired value. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Downstream Service Limit Hits	Filters subscribers by the number of times they reached 80% of their provisioned download speed within the selected time frame. The downstream service limit is based on the download speed in the subscriber's service package (from USOC).	1. Select an operator. 2. Enter the desired value. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.

Filter Category	Filter	Description	How to Apply the Filter
	Downstream Usage (GB)	Filters subscribers by the total amount of data received by the subscriber (for example: streaming video, viewing web pages) within the selected time frame.	1. Select an operator. 2. Enter the desired threshold (in gigabytes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Total Data Usage (GB)	Filters subscribers by the total amount of data used (both sent and received) within the selected time frame.	1. Select an operator. 2. Enter the desired threshold (in gigabytes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Upstream Service Limit Hits	Filters subscribers by the number of times they reached 80% of their provisioned upload speed within the selected time frame. The upstream service limit is based on the upload speed in the subscriber's service package (from USOC).	1. Select an operator. 2. Enter the desired threshold. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.

Filter Category	Filter	Description	How to Apply the Filter
	Upstream Usage (GB)	Filters subscribers by the total amount of data sent by the subscriber (for example: uploading files, sending emails) within the selected time frame.	1. Select an operator. 2. Enter the desired threshold (in gigabytes). 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
Packages and Services	Attainable Rate (Mbps)	Filters subscribers by the maximum possible internet speed they can achieve in ideal conditions.	Select an operator, and then select the desired value(s).
	Download Speed (Mbps)	Filters subscribers by their contracted download speed. This information comes from billing.	Select an operator, and then select the desired value(s).
	Last Downgraded	Identifies subscribers who downgraded their service within the selected time frame.	Select a time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Last Upgraded	Identifies subscribers who upgraded their service within the selected time frame.	Select a time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	Number of Services	Filters subscribers by the number of [SmartHome/SmartLife] services they have purchased.	Select an operator, and then select the desired value(s).

Filter Category	Filter	Description	How to Apply the Filter
	Potential Bark Subscribers	Identifies households without Bark and likely comprised of at least adult with children present to identify potential Bark upsell opportunities. This may be based on probabilistic data.	Select an operator, and then select the desired value(s).
	Service Tier	Filters subscribers by service tier.	Select an operator, and then select the desired value(s).
	Subscribed to Arlo	Identifies subscribers with or without an Arlo subscription.	Select an operator, and then select either Y or N.
	Subscribed to Bark	Identifies subscribers with or without a Bark subscription.	Select an operator, and then select either Y or N.
	Subscribed to ExperienceIQ	Identifies subscribers with or without an ExperienceIQ subscription.	Select an operator, and then select either Y or N.
	Subscribed to ProtectIQ	Identifies subscribers with or without a ProtectIQ subscription.	Select an operator, and then select either Y or N.
	Subscribed to SmartBiz	Identifies subscribers with or without a SmartBiz subscription.	Select an operator, and then select either Y or N.
	Subscribed to SmartTown	Identifies subscribers with or without a SmartTown subscription.	Select an operator, and then select either Y or N.
	Upload Speed (Mbps)	Filters subscribers by their contracted upload speed. This information comes from billing.	Select an operator, and then select the desired value(s).
	USOC	Filters subscribers by USOC.	Select an operator, and then select the desired value(s).
	Video Service	Identifies subscribers with or without video service.	Select an operator, and then select either Y or N.
	Voice Service	Identifies subscribers with or without voice service.	Select an operator, and then select either Y or N.
Systems and Devices	Camera Brands	Filters subscribers the brand of camera connected to their system: • Arlo • Nest • Other • Ring	Select an operator, and then select the desired camera brand(s).

Filter Category	Filter	Description	How to Apply the Filter
	Device Types	Filters subscribers by the type of device connected to their GigaSpire (for example: camera, computer, printer).	Select an operator, and then select the desired device type(s).
	Has Mesh System	Identifies subscribers with at least one mesh (satellite) Wi-Fi system.	Select an operator, and then select either Y or N.
	Managed System	Filters for subscribers with a system (either Calix or third-party systems) that is both managed by Calix Cloud and associated to a subscriber in Calix Cloud.	Select an operator, and then select either Y or N.
	Model	Filter subscribers by system model. This filter includes both Calix and third-party systems.	Select an operator, and then select the desired system model(s).
	Number of Devices	Filters subscribers by the number of both wired and wireless devices connected to their GigaSpire.	Select an operator, and then select the desired value(s).
	Number of Wired Devices	Filters subscribers by the number of wired devices connected to their system.	Select an operator, and then select the desired value(s).
	Number of Wireless Devices	Filters subscribers by the number of wireless devices connected to their system.	Select an operator, and then select the desired value(s).
	Onboarded Arlo Camera	Identifies subscribers who have onboarded an Arlo camera in the Arlo app.	Select an operator, and then select the desired value(s).
	SmartTown Connected Devices	Filters subscribers by the number of devices connected to their SmartTown hotspot.	1. Select an operator. 2. Enter the desired number of systems. 3. Select the desired time period: - Within Days: Choose 30, 60, or 90 days. - Since: Select a date. - Between: Select the start and end dates. - Before: Select a date.
	System Family	Filters subscribers by Calix system family.	Select an operator, and then select the desired option(s).

Note for orgs migrated to the next-generation Calix Cloud platform

Some segmentation filters from traditional Engagement Cloud were removed and/or replaced in next-generation Engagement Cloud. Affected filters include:

- Excess Bandwidth (*removed*)
- Number of Smart Cameras (*removed*)
- Servify (*removed*)
- No. of Generation (*removed*)
- Streaming (*removed; use Streaming Usage (GB)*)
- Gaming (*removed; use Gaming Usage (GB)*)
- Social (*removed; use Social Usage (GB)*)
- WFH (*removed; use Business Application Usage (GB)*)
- Has Service (*removed*)
- Usage Type (*removed; use Streaming Usage, Gaming Usage, Social Usage, or Business Application Usage*)
- Uploaded_segment_name (*removed*)
- Household Type (*removed*)
- Potential Bark Subscriber (*removed*)

Work with Campaigns

This topic provides information on campaigns in Calix Engagement Cloud.

Engagement Cloud campaigns send groups of subscribers (audiences) from Engagement Cloud to marketing channels, which then deliver marketing materials (emails, direct mailers, etc.) to the subscribers. Engagement Cloud integrates with email, direct mail, and social media marketing platforms to support various media types.

Engagement Cloud sends the following subscriber information (from billing) to each channel:

- First Name
- Last Name
- Address (combined street, street address 2, city, state, ZIP code, country)
- Email Address
- Phone

The Calix Engagement Cloud Plus entitlement enables prospect marketing via acquisition campaigns. See *Work with Demographic Segments* (on page [156](#)) for more.

From the **Campaigns** page, you can perform the following tasks:

- View existing triggered and scheduled campaigns.
- View campaign results. Engagement Cloud tracks incremental revenue and metrics for up to 60 days after the campaign end date.
- Create new scheduled and triggered email, direct mail, and mobile notification campaigns.
- Configure channel integrations and view channel results:
 - Facebook[®]
 - Mailchimp[®]
 - Constant Contact[®]

Note: The Constant Contact integration requires reconfiguration after org migration to next-generation Engagement Cloud. Go to **Campaigns > Channels** to remove and then reconfigure the integration. See *Configure and Deploy with Constant Contact* (on page [117](#)) for more.

- HubSpot[®]
- Postalytics[®]
- NISC
- Mobile notifications
- Access the Electronic Content Builder (ECB), a library of fully customizable marketing assets—such as photos, videos, and ads—you can use in your campaigns. Click the **Electronic Content Builder** button to launch the ECB.

Note: Access to the Campaigns page requires a user role with the Campaign permission assigned. To create and edit campaigns, set the Campaign permission to *Write and Read*. To view campaign results, assign the Revenue permission. Organization Administrators can configure roles and permissions from Administration > Roles.

Audience Deployment Methods

Engagement Cloud offers two methods for deploying audience members to engagement channels:

- **Triggered campaign:** An ongoing Mailchimp email or mobile notification campaign that deploys subscribers who meet a specified threshold (for example: 50 service limit hits within 30 days). On your selected start date, Engagement Cloud creates an initial audience consisting of any subscribers in the selected segment and any subscribers within the previous 30 days who meet the criteria. Then, for the duration of the campaign, Engagement Cloud automatically adds any subscribers who meet the threshold and deploys them to the selected channel(s) daily. Audience membership changes throughout the campaign as Engagement Cloud adds members to and removes members from the campaign.
- **Scheduled campaign:** A one-time campaign that deploys a segment to your preferred channel(s) on a specified start date. The channel deploys campaign materials (for example: an email) based on a schedule defined within the channel. Audience membership does not change during a scheduled campaign.

The Calix Engagement Cloud Plus entitlement enables direct mail campaigns for prospects.

Campaign Audiences versus Segments

An *audience* is a group of contacts deployed from Engagement Cloud as part of a campaign. The number of members in an audience does not change, ensuring accurate results. Each audience is associated with a campaign.

A *segment* is a subset of subscribers that shares common attributes. Engagement Cloud provides pre-built segments for specific subscriber personas (like gamers) and allows you to create custom segments. Segments are dynamic and update automatically, and segment size may fluctuate day-to-day based on which subscribers meet the filter criteria.

Note that campaigns created prior to the May 2023 (R23.2) release continue to report on dynamic segments, while campaigns created after 23.2 track static audiences.

View Existing Campaigns

This topic describes how to view scheduled and triggered campaigns in Calix Engagement Cloud.

The **Campaigns** page lists your scheduled and triggered campaigns. Click on a campaign name to edit the campaign or view campaign results. Filter the list by type and/or status to quickly find a campaign.

For scheduled campaigns, view the following details:

- Campaign name
- Status (*Draft, Pending, Deployed, Complete*)
- Type (*Subscribers, Residential Prospects, Business Prospects*)
- Audience name
- Audience size
- Selected channel(s)
- Start date
- End date

For triggered campaigns, view the following details:

- Campaign name
- Status (*Draft, Pending, Active, Paused, Complete*)
- Trigger
- Associated audience
- Number of subscribers most recently deployed
- Selected channel(s)
- Start date
- End date

Campaign Status Definitions

Engagement Cloud uses the following statuses for triggered and scheduled campaigns:

- Triggered campaign statuses:
 - Draft: Some campaign settings have been configured, but the campaign has not yet been deployed.
 - Pending: The campaign has been deployed, but the scheduled start date has not yet been reached.
 - Active: The scheduled start date has passed, and Engagement Cloud is actively deploying subscribers to the selected channel(s) and tracking results.
 - Paused: The campaign is active, but is not deploying subscribers to the selected channel(s).

-
- Complete: The campaign has ended and Engagement Cloud is no longer tracking results.
 - Scheduled campaign statuses:
 - Draft: Some campaign settings have been configured, but the campaign has not yet been deployed.
 - Pending: The campaign has been deployed, but the scheduled start date has not yet been reached.
 - Deployed: The campaign has been deployed to the selected channel(s) and Engagement Cloud is tracking results.
 - Complete: The campaign has ended and Engagement Cloud is no longer tracking results.

Create and Deploy Subscriber Campaigns

This topic describes how to navigate the Campaign Builder in Calix Engagement Cloud. Create and deploy campaigns from the **Campaigns** page.

Note: To create campaigns, users need a role with the Campaign permission assigned and set to *Write and Read*.

Using the Campaign Builder

The campaign builder organizes all settings on a single page, enabling efficient campaign creation and deployment:

Sample campaign builder for a scheduled campaign

New Scheduled Campaign

Subscribers

Cancel Save as Draft Save & Deploy

1

Name

Description (Optional)

Tracked Metric (Optional)

Select

Channels
☐ Mobile Notification

Start Date

Select Date

End Date

End Date

2

Audience

Preview Audience

Include Segment(s)

Exclude Segment(s) (Optional)

-

Estimated total recipients

3

No channel selected
Select a channel to view and configure the content being used.

Customize the design of collateral used in your campaign with the [Electronic Content Builder](#).

Sample campaign builder for a triggered campaign

The screenshot shows the 'New Triggered Campaign' interface. It features a left sidebar with campaign attributes and a main area for audience selection and channel configuration. Red circles with numbers 1, 2, and 3 highlight specific sections.

Section 1 (Left Sidebar): Contains fields for Name, Description (Optional), Tracked Metric (Optional), Trigger Type, Channels (with a checkbox for Mobile Notification), Start Date, and End Date (Optional). At the bottom, there is a link to 'Customize the design of collateral used in your campaign with the [Electronic Content Builder](#)'.

Section 2 (Audience): Includes a 'Preview Audience' button, 'Include Segment(s) (Optional)' and 'Exclude Segment(s) (Optional)' fields, and an 'Estimated next recipients' field.

Section 3 (Channel Selection): Displays a message: 'No channel selected. Select a channel to view and configure the content being used.'

At the top of the form, there are buttons for 'Cancel', 'Save as Draft', and 'Save & Deploy', along with a 'Subscribers' tab.

The following table describes each section of the Campaign Builder.

Section	Description
1. Campaign Attributes	Define the campaign name, choose metrics to track, select channels, and set the start date. For triggered campaigns, select the trigger type and configure the trigger criteria.
2. Audience	Select which segments to include in and exclude from the campaign, and view the estimated number of recipients for each selected channel.

Section	Description
3. Channel configuration	Define additional settings for the selected channels, including: • Mobile Notification: Customize the notification message and set the notification time • Mailchimp®: Enter the event name¹ (triggered campaigns only) and select personalization attributes • Constant Contact: Select personalization attributes • HubSpot: Select personalization attributes • Postalytics®: Select the corresponding Postalytics campaign² and view a preview of the campaign

¹ For triggered campaigns, Mailchimp uses automations to automatically deploy marketing materials when it receives audience members from Engagement Cloud. See *Configure and Deploy with Mailchimp* for more.

² Each campaign deployed to the Postalytics channel requires a unique triggered drip campaign. See *Configure and Deploy with Postalytics* for more.

Tracked Campaign Metrics

Tracked metrics help you gauge a campaign's impact on service adoption. For each campaign you create, you can select which metrics to track. Tracked metrics compare current and previous-day subscriber counts to highlight day-over-day changes throughout the campaign. For example, monitor how many subscribers purchase Bark and ExperienceIQ throughout an internet safety campaign.

Select from the following metrics:

- Service Tier Conversion
 - Select the desired end service tier. This metric tracked moves *into* the selected tier.
- Has GigaSpire
- Has CommandIQ
- Has SmartBiz
- Has CommandWorx
- Has SmartTown
- Has ProtectIQ
- Has ExperienceIQ
- Has Bark
- Has Arlo

You cannot add or edit metrics after a campaign deploys.

Personalization Attributes

Personalization attributes allow you to tailor campaigns to individual subscribers. For example, you can create a report that shows how many viruses and intrusions ProtectIQ detected over the last month. Personalization attributes are only available for the Mailchimp, HubSpot, and Constant Contact channels.

Create and Deploy a Triggered Subscriber Campaign

This topic describes how to create and deploy a triggered subscriber campaign in Calix Engagement Cloud.

Triggered Campaigns deploy either Mailchimp® email campaigns or mobile notifications to subscribers either after a threshold is met (for example: 50 service limit hits in 30 days) or after an event occurs (for example: system is discovered in Calix Cloud). Engagement Cloud automatically adds to the campaign any subscriber who meets the threshold and deploys them to the selected channel(s) daily until the campaign ends or is paused. You can create multiple triggered campaigns, each with its own trigger threshold.

On initial deployment of a triggered campaign, Engagement Cloud creates an audience that includes any subscribers in the selected segment(s) and any subscribers within the previous 30 days who meet the criteria. Audience membership changes during the campaign as Engagement Cloud adds members to and removes members from the campaign.

Campaign Triggers

Engagement Cloud uses two triggers for deploying triggered campaigns:

- **Threshold-based** triggers deploy a campaign based on a specific threshold (for example, 50 service limit hits within 30 days):
 - Competitor Visit Minutes (Mobile Notification Only): Number of minutes spent on competitor websites over the last 30 days.

Note: Starlink.com and satelliteinternet.com may increase the number of competitor website visits shown in Engagement Cloud and have been removed from the list of competitor websites.

- Service Limit Hits: Number of times a subscriber exceeded their service limit (either upstream or downstream) within 30 days.
 - Speed Test Minutes (Mobile Notification Only): Number of minutes spent performing speed tests over the last 30 days.
- **Milestone-based** triggers deploy a campaign after an event occurs. For example, you can deploy a welcome email two days after Calix Cloud discovers a subscriber's GigaSpire system.
 - Lifecycle event: A date related to a subscriber milestone:
 - System is Discovered: Based on the date a system first checks in to Calix Cloud. Converting a system to SmartBiz use does not count toward this trigger.

- System is Swapped: Based on the date a system swap is initiated from Calix Deployment Cloud. See the *Replace a GigaSpire with a GigaSpire* and *Replace a GigaCenter with a GigaSpire* topics in the *Calix Deployment Cloud Help* for more.
- Onboarded CommandIQ: Based on the date a subscriber onboards their system in CommandIQ.
- Created a Person in CommandIQ: Based on the date a subscriber creates a user profile (Person) in CommandIQ. See the *CommandIQ Help* for more on creating person profiles.
- Onboarded an Arlo Camera: Based on the date an Arlo camera connected to a subscriber system checks in to Calix Cloud.
- Support Ticket was Closed (*requires Calix Service Cloud*): Based on the date a Call Outcome record is saved with a 'Resolved' status in Calix Service Cloud.

Note: The 'Support Ticket was Closed' trigger only counts Call Outcome records with a Resolved status.

Considerations

- Only the Mailchimp (Standard and above) and mobile notification channels support triggered campaigns.
- Only mobile notifications support the Competitor Visit Minutes and Speed Test Minutes triggers.
- Edits made to a segment in use in a triggered campaign affect future deployments of that segment.
- You may pause a triggered campaign at any time.
- Engagement Cloud does not add subscribers to a paused campaign. After you resume the campaign, Engagement Cloud adds only new qualified subscribers to the campaign.
- Triggered campaigns may not have an associated audience at the time of deployment. While the campaign is active, Engagement Cloud automatically adds subscribers who meet the defined threshold to the campaign.
- Triggered campaigns do not require you to select a segment. With no segment selected, the trigger criteria apply to your entire subscriber base, and any subscriber who meets the criteria may receive the campaign. To focus the campaign on a smaller set of subscribers (for example: gamers), apply a segment.
- The service limit hit threshold is based on the total number of either upstream or downstream limit hits within 30 days. For example, a subscriber may reach a threshold of 50 service limits with either 50 upstream hits or 50 downstream hits.
- To calculate service limit hits, Engagement Cloud measures upstream and downstream data consumed every second (in bits/second). Every hour, Engagement Cloud takes the maximum rate (whether upstream or downstream) measured within the hour and compares it to the billing file. A service limit hits occurs when the measured rate exceeds the upstream or downstream values in the billing file.
- Competitor websites are defined during the implementation process. If you must set up a new competitor, please contact your Customer Success Manager or the TAC team.

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- Mobile notification campaigns deploy to both CommandIQ® and CommandWorx™. Before deploying a mobile notification campaign, ensure you filter for the correct subscribers (either residential with CommandIQ or business with CommandWorx).
- Different lifecycle events, such as 'System is Discovered' and 'Onboarded CommandIQ', may occur on or around the same date. If you create multiple triggered campaigns based on different lifecycle events, stagger campaign deployment to avoid sending too many materials at once to your subscribers.
- Because triggered campaigns deploy subscribers to Mailchimp on an ongoing basis, the Mailchimp integration requires additional configuration. Mailchimp uses automations to receive audience members from Engagement Cloud, and each triggered campaign requires its own automation. For automation setup instructions, see *Create a New Customer Journey* (on page [106](#)).

Note: Currently, the Mailchimp integration uses Classic Automations. However, Mailchimp is phasing out classic automations in favor of customer journeys. Existing classic automations will continue to operate and show in Mailchimp. For help converting an existing classic automation to a journey, see the following Mailchimp help topic: *Change a Classic Automation to a Customer Journey*.

Create a Triggered Campaign

The key fields to configure for a triggered campaign are:

- Name
- Trigger Type
- Channels
- Start Date

To create a triggered campaign

1. Go to **Campaigns**.
2. Select **New Campaign > Subscribers - Triggered**.
3. Enter a helpful name and description (optional) for the campaign.
4. (Optional) Select metrics to track during the campaign. Engagement Cloud tracks data for the selected metric(s) for up to 60 days after the campaign end date.
5. Select a trigger from the **Trigger Type** drop-down menu:
 - **Competitor Web Visits:**
 - Enter the amount of time (in minutes) spent on competitor websites.
 - Enter the number of days to delay resending the campaign to subscribers who continue to meet the threshold. This value ensures subscribers do not receive the campaign multiple times within a short period.
 - **Service Limit Hit:**

- Enter the number of service limit hits.
- Enter the number of days to delay resending the campaign to subscribers who continue to meet the threshold. This value ensures subscribers do not receive the campaign multiple times within a short period.
- Speed Test Minutes:
 - Enter the amount of time (in minutes) spent running speed tests.
 - Enter the number of days to delay resending the campaign to subscribers who continue to meet the threshold. This value ensures subscribers do not receive the campaign multiple times within a short period.
- Lifecycle Event:
 - Select an event from the list.
 - Enter the number of days after the event to send the campaign.

6. Select the desired channels.

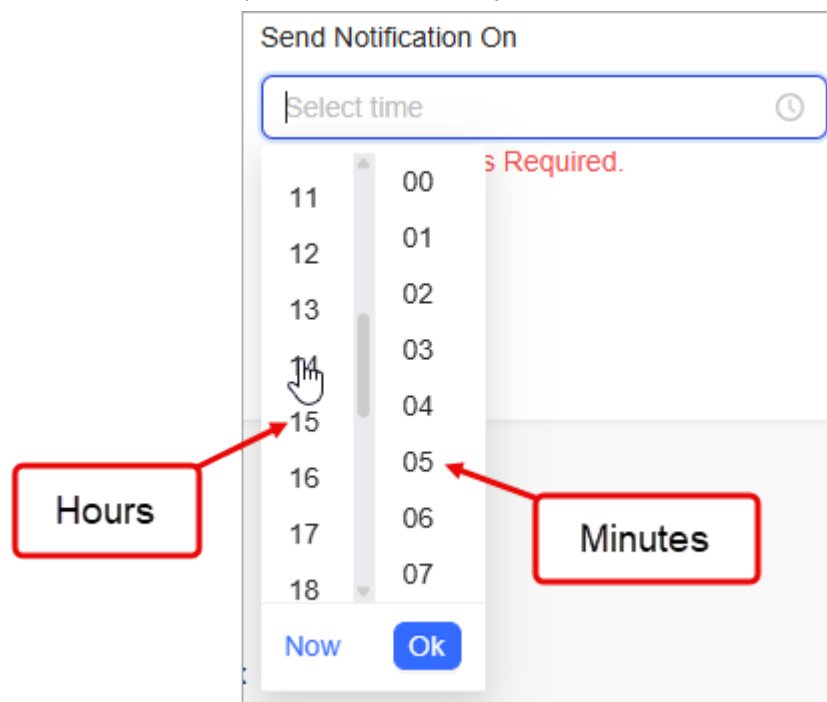
Note: The Channels section displays only channels configured in Campaigns > Channels.

7. Click into the **Start Date** field to choose a start date. Use the left and right-facing arrows to choose a month. Single arrows advance one month and double arrows advance 12 months.

The screenshot shows a date picker interface titled "Start Date". At the top is a text input field with the placeholder "Select Date" and a calendar icon. Below this is a calendar for February 2026. The calendar has a header with navigation arrows (double arrows for 12 months, single arrows for 1 month) and the text "2026 Feb". The days of the week are listed as Su, Mo, Tu, We, Th, Fr, Sa. The dates are arranged in a grid. The date "3" is highlighted with a blue border, and a hand cursor is pointing at it. At the bottom of the calendar, there is a button labeled "Today".

8. (Optional) Click into the **End Date** field to choose an end date.
9. In the Audience section, select segment(s) to include in and exclude from the campaign.
10. Configure additional settings for each selected channel:

- For mobile notifications:
 - a. In the **Message** field, enter text for the notification message (178 character limit).
 - b. (Optional) In the **Link** field, enter a URL to add a hyperlink to the notification message.
 - c. (Optional) Click **Upload** to add an image to the notification.
 - ♦ Supported file types: *.jpg, *.jpeg, *.png
 - ♦ Image size: <1 MB
 - d. Click into the **Select time** field to select the notification time. The hour component uses a 24-hour clock (for example: to deploy at 5:00 PM, choose 17 for the hour).



- e. Select the time zone from the drop-down menu.
- f. Click **Send Test** to ensure the notification appears and functions correctly.

The 'Send Test' popup dialog opens.

- g. Enter the account number for each account that should receive a test notification.
- h. Click **Send Test**.

- For Mailchimp campaigns:
 - a. Enter the **Event Name** for the corresponding automation in Mailchimp. Enter this name exactly as it appears in Mailchimp. For more, see *Configure Mailchimp for Triggered Campaigns* (on page [103](#)).
 - b. (Optional) Customize the campaign with personalization attributes:
 - ♦ Click **Add Attributes**.
 - ♦ Select the checkbox for each attribute to add.
 - ♦ Click **Save**.

11. Click **Save as Draft** to save your settings and edit later, or click **Save & Deploy** if ready to deploy.

Once deployed, you can monitor campaign progress and results on the campaign *Results* (on page [96](#)) tab.

Create and Deploy a Scheduled Subscriber Campaign

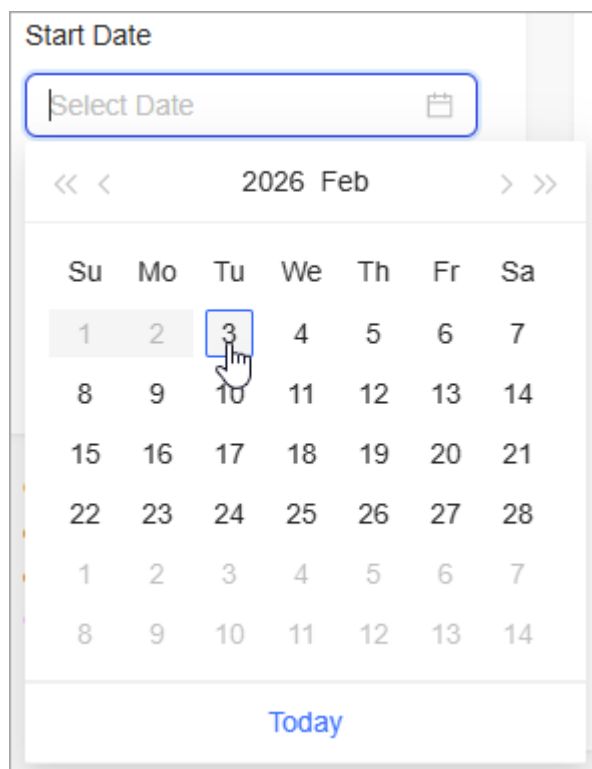
This topic describes how to create and deploy a scheduled subscriber campaign in Calix Engagement Cloud.

From **Campaigns**, you can create and deploy new scheduled campaigns and gain insights from results. Scheduled campaigns deploy a segment to selected channels on the specified start date. The channel(s) then deploy campaign materials based on a schedule defined within the channel. Audience membership remains static throughout the campaign to ensure accurate tracking and results.

To create a scheduled subscriber campaign

1. Go to **Campaigns**.
2. Select **New Campaign > Subscribers - Scheduled**.
3. Enter a helpful name and description (optional) for the campaign.
4. (*Optional*) Select metrics to track for the campaign. Engagement Cloud tracks metrics for up to 60 days after the campaign end date.
5. Select the desired channel(s).
6. Click into the **Start Date** field to choose a start date on the calendar. Use the left and right-facing single arrows to advance by month and the double arrows to advance by year.

Alternately, click **Today** to select the current date.



7. Choose the end date.

8. Select segments to include in and exclude from the campaign.

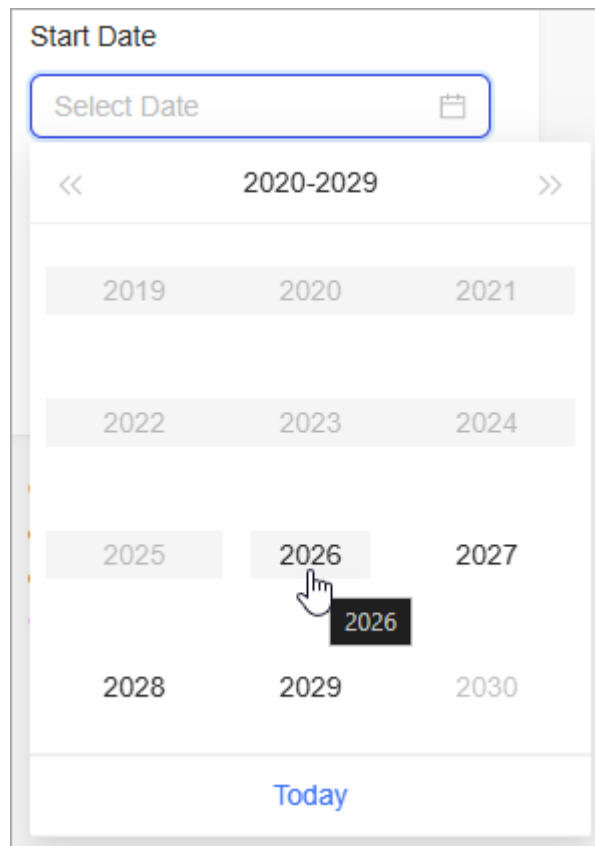
Selecting a segment enables the Preview Audience option. Click to view a list of audience members and to download the list to CSV.

9. Configure additional options for each selected channel:

- For mobile notification campaigns:
 - a. Enter message text for the notification.
 - b. (Optional) In the **Link** field, enter a URL to add a hyperlink to the notification message.
 - c. (Optional) Click **Upload** to add an image to the notification.
 - ♦ Supported file types: *.jpg, *.jpeg, *.png
 - ♦ Image size: up to 1 MB
 - d. Select the deployment **Schedule**:
 - ♦ **Scheduled**: Deploys the campaign on the selected start date. This option is selected by default for start dates later than the current date.
 - ♦ **Immediate**: Deploys the campaign immediately upon clicking Save & Deploy. This option is selected by default if the start date equals the current date.
 - e. Click into the **Select date** field to choose the deployment date and time. The date and time picker opens.
 - ♦ Click **Now** to select the current date and time.

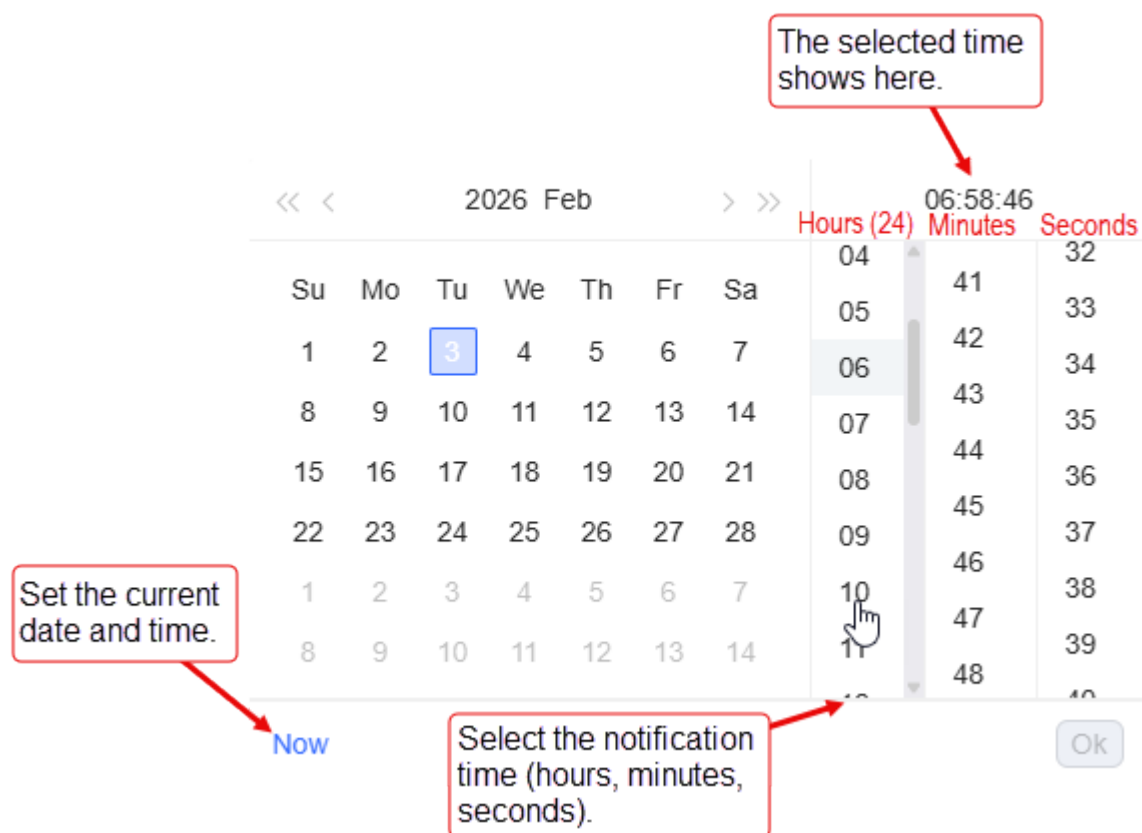
-- or --

- ◆ Click on the year and date at the top of the calendar to choose a year. Alternately, click **Today** to select the current date.



- ◆ Use the left and right-facing arrows to scroll to the desired month. Single arrows advance one month and double arrows advance 12 months.
- ◆ Click the desired date on the calendar.

- ♦ Select the deployment time: Use the scroll menus to the right of the calendar to select the hour, minute, and second to deploy the notification. The hour component is based on a 24-hour clock.



- ♦ Click **Ok** when finished.
 - f. Select the time zone.
 - g. (Optional) Click **Send Test** to deploy the notification to selected individuals and ensure the notification appears and functions correctly.
- The 'Send Test' popup dialog opens.
- h. Enter the account number of each test recipient. If adding multiple account numbers, separate each with a comma.
 - i. Click **Send Test**.
 - (Optional) For Mailchimp, Constant Contact, and HubSpot campaigns:
 - a. Select personalization attributes:
 - ♦ Click **Add Attributes**.
 - ♦ Select the checkbox for each attribute to add.
 - ♦ Click **Save**.

10. Click **Save as Draft to save your settings and edit later, or click **Save & Deploy** if ready to deploy.**

Once deployed, you can monitor campaign progress and results on the campaign *Results* (on page [96](#)) tab.

View Campaign Results

This topic describes how to view results for triggered and scheduled campaigns in Calix Engagement Cloud.

Results data can help you assess a campaign's impact on revenue and managed service conversions.

Note: To view campaign results, users require an Engagement Cloud role with the 'Revenue' permission. Organization Administrators can assign roles and permissions from the Administration section of Calix Cloud. See [Set Up Roles for Calix Engagement Cloud](#) for more.

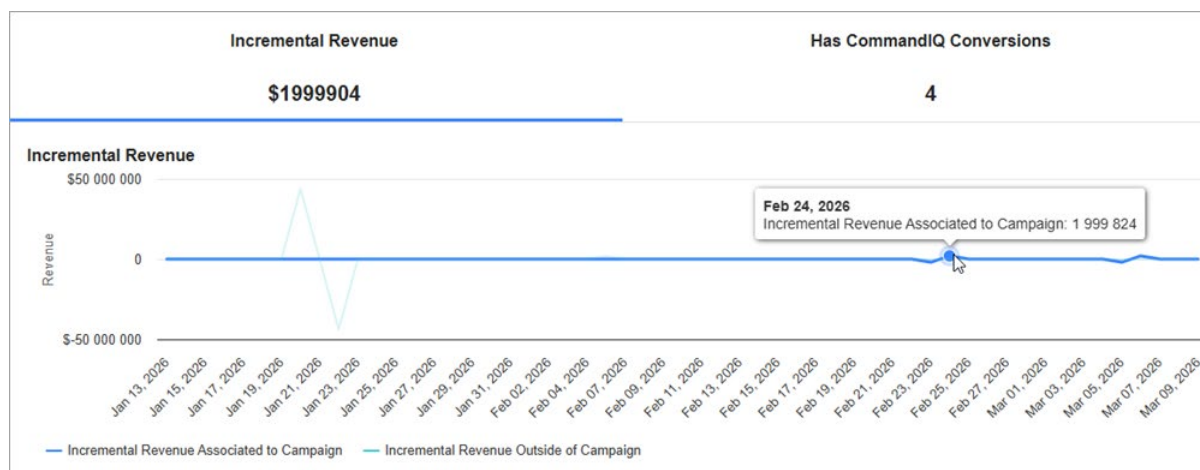
The campaign **Results** page displays the following data:

- Channel Results: Shows the deployment status and most recent date for each selected channel.
- Audience Distribution: Displays the number of audience members deployed to each channel.
- Campaign Data:
 - Incremental Revenue: Shows the change in total revenue for the campaign audience from the campaign start date to the current date. This value may be negative if audience members either churned or downgraded their service tier during the campaign.
 - Results for tracked metrics:
 - Service conversions associated to the campaign.
 - Service conversions outside of the campaign.

To view campaign results

1. Go to **Campaigns**.
2. Select either the **Scheduled** tab or the **Triggered** tab.
3. Click on the desired campaign.
4. View campaign results.

Hover over a data marker on a chart to see additional details.



To view subscriber campaign audience members

For subscriber campaigns, you can view a list of audience members and the channels to which they were deployed.

1. Go to **Campaigns**.
2. Select either the **Scheduled** tab or the **Triggered** tab.
3. Click on the desired campaign.
The Results page opens by default.
4. Click **View Audience Members**.

The View Audience Members dialog opens.

5. Review audience member details.

From the View Audience Members dialog, you can perform the following actions:

- Search for audience members by first, last, or full name
- Filter the list by channel
- Export the list to CSV
- Sort the list in ascending or descending order by column

To re-deploy a campaign

Note: Read-only users cannot re-deploy campaigns.

1. Go to **Campaigns** and click on the campaign.

2. Click on **More Details** to view the error message.

The screenshot shows a table titled "Channel Status" with columns "Channel" and "Status". A tooltip is displayed over the "More details" link for the Facebook channel, stating: "Could not create segment tag in Facebook, please check channel configuration is active or not".

Channel	Status	More details	Re-Deploy
Facebook	Error	More details	Re-Deploy
Mobile Notification	Error	More details	Re-Deploy
Mailchimp	Complete		

3. Take any necessary actions, and then click **Re-Deploy**.

Manage Campaigns

This topic describes how to edit, duplicate, and delete campaigns and how to review campaign settings in Calix Engagement Cloud.

Note: To edit, duplicate, and delete campaigns, users require an Engagement Cloud role with the Campaigns permission enabled and set to *Read and Write*.

To edit a draft campaign

1. Go to **Campaigns**.
2. Select either the **Scheduled** or the **Triggered** tab.
3. (Optional) Filter for draft campaigns.
4. Click on the *ellipsis* menu for the desired draft campaign and select **Edit**.

The screenshot shows a table with columns: Campaigns, Status, Trigger, Audience, Audience Size (Last Sent), Channel(s), Start Date, and End Date. A dropdown menu is open for the "CommandIQ" campaign, showing options: Edit, Duplicate, and Delete.

Campaigns	Status	Trigger	Audience	Audience Size (Last Sent)	Channel(s)	Start Date	End Date
CommandIQ	Draft	Lifecycle Event	-		Mobile Notification	Dec 08, 2025	

Showing 1 - 1 of 1 records

The Campaign Builder opens.

5. Edit the desired settings.
6. Click **Save as Draft** to save your settings and edit later, or click **Save & Deploy** if ready to deploy.

To edit a pending campaign

1. Go to **Campaigns**.
2. Select either the **Scheduled** or the **Triggered** tab.
3. (Optional) Filter for pending campaigns.
4. Click on the *ellipsis* menu for the desired draft campaign and select **Revert to Draft**.
The Campaign Builder opens.
5. Edit the desired settings.
6. Click **Save as Draft** to save your settings and edit later, or click **Save & Deploy** if ready to deploy.

To duplicate a campaign

The Duplicate action allows you to quickly create a new campaign from an existing campaign. Duplicate copies all settings, including the start date, into the new campaign.

1. Go to **Campaigns**.
2. Select either the **Scheduled** or the **Triggered** tab.
3. (Optional) Filter for draft campaigns.
4. Click on the *ellipsis* menu for the desired draft campaign and select **Duplicate**.

Campaigns	Status	Trigger	Audience	Audience Size (Last Sent)	Channel(s)	Start Date	End Date	
CommandIQ	● Draft	Lifecycle Event	-		Mobile Notification	Dec 08, 2025		⋮
Showing 1 - 1 of 1 records								Edit Duplicate Delete

The new campaign opens in the Campaign Builder.

5. Modify campaign settings as desired.
6. Click **Save as Draft** to save your settings and edit later, or click **Save & Deploy** if ready to deploy.

To delete a campaign

1. Go to **Campaigns**.
2. Select either the **Scheduled** or the **Triggered** tab.
3. (Optional) Filter for draft campaigns.

- Click on the *ellipsis* menu for the desired draft campaign and select **Delete**.

Campaigns	Status	Trigger	Audience	Audience Size (Last Sent)	Channel(s)	Start Date	End Date	
CommandIQ	Draft	Lifecycle Event	-		Mobile Notification	Dec 08, 2025		...
Showing 1 - 1 of 1 records								Edit Duplicate Delete

A confirmation dialog displays.

- Click **Delete Campaign** to confirm.

To review campaign settings

After you deploy a campaign, you can open the campaign to review the campaign's settings.

- Go to **Campaigns**.
- Click on the name of the campaign.
- Select the **Details** tab.
- Review the campaign configuration.

View and Configure Channel Integrations

The topics in this section describe how to configure channel integrations with Calix Engagement Cloud. Channels receive campaign audiences from Engagement Cloud and deploy marketing materials (emails, postcards, etc.) to audience members.

Engagement Cloud integrates with the following marketing tools for email, direct mail, and social campaigns:

- Constant Contact® (email)
- Facebook® (social media)
- HubSpot® (email and direct mail; direct mail for residential prospects)
- Mailchimp® (email; direct mail for residential prospects)
- Mobile notification (CommandIQ and CommandWorx)
- NISC
- Postalytics® (direct mail for prospects)

Configure channels and view channel results from **Campaigns > Channels**.

Considerations

- After a subscriber opts out from digital marketing emails, Engagement Cloud removes the email address from all existing audience segments and excludes the email address from mobile messaging campaigns and from segments sent to Mailchimp, HubSpot, and Constant Contact in future campaigns. However, the address remains in segments that have already deployed. Service providers may wish to resend audience segments to engagement channels to account for any opted-out email addresses part of segments prior to August 10, 2022.

Configure and Deploy with Mailchimp

About Mailchimp

Mailchimp® is an email-based marketing platform that partners with Calix Engagement Cloud via API integration. The Mailchimp integration with Engagement Cloud allows marketers to engage subscribers via Mailchimp email campaigns. To learn more about Mailchimp, visit <https://mailchimp.com/>.

Prerequisites: To use Mailchimp, Organizations must have an active account that they authorize Calix Engagement Cloud to manage by following the Configuration steps. Calix Engagement Cloud provides email and mailing addresses and groups them using tags to enable Calix Engagement Cloud segments to be employed in Mailchimp email and direct mail campaigns.

Mailchimp uses the following terminology:

- **Audience:** A group of email contacts, a subset of which may be flagged as subscribers.
- **Tag:** A logical identifier applied to a contact to create smaller segmentation within the Mailchimp audience.

Mailchimp Integration with Calix Engagement Cloud

When Calix Engagement Cloud deploys a campaign to Mailchimp®, the following occurs:

- Engagement Cloud pushes subscriber information (name, email address, mailing address, and phone number) from the campaign segment to the configured Mailchimp audience to create contacts.

Note: Engagement Cloud does not sync phone number for prospect records.

- If a contact is already present, it will not be added to the list to prevent duplication.
- Any newly created contacts will be flagged as **Subscribed**.
- Addresses are pushed in batches via API and could take up to 12 hours to completely populate in Mailchimp.

- After the entire segment is pushed to Mailchimp, email and mailing addresses present in the segment are tagged in Mailchimp with the campaign name.
- Marketers can then use tags to execute email and direct mail campaigns against Calix Engagement Cloud segments with Mailchimp.

Considerations

- Organizations using the API for billing integration must populate the *first_name* and *last_name* fields. These fields populate the first name and last name fields in Mailchimp.
- Mailchimp Standard accounts limit audience data fields to 30 per contact. The limit does not apply on a per-campaign basis; fields from all campaigns synced from Engagement Cloud to Mailchimp count toward the limit.

Marketing Communication Opt-Outs

After a subscriber opts out from digital marketing communications, Engagement Cloud removes the email address from all existing audience segments and excludes the email address from segments sent to Mailchimp in future campaigns. Opt-outs may take up to 14 days to register in Engagement Cloud, though they typically take less. However, the address remains in segments that have already deployed to Mailchimp. BSPs may wish to resend audience segments to Mailchimp to account for subscribers who opted out from marketing communications prior to August 10, 2022.

Configure the Mailchimp Integration

From **Campaigns > Engagement Channels**, you can configure Mailchimp.

Considerations

- Only one Mailchimp account is permitted per Organization. After Mailchimp is configured within Calix Engagement Cloud, the configuration persists upon future user logins.

To configure the Mailchimp integration

1. Go to **Campaigns > Channels**.
2. Click **Configure Channel** for Mailchimp.
3. Review the integration instructions.
4. Click **Connect to Mailchimp**.
5. Click **Continue**.
The browser redirects to the Mailchimp login page.
6. Log in to your Mailchimp account.
7. Click **Allow** to authorize account access.
8. Click **Test Connection** to ensure Engagement Cloud can retrieve your audience list from Mailchimp.

9. Select your desired audience and then click **Save Audience**.

Remove the Mailchimp Configuration

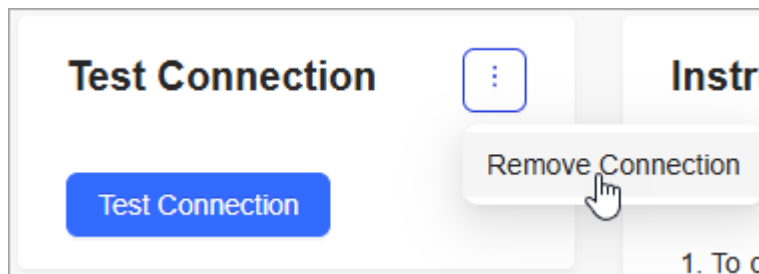
From **Campaigns > Channels**, you can remove the Mailchimp configuration directly within Calix Engagement Cloud.

Considerations

- When Mailchimp deauthorization occurs, Engagement Cloud deletes the locally stored Mailchimp access token, and the organization's Mailchimp campaign history is deleted from Engagement Cloud. Additionally, Channel Results are cleared as Engagement Cloud is no longer authorized to access this data. The data that resides within Mailchimp is not modified.
- If Mailchimp is reauthorized, the marketing channel is immediately available for use in Campaigns. Channel Results populate at midnight PST the following day per normal functionality.

To remove the Mailchimp integration

1. Go to **Campaigns > Channels**.
2. Click the Mailchimp **Settings**.
3. Click the *three dots* icon and select **Remove Connection**.



Removing the Mailchimp connection clears the connection data and campaign history.

4. Click **Remove Connection** to confirm.

After the configuration is removed, the application returns to Channels page.

Configure Mailchimp for Triggered Email Campaigns

This topic describes how to configure the Mailchimp integration with Calix Engagement Cloud.

Before you can deploy a triggered campaign from Calix Engagement Cloud, you must configure an automation in Mailchimp®. During a triggered campaign, Engagement Cloud deploys new subscribers to Mailchimp daily. Automations ensure Mailchimp can receive and deploy to new subscribers for the duration of a triggered campaign. Each triggered campaign you create in Engagement Cloud requires its own automation in Mailchimp.

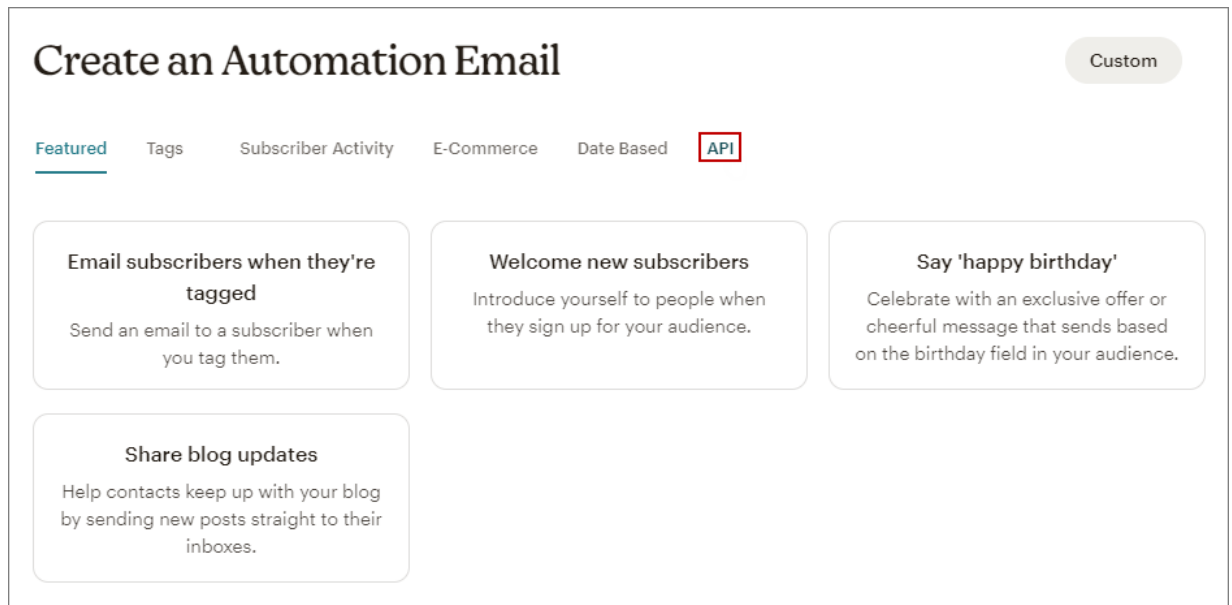
Considerations

- This method uses Mailchimp Classic Automations. Mailchimp is retiring Classic Automations in favor of Customer Journey automations, though existing Classic Automations will continue to operate. If you choose, you can convert a Classic Automation to a Customer Journey. For new triggered campaigns, use Customer Journey automations. See *Create a New Customer Journey Automation* (on page [106](#)) for more.
- If you need to create a new automation for a triggered campaign, create a Customer Journey automation. See *Create a New Customer Journey Automation* (on page [106](#)) for instructions.
- Mailchimp does not assign tags to triggered campaigns.

To create an automation in Mailchimp

Create the campaign

1. Log in to Mailchimp.
2. Go to the following URL: <https://us13.admin.mailchimp.com/customer-journey/#/create-campaign/explore/emailCampaign:custom>.
3. Click **API**.



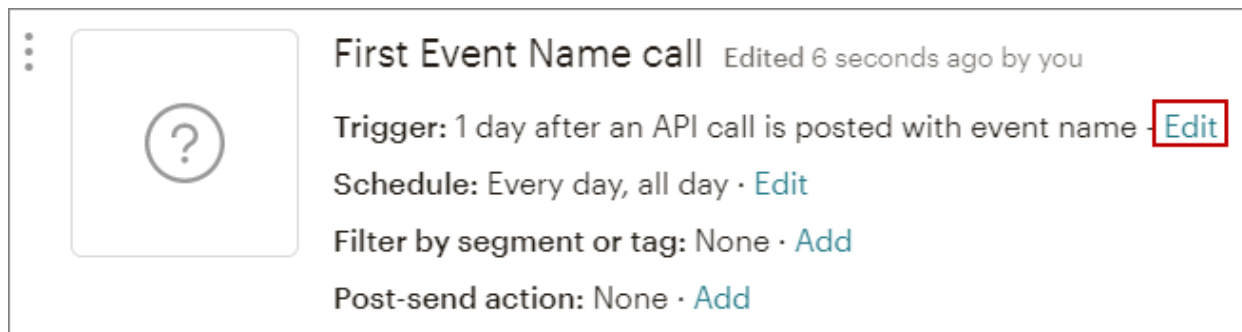
4. Click **Event API**.
5. Enter a **Campaign name**.
6. Select the same audience selected during the Mailchimp integration setup in Engagement Cloud.

Note: For triggered campaigns to work properly, this audience must match the Mailchimp integration audience in Engagement Cloud. If the audiences differ, Mailchimp cannot send the campaign. See *Configure the Mailchimp Marketing Channel* for more information.

7. Click **Begin**.

Configure trigger and schedule settings

1. Click **Edit** to configure the Trigger.



First Event Name call Edited 6 seconds ago by you

Trigger: 1 day after an API call is posted with event name [Edit](#)

Schedule: Every day, all day · [Edit](#)

Filter by segment or tag: None · [Add](#)

Post-send action: None · [Add](#)

2. Select **Immediately** from the **Delay** drop-down menu.
3. Enter an **Event Name** (for example, service-limit-hit).

Note: Ensure the event name is specific to the campaign. For example, service-limit-hit-gamers or service-limit-hit-wfh. Enter this exact event name when you create the triggered campaign in Engagement Cloud.

Delay	Settings
immediately ▼	Event Name: service-limit-hit

4. Click **Update Trigger**.
5. Click **Edit** to configure the Schedule.
6. Configure the schedule as desired.
7. Click **Save Schedule**.

Design the email

1. Click **Design Email**.

Tip: Use the *Electronic Content Builder (ECB)* to customize pre-made marketing assets for the campaign.

2. Configure email settings as desired, then click **Next**.
3. Select a Template for the email, then click **Next**.
4. Edit the email template as desired.
5. Click **Save And Continue**.

Send the email

1. Click **Next**.
2. Review the workflow.
3. Click **Start Sending**.
The **Start Automation Workflow** popup appears.
4. Click **Start Sending** to confirm.

To view triggered campaign recipients in Mailchimp

1. From Mailchimp, go **Campaigns > All Campaigns**.
2. Click on the **Service Limit Hits** campaign.
3. Scroll to the **1 Workflow Email** section.
4. Click the **View Report** button for the Workflow Email.
5. Click the number of recipients at the top of the report.

Create a New Customer Journey Automation

Before you can deploy a triggered campaign from Calix Engagement Cloud, you must create a Customer Journey automation in Mailchimp®. During a triggered campaign, Engagement Cloud deploys new subscribers to Mailchimp daily. Automations ensure Mailchimp can receive subscribers from Engagement Cloud and deploy campaigns for the duration of a triggered campaign.

Considerations

- Before you proceed, ensure the Mailchimp channel configuration in Engagement Cloud is configured with the correct audience. See [Configure the Mailchimp Integration](#) for more.
- Each triggered campaign requires its own Customer Journey automation in Mailchimp.
- Converting an existing Classic Automation to a Customer Journey erases all activity tracking for the associated campaign. Campaign activity tracking starts over after conversion.
- Converting Classic Automations to Customer Journeys is optional, as existing Classic Automations will continue to operate in Mailchimp. However, it is recommended to use Customer Journeys for new campaign automations.

To create a new Customer Journey automation

1. Log in to your Mailchimp account.
2. Select **Automations** from the left side menu.
3. Click **Build from Scratch**.
4. Enter a campaign name.

5. Select the audience list.

Note: This list must match the audience selected on the Mailchimp integration page in Engagement Cloud.

6. Click **Start Building**.
7. Click **Choose a Starting Point**.

A popup appears.

8. Select **Event API**.
9. Select the Event name for the campaign, then click **Save Starting Point**.

Note: The event name must match the event name selected in the Mailchimp configuration in Engagement Cloud.

The journey map displays.

10. Click **Add a journey point**.
 11. Select **Send email**.
- The Send Email builder opens.
12. Configure the email message, then click **Save and Return to Journey**.
- The journey now includes a trigger step and an action step.
13. Click **Continue**.
 14. *Optional:* Select a goal for the journey.
 15. Select **Turn On**, and then click **Continue**.

The new journey appears in the Automations section of Mailchimp.

To convert an existing Classic Automation to a Customer Journey automation

Note: This action optional, as existing Classic Automations will continue to operate in Mailchimp.

1. Log in to your Mailchimp account.
 2. Go to **Automations > All journeys** in the left side menu.
 3. Select the **Classic Automations** tab.
 4. Click the *three dots* menu and select **Convert to Journey**.
- A popup appears.
5. Click **Continue to Journey**.
 6. The customer journey builder opens. Mailchimp appends 'Journey' to the existing automation name.
 7. Click **Continue**.
 8. *Optional:* Select a goal for the journey.

9. Click Turn On.

Mailchimp enables the new automation, and the automation shows on the All journeys > Customer Journeys page with an 'Active' status. The converted classic automation appears on the All journeys > Classic Automations tab with an 'Archived' status.

View Mailchimp Channel Results

From Calix Engagement Cloud, you can view analytics for Mailchimp email and direct mail campaigns, including:

- Number of emails sent
- Number of emails opened
- Number of emails bounced
- Number of contacts who unsubscribed

Considerations

- Campaigns are executed via Mailchimp.
- Engagement Cloud provides the audience and tags to Mailchimp, and then retrieves and displays the results.

Note: All Mailchimp campaigns are listed, in addition to the campaigns associated with Calix Engagement Cloud.

- Results are updated daily at midnight Pacific time. The timestamp reflects when the update occurs in the user's time zone.

To view Mailchimp channel results

1. Go to **Campaigns > Channels**.
2. Click the Mailchimp **Channel Results**.

Click on a column header to sort the table in ascending or descending order by that column. Search for campaigns in the Search box and view channel results from the previous 1, 3, 6, or 12 months.

Configure and Deploy with Facebook

The Facebook[®] channel enables service providers to connect Calix Engagement Cloud to the Facebook platform via the API for delivery of micro-segments and to automatically retrieve insights from executed campaigns. With the Facebook channel, you can achieve the following tasks:

- Deploy campaigns to push contact data to Facebook's platform as a custom audience.
- Execute a campaign within Facebook using the custom audience(s).
- View the results in Engagement Cloud.

To create the custom audience, Facebook matches email addresses with Facebook profiles. If a record does not have an associated email address, no match is found.

Requirements

To configure the Facebook[®] channel, your organization requires the following account and application setup:

- A Facebook Business Manager Account
- A Facebook Developer Account
- Access to Facebook Apps

For information on creating your Facebook Business account, follow the steps outlined in the *How to Set Up your Facebook Business Page and Business Manager Account Success Best Practice* article.

Configuring the channel requires providing two account-specific values:

- Facebook API Key
- Facebook Business Account ID

Note: To utilize Facebook's Custom Audience feature, you must accept the *Facebook Custom Audience Terms of Service*.

Facebook Integration with Calix Engagement Cloud

When you deploy a campaign configured with Facebook[®] as a marketing channel, the following occurs:

- Calix Engagement Cloud creates a new Facebook Custom Audience named after the campaign.
 - The Facebook API returns the audience ID, which is stored for later use.
 - The audience is visible on the Facebook Business Audiences page.
- Calix Engagement Cloud retrieves the subscriber record data by SegmentId. Each record contains, if available, the following data: First Name, Last Name, Email Address, Phone (subscribers only), Zip, State, and Country.

Note: Engagement Cloud does not sync phone number for prospect records.

- Calix Engagement Cloud pushes the data to Facebook via the API.
- Facebook matches subscriber record email addresses with corresponding Facebook profiles to populate the Custom Audience. If a record does not have an associated email address, no match is found.

A custom audience now exists within Facebook with Calix Engagement Cloud customer data that can be used to execute campaigns within the Facebook ecosystem.

Guidelines

- Facebook API requires that all received subscriber data is lowercase and hashed. A maximum of 10,000 records may be merged to an audience at one time. An unlimited number of records may be attributed to a single audience. Multiple pushes may be required. Changes to custom audiences are not immediate and may take up to 24 hours.
- If a campaign name is reused and a customer audience is already present, a new audience is not created. The current subscriber segment is merged with the existing custom audience. Custom audience membership is manageable in Facebook.

Configure the Facebook Integration

This topic describes how to configure the Facebook integration with Calix Engagement Cloud.

Before you begin, log in to your Facebook Business account to obtain your Facebook API Key and your Facebook Account ID.

To configure the Facebook integration

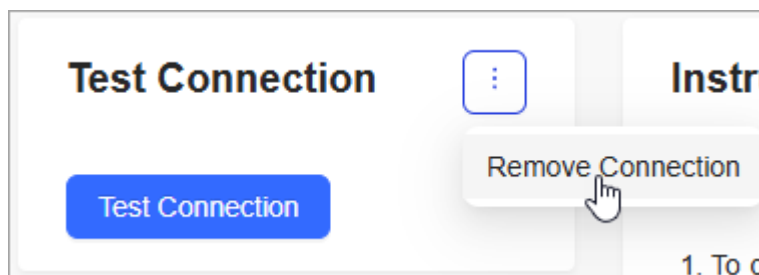
1. Go to **Campaigns > Channels**.
2. Click **Configure Channel** in the Facebook tile.
3. Review the integration instructions.
4. Enter the Facebook API Key and **Facebook Ad Account ID**.
5. Click **Test Connection**.

Remove the Facebook Integration

If your organization would like to remove the Facebook configuration, you can do so directly within Calix Engagement Cloud.

To remove the Facebook configuration

1. Go to **Campaigns > Channels**.
2. Click the Facebook **Settings**.
3. Click the *three dots* icon and select **Remove Connection**.



- Removing the Facebook connection clears the connection data and campaign history.
4. Click **Remove Connection** to confirm.

After the configuration is removed, the application returns to the Channels page.

Create a Facebook Lookalike Audience

Facebook® lookalike audiences allow you to find and market to users who share demographics or behaviors with your current subscribers. After you send an audience segment to Facebook from Calix Engagement Cloud, Facebook analyzes the subscribers in the segment and builds a profile of similar users. When you deploy a Facebook ad, Facebook targets the ad to users who fit the profile.

Creating a lookalike audience consists of the following steps:

1. Create an audience segment containing all subscribers in Engagement Cloud. This list is used to exclude current subscribers from the campaign.
2. Create a campaign to deploy the segment to Facebook.
3. Create a 'high value' audience segment in Engagement Cloud as a model for the lookalike segment.
4. Create a campaign to deploy the segment to Facebook.
5. Create the lookalike audience in Facebook.
6. Create the marketing ad in Facebook.

To create an all-subscriber segment in Engagement Cloud

1. Navigate to **Dashboards > Subscribers**.
2. Click **Save as New Segment**.
3. Enter a **Segment Name** (e.g., All Customer List).
4. Click **Save**.

Next, create a Facebook campaign to send the segment to Facebook.

To create a Facebook campaign

1. Select **Campaigns**.
2. Click **New Campaign**.
3. Enter a **Campaign Name**.
4. Select the newly-created segment.
5. Complete the additional fields and click **Next**.
6. Select the Facebook marketing channel and click **Next**.
7. Click **Deploy Campaign**. Engagement Cloud pushes the segment to your Facebook Audiences. Deployment may take up to 60 minutes.

Next, create a high value segment in Engagement Cloud.

To create a high value segment in Engagement Cloud

1. Navigate to **Dashboards > Subscribers**.
2. Click **+ Add Filter**.
3. Apply the necessary filters to create your desired segment.
4. Enter a **Segment Name** (e.g., Facebook Lookalike Segment).
5. Select a **Segment Type** and click **Save**.

After you create the segment, create a new Facebook campaign to send the segment to Facebook.

Next, create the lookalike audience in Facebook.

To create the lookalike audience in Facebook

You can create the audience from either your Facebook Ads Manager or your Facebook Business Page.

1. Navigate to **Audiences**.
2. Select the checkbox for your lookalike audience.
3. Select **Actions > Create Lookalike**.
4. Select your lookalike campaign from the **Select Your Lookalike Source** menu.
5. Select the **Audience Location** (e.g., United States).
6. Select how closely you want Facebook to match the audience. It is recommended to start at 3% or 4%.
7. Click **Create Audience**. Facebook may take up to 24 hours to create the audience.

Next, create your ad in Facebook.

To create your ad in Facebook

1. Navigate to **Audiences**.
2. Select the checkbox for the newly-created lookalike audience.
3. Select **Actions > Create Ad**.
4. Select a campaign objective and click **Continue**.

Note: The Traffic campaign objective assumes that the advertising call-to-action (CTA) links back to your website for prospective customers to get more information or enroll.

5. Enter a **Campaign name** and click Next.

Note: If this is your first time using a lookalike audience, it is recommended to leave the Campaign Budget Optimization toggle disabled.

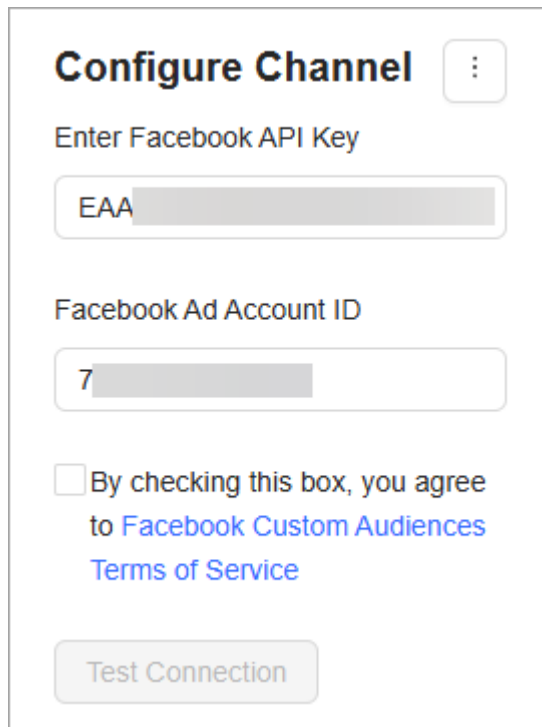
6. Enter your **Budget**, **Start date**, and **End date**.
7. Select **Use Saved Audience**.
8. Search for and select your lookalike audience in the **Custom Audiences** field.
9. Click **Exclude**.
10. Select your full customer list.
11. Click **Save This Audience**.
12. In the Placements section, select either Automatic placements or Manual placements.
 - **Automatic placements (recommended)** permit Facebook's delivery system to allocate your ad set budget across multiple placements based on where they perform best. Select this option if you do not have any campaign metrics to guide you.
 - **Manual placements** allow you to choose where users see the ad. Select this option if you can leverage performance from similar campaigns to target where users engage with your ads.
13. Click **Next** to finish your ad.

Regenerate the Facebook Configuration API Key

The Facebook® integration API key may occasionally expire, breaking the connection between Engagement Cloud and Facebook. To continue using the Facebook channel, regenerate the API key. When the key expires, a message displays on the Calix Engagement Cloud Home page. Engagement Cloud displays the message only on the first login after the key expires, and only for that one user. Users may view subsequent errors on the campaign Results page.

To regenerate the Facebook API key

1. Log in to your *developers.facebook.com* account.
2. Click on the **My Apps** tab.
3. Click on the Facebook app you wish to connect to Engagement Cloud.
4. Select **Marketing API > Tools** in the left sidebar.
5. Select all three options below 'Select Token Permissions'.
6. Click **Get Token**.
7. Copy the token.
8. From Engagement Cloud, navigate to **Campaigns > Engagement Channels**.
9. Click the Facebook **Configuration Settings**.

10. Enter the Facebook API Key.

The screenshot shows a 'Configure Channel' form with a three-dot menu icon in the top right corner. The form contains the following elements:

- A label 'Enter Facebook API Key' above a text input field containing 'EAA' followed by a greyed-out area.
- A label 'Facebook Ad Account ID' above a text input field containing '7' followed by a greyed-out area.
- A checkbox that is currently unchecked, followed by the text 'By checking this box, you agree to [Facebook Custom Audiences Terms of Service](#)'.
- A 'Test Connection' button at the bottom.

11. Click Test Connection. After a successful connection, the Facebook integration is available for use.**View Facebook Channel Results**

From Calix Engagement Cloud, you can view analytics regarding campaigns from Facebook[®], including:

- Number of impressions
- Total clicks
- Total unique clicks

Facebook campaigns works as follows:

- Campaigns are executed by Facebook.
 - The provided subscriber data attempts to locate high probability matches.
 - With more data, better matches are obtained.
- Engagement Cloud provides the data for Facebook to use when executing campaigns.
- Campaign results are provided to Engagement Cloud nightly at Midnight PST.
- Results are then viewable within Engagement Cloud.

Guidelines

- Results data displayed is generated by and then retrieved from Facebook.
- You cannot modify the results in Engagement Cloud.

- All Facebook campaigns are listed, in addition to the campaigns associated with Engagement Cloud.

To view Facebook campaign results

1. Go to **Campaigns > Channels**.
2. Click the Facebook **Channel Results**.

Click on a column header to sort the table in ascending or descending order by that column. Search for campaigns in the Search box and view channel results from the previous 1, 3, 6, or 12 months.

Understanding Facebook Terminology

To understand the results, see the following Facebook terminology:

- Impressions: The number of times an add is on screen.
- Total Clicks: The number of times an ad is clicked.
- Total Unique Clicks: The number of people that click an ad.
- Reach: The number of distinct people that see an ad.
- Spend: Amount that Facebook has charged for the campaign.
- CPM: Cost per 1000 impressions.
 - $CPM = 1000 * (Spend / Impressions)$.
- CPC: Cost Per Click.
 - $CPC = (Spend / Clicks)$.
- Link Clicks: The number of clicks on links within the ad that led to advertiser-specified destinations, on or off Facebook.
- Frequency: The estimated average number of times each person sees an ad.

For more information, refer to *Facebook documentation*.

Configure and Deploy with HubSpot

The HubSpot® channel enables service providers to integrate Calix Engagement Cloud with the HubSpot platform to broaden marketing capabilities, including segmentation, targeting, and measurement. This integration enables sending audience lists created in Engagement Cloud directly to HubSpot for campaign execution (social, email, other channels) and providing automated measurement to Engagement Cloud from HubSpot.

Prerequisite: To configure Engagement Cloud with HubSpot, you must have a HubSpot account with Super Admin permissions.

Note: To execute a direct mail campaign with HubSpot, you must configure a direct mail integration within your HubSpot account. Visit the *HubSpot App Marketplace* for a full list of direct mail campaign integrations.

Integration/Authentication Overview

- Engagement Cloud and HubSpot use an OAuth cut through to allow users to authenticate. The result of this authentication is an Access Token and a Refresh Token to be used for Hubspot APIs.
- When the login sequence is completed and the tokens are received, the Hubspot configuration is considered Active. This authorization may be revoked by the user through the Hubspot UI, and the HubSpot configuration becomes Inactive in Engagement Cloud.
- Engagement Cloud integrates with the Contacts and Marketing Email object points within HubSpot.

HubSpot Integration with Calix Engagement Cloud

After you successfully configure the HubSpot integration in Calix Engagement Cloud, you can send your Engagement Cloud segments directly to the segment you have selected in HubSpot. Engagement Cloud campaigns appear as segments in HubSpot.

After you deploy a campaign in HubSpot, you can view email interaction results in Engagement Cloud. Results detail the number of emails Sent, Delivered, Opened, and Processed.

Configure the HubSpot Integration

This topic describes how to configure the HubSpot integration with Calix Engagement Cloud.

To configure the HubSpot integration

1. Go to **Campaigns > Channels**.
2. Click **Configure Channel** for HubSpot.
3. Click **Connect To HubSpot**.
The HubSpot Log in dialog opens.
4. Click **Continue**.
The page redirects to the HubSpot login page.
5. Click **Sign in to your HubSpot account**.
6. Enter your HubSpot login credentials.

Once set up, return to the Engagement Cloud Channels page to test the configuration.

7. Click **Configuration Settings** (currently Active).
8. Click **Test Configuration**.

If successful, the integration is available for use.

View HubSpot Channel Results

From Calix Engagement Cloud, you can view analytics for HubSpot campaigns, including the number of emails sent, delivered, and opened.

To view HubSpot channel results

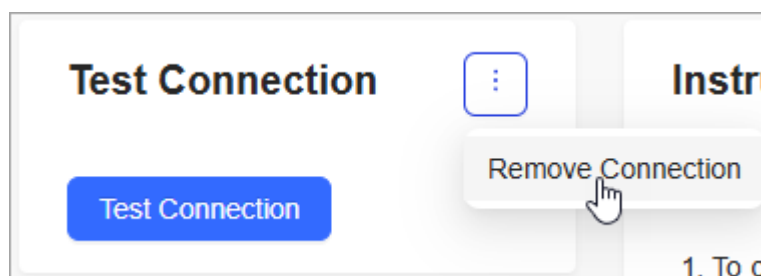
1. Go to **Campaigns > Channels**.
2. Click the HubSpot **Channel Results**.
Click on a column header to sort the table in ascending or descending order by that column. Search for campaigns in the Search box and view channel results from the previous 1, 3, 6, or 12 months.

Remove the HubSpot Configuration

If your organization would like to remove the HubSpot configuration, you can do so directly within Calix Engagement Cloud.

To remove the HubSpot configuration

1. Go to **Campaigns > Channels**.
2. Click the HubSpot **Settings**.
3. Click the *three dots* icon and select **Remove Connection**.



Removing the HubSpot connection clears the connection data and campaign history.

4. Click **Remove Connection** to confirm.

After the configuration is removed, the application returns to the Channels page.

Configure and Deploy with Constant Contact

The Constant Contact® channel enables service providers to integrate Calix Engagement Cloud with Constant Contact to broaden marketing capabilities, including segmentation, targeting, and measurement. This integration enables sending audience lists created in Engagement Cloud directly

to Constant Contact for campaign execution (social, email, direct mail, other channels) and providing automated measurement to Engagement Cloud from Constant Contact.

Note: For orgs migrated to the next-generation Engagement Cloud, the Constant Contact integration requires reconfiguration after migration. Go to **Campaigns > Channels** to remove and then reconfigure the integration.

Marketing Communication Opt-Outs

After a subscriber opts out from digital marketing emails, Engagement Cloud removes the email address from all existing audience segments and excludes the email address from segments sent to Constant Contact in future campaigns. Opt-outs may take up to 14 days to register in Engagement Cloud, though they typically take less. However, the address remains in segments that have already deployed to Constant Contact. BSPs may wish to resend audience segments to Constant Contact to account for subscribers who opted out from marketing communications prior to August 10, 2022.

Configure the Constant Contact Integration

This topic describes how to configure the Constant Contact integration with Calix Engagement Cloud.

From **Campaigns > Channels**, you can configure Constant Contact. Follow the steps outlined in this help topic or the Instructions available on the Engagement Channels application page.

To configure the Constant Contact integration

1. Go to **Campaigns > Channels**.
2. Click **Configure Channel** for Constant Contact.
3. Click **Connect to Constant Contact**.
4. Click **Log In Constant Contact**. You are redirected to the Constant Contact login page.
5. Enter your Constant Contact login credentials.
6. Click **Sign In**.
7. Click **Allow** to authorize account access.
8. Test the connection to ensure Engagement Cloud can send your audience segments to Constant Contact. Engagement Cloud populates your audience segments, allowing you select the desired group.

View Constant Contact Channel Results

From Calix Engagement Cloud, you can view analytics for Constant Contact campaigns, including:

- Number of emails sent
- Number of emails opened

- Number of email links clicked
- Number of emails forwarded
- Number of emails bounced
- Number of contacts opted out from marketing communications
- Number of emails reported for abuse

To view Constant Contact channel results

1. Go to **Campaigns > Channels**.
2. Click the Constant Contact **Channel Results**.

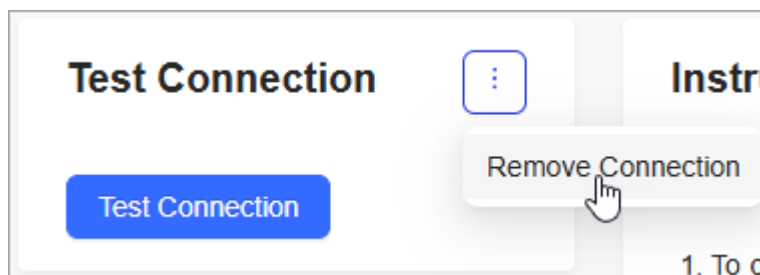
Click on a column header to sort the table in ascending or descending order by that column. Search for campaigns in the Search box and view channel results from the previous 1, 3, 6, or 12 months.

Remove the Constant Contact Integration

You can remove the Constant Contact integration directly within Calix Engagement Cloud.

To remove the Constant Contact integration

1. Navigate to **Campaigns > Channels**.
2. Click the Constant Contact **Configuration Settings**.
3. Click the *three dots* icon and select **Remove Connection**.



4. Click **Remove Connection** to confirm.

After the configuration is removed, the application returns to the Channels page.

Configure and Deploy with Postalystics

The Postalystics[®] channel allows service providers to integrate Calix Engagement Cloud with the Postalystics platform to broaden direct mail marketing capabilities. This integration enables sending prospect (*Calix Engagement Cloud Plus entitlement required*) audience lists created in Engagement Cloud Plus directly to Postalystics for direct mail campaign execution.

Engagement Cloud sends the following prospect contact information to Postalystics:

- First name

- Last name
- Address (street, city, state, and ZIP from the contact's third-party demographic data record)

Considerations

- The Postalytics integration supports only prospect campaigns (both residential and business).
- Multiple prospect records may share a mailing address. To prevent that address from receiving multiple mail pieces from the same campaign, Engagement Cloud de-duplicates prospect records with the same mailing address.
- Because the Postalytics integration supports only direct mail campaigns, audience deployments from Engagement Cloud omit contacts without address information.

Configure the Postalytics Integration

This topic describes how to configure the Postalytics integration with Calix Engagement Cloud.

From **Campaigns > Channels**, you can configure the Postalytics integration. Follow the steps outlined in this help topic or the Instructions available on the Engagement Channels application page. For a more efficient setup, open each application (Postalytics and Engagement Cloud) in a separate browser tab.

To obtain your Postalytics API Key

1. Log in to your Postalytics account.
2. Go to **Connect Hub > Connect > API Key**.
3. Copy the Authentication Header.

Note: Be sure to include two trailing equals signs (==) at the end of the key.

To configure the Postalytics integration

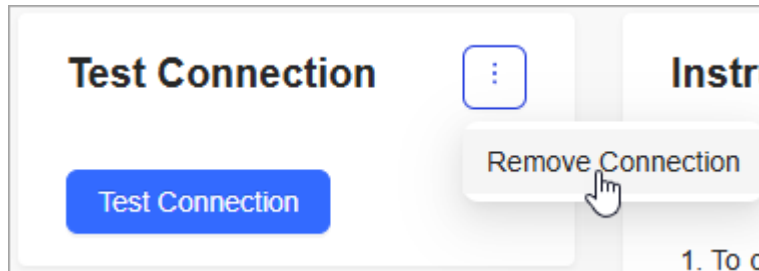
1. Go to **Campaigns > Channels**.
2. Click **Configure Channel** for Postalytics.
3. Paste or enter the Postalytics key into the **Authorization Header** field.
4. Test the connection to ensure Engagement Cloud can send your campaign audiences to Postalytics.

Remove the Postalytics Integration

From **Campaigns > Channels**, you can remove the Postalytics configuration directly within Calix Engagement Cloud.

To remove the Postalytics integration

1. Go to **Campaigns > Channels**.
2. Click the Postalytics **Configuration Settings**.
3. Click the *three dots* icon and select **Remove Connection**.



Removing the Postalytics connection clears the connection data and campaign history.

4. Click **Remove Connection** to confirm.

After the configuration is removed, the application returns to the Channels page.

Configure and Deploy with NISC

The NISC channel enables service providers to integrate Calix Engagement Cloud with and deploy subscriber segments to NISC.

Configure the NISC Integration

From **Campaigns > Channels**, you can configure the NISC integration.

To obtain your NISC UUID

NISC supplies your organization UUID upon request. Contact NISC support for assistance.

To configure the NISC integration

1. From Engagement Cloud, go to **Campaigns > Channels**.
2. Click **Configure Channel** for NISC.
3. Enter your **NISC UUID**.
4. Test the connection to ensure Engagement Cloud can send your campaign audiences to NISC.

View Mobile Notification Channel Results

From Calix Engagement Cloud, you can view analytics for mobile notification campaigns, including:

- Audience
- Campaign status

- Number of subscribers in the audience
- Number of notifications sent
- Campaign start date
- Campaign end date

To view mobile notification channel results

1. Go to **Campaigns > Channels**.
2. Click on the mobile notification **Channel Results**.

Click on a column header to sort the table in ascending or descending order by that column. Search for campaigns in the Search box and view channel results from the previous 1, 3, 6, or 12 months.

Engagement Channel Error Strings

The following tables list error codes and provide troubleshooting tips for the Constant Contact, Facebook, HubSpot, and Mailchimp, and mobile notification engagement channels.

Constant Contact

Error String	Notes	Action
Unauthorized	Channel Authorization is not valid.	Go to Campaigns > Channels > Constant Contact and re-authorize the integration.
Could not create Constant Contact segment	Constant Contact API call to add named segment list failed.	
Cannot exceed max limit of 1000 lists	Constant Contact API call to add named segment list failed due to resource limit.	Log in to Constant Contact with the integration account and remove audience lists. Alternatively, authorize a different Constant Contact account.
Constant Contact: Add subscriber to static list failed	Constant Contact API call to add contacts to segment list failed.	
Could not create segment tag in Constant Contact	Constant Contact API call to add named segment list failed.	

Error String	Notes	Action
No subscriber data from Campaign Segment	At time of campaign deployment, given segment matched no contacts in Engagement Cloud data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only non-opt-out subscribers and prospects with a non-null Email Type can be included for campaign distribution.
Manually canceled due to segment size	Deployment to Constant Contact was stuck due to large segment size. Manually canceled.	Examine the segment to determine if the segment can be smaller. Re-deploy the campaign to Constant Contact.

Facebook

Error String	Notes	Action
No Subscriber data from Campaign Segment	At time of campaign deployment, given segment matched no subscriber contacts in Engagement Cloud data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only prospects with a non-null Email Type can be included for campaign distribution.
Facebook: (#2635) Your app has been upgraded to version v17.0, please use this version or newer. This can be verified in the settings tab on the App Dashboard.	The Facebook API version has changed, rendering the stored API version out-dated.	Go to Campaigns > Channels > Facebook and re-run the authorization test. This should clear the issue and allow for re-deployment. If not, check the Facebook account for valid authorization values and re-enter in Engagement Cloud.

Error String	Notes	Action
No contact data from Campaign Segment	At time of campaign deployment, given segment matched no prospect contacts in Engagement Cloud data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only prospects with a non-null Email Type can be included for campaign distribution.
Facebook: (#2654) No Write Permission to This Ad Account: User must have write permission to create audience under this ad account	At time of campaign deployment, authorized Facebook account does not have appropriate write permission.	Check the Facebook account configuration. Update the integration settings in Campaigns > Engagement Channels > Facebook, if needed.
Unsupported post request. Object with ID '120205891169540018' does not exist, cannot be loaded due to missing permissions, or does not support this operation.	During campaign deployment, the Engagement Cloud API call referenced a Facebook audience list that could not be found. This is unexpected, but could happen if audience lists are manually deleted from Facebook.	Re-deploy the campaign to Facebook to re-create the audience list and re-sync the contacts.
Facebook: Error validating application. Application has been deleted.	At time of campaign deployment, authorized Facebook application is no longer valid.	Check the configuration in Facebook and re-authorize the app if necessary. If authorization values are updated, re-enter them on the Campaigns > Engagement Channels > Facebook page.
Could not create segment tag in Facebook	Facebook API call to add named segment list failed.	Check the Facebook account to see if resource limits have been reached.

Error String	Notes	Action
Unsupported post request. Object with ID '23853368885170151' does not exist, cannot be loaded due to missing permissions, or does not support this operation.	During campaign deployment, the Engagement Cloud API call referenced a Facebook audience list that could not be found, or a similar error occurred. This is unexpected.	Check the Facebook account configuration and/or re-deploy the campaign to Facebook to re-create the audience list and re-sync the contacts.
Business Account Needed to Create/Edit This Audience	The Facebook account is not properly configured as a Business Account.	Create a Facebook Business Account and update the integration configuration in Engagement Cloud.
Facebook: (#2654) Customer List Audiences Are Not Available: Ad accounts owned by businesses new to Facebook Products can create customer file Custom Audiences after several weeks of following our policies.	Facebook imposes some limitations on which accounts can use features required by Engagement Cloud.	
The user doesn't have the permission to create ads with this ad account	The Facebook user associated to the authorization information does not have the proper permissions in Facebook.	Check the configuration in Facebook.
Facebook: (#200) You can not create CUSTOM audience because you can only have up to 500 audiences of this type.	The Facebook API call to add named segment list failed due to resource limit.	Log in to the Facebook account and remove audience lists. Alternatively, authorize a different Facebook account in Engagement Cloud.
Custom Audience Terms Not Accepted	The Facebook user/account authorized in Engagement Cloud has not accepted the Facebook terms required for Engagement Cloud functionality.	Check and update the configuration in Facebook.

Error String	Notes	Action
Manually canceled due to segment size	Deployment to Facebook was stuck due to large segment size and manually canceled.	Examine the segment to determine if the segment can be smaller. Re-deploy the campaign to Facebook to try again.

HubSpot

Error String	Notes	Action
Hubspot: Add email to list failed	Channel Authorization is not valid.	Go to Campaigns > Channels and re-authorize the integration.
No contact data from Campaign Segment	At time of campaign deployment, given segment matched no contacts in CEC data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only non-opt-out subscribers and prospects with a non-null email type can be included for campaign distribution.
Errors found processing batch update	Batch API calls used to add contacts to HubSpot and tag them with segment list ran into error(s).	
Any of the listed authentication credentials are missing	The Hubspot User authorized in CEC does not have the proper permissions or credentials in Hubspot.	Log in to HubSpot and check the configuration.
No Subscriber data from Campaign Segment	At time of campaign deployment, given segment matched no contacts in Engagement Cloud data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only non-opt-out subscribers and prospects with a non-null email type can be included for campaign distribution.

Error String	Notes	Action
Manually cancelled due to segment size	Deployment to HubSpot was stuck due to large segment size and manually canceled.	Examine the segment to determine if the segment can be smaller. Re-deploy the campaign to HubSpot to try again.

Mailchimp

Error String	Notes	Action
Could not create segment tag in Mailchimp	Mailchimp API call to add named segment list failed.	
You have exceeded the maximum number of 30 merge fields for this list	Mailchimp only allows up to 30 unique personalization fields (merge tags) for a campaign audience. This campaign deployment attempted to add additional personalization fields, pushing the total beyond 30.	Check the selected audience in the Mailchimp account. - Remove some merge fields from the selected audience. - Select a different Mailchimp audience in the Engagement Cloud Mailchimp channel configuration.
You cannot have more than 30 merge fields on a list.	Mailchimp only allows up to 30 unique personalization fields (merge tags) for a campaign audience. This campaign deployment attempted to add additional personalization fields pushing the total beyond 30.	Check the selected audience in the Mailchimp account. - Remove some merge fields from the selected audience. - Select a different Mailchimp audience in the Engagement Cloud Mailchimp channel configuration.

Error String	Notes	Action
No Subscriber data from Campaign Segment	At time of campaign deployment, given segment matched no subscriber contacts in CEC data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only non-opt-out subscribers can be included for campaign distribution.
Mailchimp: No content returned	A Mailchimp API called by Engagement Cloud during campaign deployment was unable to return content. This is unexpected.	Re-deploy the campaign to Mailchimp from the Campaign Results page.
Unable to add contacts to audience	The Mailchimp API to add contacts to the list tag within the audience has failed.	
Couldn't find campaign channel	The Engagement Cloud configuration for the channel under the campaign was not found at the time of campaign deployment. This is unexpected and should only be possible if database objects are manually deleted.	Create a new campaign to match the existing campaign configuration. This also creates new campaign channel object(s). File a TAC ticket via MyTouchpoint.
API Key Invalid	Channel Authorization is not valid.	Go to Campaigns > Channels > Mailchimp and re-authorize the Mailchimp account.
No contact data from Campaign Segment	At time of campaign deployment, given segment matched no prospect contacts in Engagement Cloud data.	Examine the segment in Segments. Change segment filters to match contacts in the segment, or create a new campaign with a different segment. Only prospects with a non-null email type can be included for campaign distribution.

Error String	Notes	Action
Unknown error	Unknown	File a TAC ticket via MyTouchpoint.
Manually cancelled due to segment size	Deployment to Mailchimp was stuck due to large segment size and manually canceled.	Examine the segment to determine if the segment can be smaller. Re-deploy the campaign to Mailchimp to try again.

Mobile Notification

Error String	Notes	Action
No MAC addresses for campaign segment	At time of campaign deployment, given segment matched no mobile app subscribers in Engagement Cloud data.	Examine the segment in Segments. Change segment filters to match some mobile user contacts, or create a new campaign with a different segment. Only subscribers marked as mobile app users and with a MAC address can be included for mobile notification campaign.
Error on configuring Mobile Messaging	The Engagement Cloud interaction with MAP subsystem has returned an error.	File a TAC ticket via MyTouchpoint.

Configure Engagement Cloud Settings

From the **Configuration** page, you can assess billing data quality, manage ZIP code filter and contact file uploads, and configure data mapping.

The Configuration page has four tabs:

- **Billing Data Quality** helps you identify and correct discrepancies between the number of subscribers in billing (either pipeline or API) and the number of subscribers visible in Engagement Cloud.
- **Subscriber/Prospect Lists** allows you to upload address lists to populate the Subscriber Lists and Prospect Lists segment attribute filters.
- **ZIP Code Filter Lists** allows you to upload lists of ZIP codes to populate the ZIP/ZIP+4 segment attribute filter. The ZIP/ZIP+4 filter enables filtering and segmentation by geographic area.
- **Data Mapping** allows you to modify package mappings and rider field mappings.

View Billing Data Quality

This topic provides information on the **Configuration > Billing Data Quality** page in Calix Engagement Cloud.

The Billing Data Quality page helps you identify and correct discrepancies between the number of subscribers in billing (either pipeline or API) and the number of subscribers visible in Engagement Cloud. A lower subscriber count in Engagement Cloud means that insights KPIs and charts show incomplete data and that segments do not include all possible subscribers.

From this page, you can also download lists of subscribers with incomplete records, subscribers without a matched endpoint, and endpoints without a matched subscriber record. Use this information to close data gaps and increase the number of subscribers visible in Engagement Cloud.

Billing Data Quality Charts/KPIs

See the following table for a description of each KPI and chart on the Billing Data Quality page.

Chart/KPI	Description
Total Subscribers in Billing KPI	Shows the total number of subscribers in billing (either API or pipeline).

Chart/KPI	Description
Subscribers Visible in Engagement Cloud KPI	Shows the percentage and the number of subscribers who have both service and technology information in billing. This information is required for a successful endpoint match and to ensure Engagement Cloud includes insights data for each billing subscriber. A low percentage may indicate incomplete or inaccurate billing information for some subscribers. To see which subscribers, click Download incomplete records .
Billing Subscribers vs Visible Subscribers chart	Tracks the number of subscribers in billing and the number of subscribers visible in Engagement Cloud over the last year.
Subscriber to Endpoint Match Rate KPI	Displays both the percentage and the number of subscribers with a valid endpoint match. Successful subscriber-to-endpoint matching ensures Engagement Cloud can gather and present insights and usage behavior for each subscriber. A low match rate means some subscriber data is missing in Engagement Cloud. This KPI presents two data download options: • Download unmatched subscriber records: Click to view subscriber records without a matched endpoint. • Download unmatched endpoints: Click to view endpoints with a matched subscriber record.
Subscribers with Managed Systems KPI	Shows the percentage and the number of subscribers who have a system (either Calix or third-party) that is both managed by Calix Cloud and associated with a subscriber in Calix Cloud. A low percentage may indicate a low subscriber to endpoint match rate.
Demographic or Business Data Match Rate KPI	Shows the percentage and the number of subscribers matched to <i>either</i> a demographic data record <i>or</i> a business data record. Engagement Cloud matches subscriber records to demographic data records by address (either service or billing). The address must be able to receive mail. A low match rate may indicate that some subscribers have inaccurate or incomplete address information. To see a list of unmatched subscribers, click Download unmatched records.

How can I ensure my billing data is complete and correct?

Check the following for each billing subscriber.

What to Check	Potential Impact of Incomplete/Incorrect Data
Service and technology fields have data	Subscriber not counted in Engagement Cloud
Device IDs correct and not reused	System mapping fails
serviceType values all lowercase (for example: data, voice, video)	Subscriber not counted in Engagement Cloud
Subscriber names use correct capitalization and omit special characters	Campaigns may look unprofessional
Addresses are valid, use USPS-standard formatting, and include a ZIP code	• Direct mail undeliverable • Subscriber not matched to third-party demographic data record • GeoMapping fails
Email address field populated with valid email address	Subscriber unable to receive email campaigns
customerType field contains exactly either Residential or Business (examples of incorrect data: res, biz, resi, residential)	Segmentation breaks

Upload Residential Subscriber and Residential Prospect Lists

This topic describes how to upload lists of subscribers and prospects into Calix Engagement Cloud.

From **Configuration > Subscriber/Prospect Lists**, you can upload residential subscriber and residential prospect* contact lists into Engagement Cloud. Engagement Cloud matches subscriber contacts to existing subscriber records and prospect contacts to third-party demographic records. These lists populate the Subscriber Lists and Prospect Lists segment attribute filters.

Use Case: Prospect List Upload

Rebecca, the marketing specialist, wants to see how ABC Electric Co-op's electric and video customers differ from their data subscribers. Additionally, she would like to market ABC's data services to their electric and video customers.

To accomplish this task, Rebecca does the following:

1. Create two address lists: one for electric and video customers and one for data subscribers.
2. Upload the two lists into Engagement Cloud.
3. Go to **Segments** and create a new subscriber segment

4. Apply the Subscriber Lists segment attribute filter and select the two uploaded lists.
5. Create a campaign and deploy the segment.

Guidelines

General

- Prospect File Upload requires Calix Engagement Cloud Plus entitlement.
- Engagement Cloud only uses the Email(s) column to improve matching and does not send email addresses to your marketing channels.
- File processing occurs hourly. After processing, the file status changes to Ready.
- Engagement Cloud ignores file entries that do not match a subscriber record.
- Engagement Cloud does not add file data to the SmartTable.
- Some fields convert to all lower-case during processing.
- File uploads may only contain ZIP codes within your base area. Engagement Cloud disregards contacts with ZIP codes outside of the base area. If you later add a ZIP code to your base area, you must re-upload any list that includes that ZIP code; Engagement Cloud does not automatically re-match contacts.

File Formatting

- Subscriber/Prospect File Upload supports only CSV files; the CSV UTF-8 file type is not supported.
- Ensure your file uses the following format. Label column headers exactly as they appear in the following example; mis-spelled column headers result in an upload error.

File Example							
First Name	Last Name	Street Address 1	Street Address 2	City	State	ZIP	Email(s)
Tom	Jones	123 East Lane	Ap # 6	San Jose	CA	23137	tom.j@email.com; tom2@email.com
Bailey	Thompson	P.O. Box 132	77 Sunny St.	Omaha	Idaho	32140	
Tina	Hawkins	22-109 Great Avenue		Chicago	IL	00011	tina@email.com

- The **Street Address 1**, **City**, **State**, and **ZIP** columns are required and may not contain blank cells. Engagement Cloud uses these fields to match list entries to subscriber records.
- For the street address, use the subscriber's physical/service address and not the billing/mailling address.
- The **First Name**, **Last Name**, **Street Address 2**, and **Email(s)** columns are optional.
- When entering multiple email addresses, separate email addresses with a semi-colon (;) followed by a space (for example: email@example.com; email2@example.com).
- Files may contain up to 50,000 lines, including the column header.
- File uploads do not support ZIP+4.

File Management

- You cannot edit uploaded lists.
- Calix Cloud stores uploaded list files indefinitely.

To upload a subscriber or prospect list

1. Go to **Configuration > Subscriber/Prospect Lists**.
2. Click **Upload File** and select the list type.
The Upload List dialog opens.
3. Review the upload requirements and the file format. Click the template link to download a pre-formatted CSV template.
4. Click **Browse** to select the file from your computer.

To view contacts in an uploaded list

1. Go to **Configuration > Subscriber/Prospect Lists**.
2. Click on the desired list name.

View the contact list. The Record Match column shows whether each contact matched to either a subscriber record or a demographic record, depending on the list type.

First Name	Last Name	Street Address	Email(s)
Bailey	Thompson	1021 MAIN PL,,Azalea,OR,97410	
Tina	Hawkins	1021 MAIN PL APT 100,,Azalea,OR,97410	

To search a list, type a first name, last name, street address (street, city, or state), or email address into the Search box. The list dynamically updates results as you type.

Subscriber_Prospect_Template (4).csv

0 uploaded rows successfully matched to prospects records

First Name	Last Name	Street Address
Bailey	Thompson	P.O. Box 132,77 Su

To use an uploaded list for segmentation

After a list is processed (status = *Ready*), it populates in either the Subscriber Lists or Prospect Lists segment attribute filter.

1. Go to **Segments**.

2. Click **New Segment** and select the segment type.
3. Enter a name and optional description for the segment.
4. Click **Manage Filters**.
5. Click **+ Add Filter**.
6. Select either the **Subscriber Lists** or **Prospect Lists** filter. Alternately, search for the filter by typing into the Search field.
7. Select an operator. Operators determine whether the segment includes (Equals, In) or excludes (Not Equals, Not In) list contacts.
8. Select a list from the **Select value(s)** drop-down menu.
9. Click **+ Add Filter** to add more filters, or click **Save** to apply the filters.
10. Click **Save Segment**.

Once saved, you can view segment details in Dashboards or use the segment in a campaign.

Manage ZIP Code Filter Lists

This topic describes how to upload ZIP code filter lists into Calix Engagement Cloud.

From **Configuration > ZIP Code Filter Lists**, you can upload lists of ZIP codes to populate the ZIP Code List segment attribute filter. Upload lists to help identify potential upsell and acquisition opportunities, to locate new market areas for subscriber acquisition, and to plan service expansions.

Define Your Base Area

The base area is the collection of ZIP codes in which you offer your services. Engagement Cloud automatically adds ZIP codes in existing subscriber records to the base area. For prospect marketing, Engagement Cloud only provides data for prospects within the base area.

Update Your Base Area

Currently, you cannot view or modify your base area directly from within Calix Cloud. If you would like to update your base area, please log a case in MyCheckpoint.

Guidelines

General

- ZIP Code File Upload accepts CSV and XLSX files. The CSV UTF-8 file type is not supported.
- The file must include ZIP codes within your pre-defined base area.
- ZIP code lists may not be used to define your base area.
- Files may take up to 24 hours to become available.

- ZIP+4 only applies to prospect households.
- Prospect data requires Calix Engagement Cloud Plus entitlement.

File Formatting

Tip! To ensure your files are correctly formatted, download the pre-formatted template. Go to **Configuration > ZIP Code Filter Lists**, select **Upload List**, and then click the template link.

- Files must not contain column headers.
- File names may only include alphanumeric characters and spaces.
- Files may contain ZIP code only, ZIP and +4, ZIP and service availability, or ZIP, +4, and service availability.
- The ZIP code column may not contain empty cells and only accepts five-digit numerical values as inputs (e.g., 12345, 00678). For ZIP codes with fewer than five digits, precede the ZIP code with zeros (for example: 1234 > 01234).
- The +4 column only accepts four-digit numerical values as inputs (e.g., 6789). Cells in this column may be empty.
- The service availability column accepts only the text strings 'Yes' and 'No' as inputs. Cells in this column may be empty. Empty cells count as 'No'.
- Files may include three pieces of data:
 - Column 1 (required): ZIP code (e.g., 12345)
 - Column 2 (optional): +4 (e.g., 6789)
 - Column 3 (optional): Service availability (Yes/No)
- Columns must be arranged in the following order:

ZIP code	+4	Service availability
----------	----	----------------------

Example good file with ZIP code, +4, and service availability and no column headers

90210	2468	
12345		No
54321	9876	Yes

Example bad file with column headers (results in error)

ZIP	+4	Service
12345		No
54321	9876	Yes

Example bad file with empty ZIP code cell (results in error)

90210	2468	
-------	------	--

12345		No
	9876	Yes

Example bad file with service data in +4 column (results in error)

90210	2468	
12345	No	
54321	9876	Yes

Upload and Use ZIP Code List Files

To upload a ZIP code filter list

After you create your ZIP code list file, you may upload it to Calix Cloud.

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click **Upload List**.
The Upload File dialog window opens.
3. Click **Browse** to select the file from your computer.

After you click save, the page refreshes to display the list of ZIP codes imported from the list file. You may modify the ZIP code list and add, edit, or remove ZIP codes from the list. See Manage ZIP Code Lists for more information.

To use a ZIP code list for segmentation

After a list is processed (status = *Ready*), it populates in either the ZIP Code List segment attribute filter.

1. Go to **Segments**.
2. Click **New Segment** and select the segment type.
3. Enter a name and optional description for the segment.
4. Click **Manage Filters**.
5. Click **+ Add Filter**.
6. Select the **ZIP Code List** filter. Alternately, search for the filter by typing into the Search field.
7. Select an operator. Operators determine whether the segment includes (Equals, In) or excludes (Not Equals, Not In) ZIP codes in the list.
8. Select the desired list(s).

Selecting the checkbox for a list includes all ZIP codes within that list. To select specific ZIP codes within a list, click the right-facing arrow to expand the list, then select the checkbox for the desired ZIP code(s).

9. Click **+ Add Filter** to add more filters, or click **Save** to apply the filters.
10. Click **Save Segment**.

Once saved, you can view segment details in Dashboards or use the segment in a campaign.

Manage ZIP Code Lists

From **Configuration > ZIP Code Filter Lists**, you can add, edit, and delete ZIP code lists.

To edit ZIP code list name

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click on the *pencil* icon for the list to edit.
Alternately:
 - a. Click on the name of the desired list.
 - b. Click on the *three dots* menu and select **Rename**.
3. Edit the list name, then click **Save**.

To delete a ZIP code list

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click on the *trashcan icon* for the desired list.
Alternately:
 - a. Click on the list name.
 - b. Click on the *three dots* menu and select **Delete List**.
3. Click **Delete** to confirm.

To add a ZIP code to a ZIP code list

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click on the desired list name.
3. Click **Add ZIP**.
4. Enter the new **ZIP Code** and optional **+4**.
5. Select whether you offer your services within the ZIP code.
6. Click **Save**.

To edit a ZIP code within a ZIP code list

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click on the desired list name.
3. Click on the *pencil* icon for the ZIP code to edit.
4. Modify the **ZIP Code**, **+4**, and **Has Service** option as needed.
5. Click **Save Changes**.

To delete multiple ZIP codes from a ZIP code list

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click on the desired list name.
3. Select the checkbox for each ZIP code to delete.
4. Click **Delete Selection**.
5. Click **Delete** to confirm.

To delete a single ZIP code from a ZIP code list

1. Go to **Configuration > ZIP Code Filter Lists**.
2. Click on the desired list name.
3. Click on the *trashcan* icon for the desired ZIP code.
4. Click **Delete** to confirm.

Configure Data Mapping

This topic describes how to modify package mappings and rider field mappings in Calix Engagement Cloud.

Note that these settings have no effect on data in Engagement Cloud. Rather, they function as a reference for or reminder of your USOC and rider field configurations.

To add a package mapping

Package mappings associate USOCs (service tiers) with pre-built subscriber segments.

1. Go to **Configuration > Data Mapping**.
2. Click **Add Package Mapping** in the Package Mapping section.
3. Enter a **Name** for the package.

4. Select a **USOC** code and a **Target Segment**.
5. Click **Save**.

To edit rider field mappings

Rider fields allow you to add unique subscriber data (for example: birthday, onboarding date) to your billing file.

1. Go to **Configuration > Data Mapping**.
2. Click on the *down-facing arrow* in the Metrics column.
3. Select an option from the drop-down menu:
 - None
 - Onboarding Date
 - ARPU
 - Number of Services
 - NPS
 - Package
 - Birthday
 - <another option>
 - a. Enter the desired metric and an optional description.

Chapter 4

Using Calix Engagement Cloud for Subscriber Acquisition

The Calix Engagement Cloud Plus entitlement enables prospect marketing with the addition of demographic data for residential subscribers and prospects and business data for business subscribers and prospects.

Topics covered

This chapter covers the following topics:

- Understanding demographic data
- Working with demographic data for residential subscribers and prospects
- Working with business data for business subscribers and prospects
- Working with demographic and business segments

Considerations

- Demographic data, business data, and prospect marketing require Calix Engagement Cloud Plus entitlement. For more information or to purchase, please contact your Customer Success Manager.
- Currently, demographic data are available only for the United States (all 50 states, no U.S. territories).
- Engagement Cloud only provides demographic and business data for prospects within your defined base area.

Understanding Demographic Data

This topic provides an introduction to demographic data in Calix Engagement Cloud.

Demographic data describe who a subscriber is ("quantitative" attributes like age, family size, marital status) and how a subscriber behaves ("qualitative" characteristics like spending and technology usage habits). From Engagement Cloud, marketers can use demographic data and the provided pre-built segments to gain insights into specific portions of their current subscriber base, to gather detailed information on prospective subscribers (prospects), and to guide the creation of highly targeted marketing campaigns for subscriber acquisition. Demographic data update quarterly.

Engagement Cloud serves prospect data (demographic and business) for only prospects within your base area.

Residential Demographic Data

Engagement Cloud includes three unique segmentation categories for residential households:

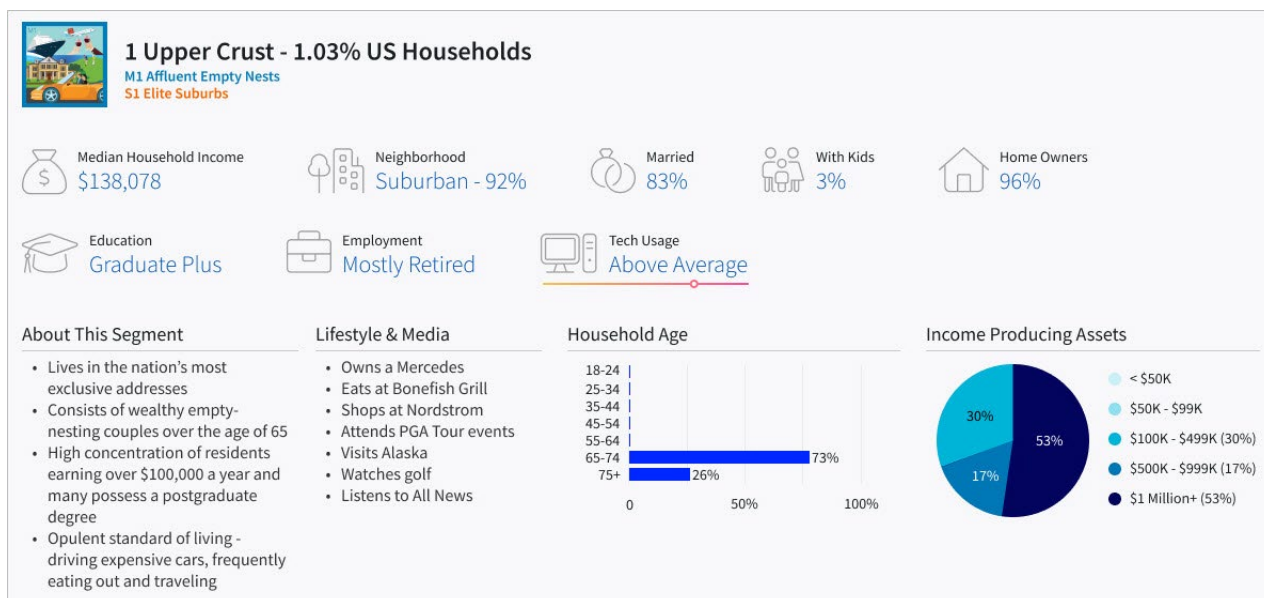
- *Lifestyle* provides comprehensive socioeconomic insights and defines 68 segments by socioeconomic rank from most (Upper Crust) to least (Bedrock America) affluent.
- *Finance* provides financial behaviors insights and defines 60 segments ranked from most (The Wealth Market) to least (Foundational Occupant) affluent.
- *Tech* provides technology usage insights and defines 53 segments ranked from highest (Technovators) to lowest (Last to Adapt) technology usage and adoption.

Residential Segment Profiles

Each Lifestyle, Finance, and Technology segment includes a profile highlighting key attributes of the segment.

You can view demographic profile data in the Top Profiles tables on **Dashboards > Subscribers > Demographic** and **Dashboards > Residential Prospects**. To view the profile, click on a segment.

Sample Lifestyle profile



Business (SMB) Data

Engagement Cloud serves insights data specifically for small and medium-sized businesses (SMBs), including annual revenue, number of employees, and business type.

- View insights data for business subscribers on **Dashboards > Subscribers > Business**.
- View insights data for business prospects on **Dashboards > Business Prospects**.

Engagement Cloud matches subscribers to SMB data records based on address. Engagement Cloud first checks the service address, and if no match is found, checks the billing address.

Explore Subscriber Demographic Data

See the topics in this section for viewing residential subscriber and business subscriber data in Calix Engagement Cloud.

The Subscribers dashboard includes two sub-tabs:

- **Demographic** presents detailed information on residential subscriber composition, including subscriber to demographic data match rate, percentage of subscribers who are homeowners, and household income distribution.
- **Business** presents key metrics for your business subscribers, including SmartBiz adoption, subscriber to business record match rates, and annual revenue distribution.

View Residential Subscriber Insights

The **Dashboards > Subscribers > Demographic** page presents detailed information on subscriber composition, including subscriber to demographic data match rate, percentage of subscribers who are homeowners, and household income distribution. Data come from third-party demographic records. Charts and KPIs show data for the previous 30 days.

This topic describes each chart and KPI on the **Dashboards > Subscribers > Demographic** dashboard.

Residential Subscriber Filters

Use the following filters to refine demographic residential subscriber data:

- Segment
- Region
- Location
- Customer Type
- Service Tier
- Campaign Audience

Insights Charts and KPIs

See the following table for description of each chart and KPI on Subscribers > Demographic.

Chart/KPI	Description
Subscribers with Demographic Data KPI	Displays two metrics: - The count of subscribers matched to a demographic data record - The percentage of total subscribers represented by the count
Families KPI	Shows the percentage of subscribers within the selected dataset categorized as a family.
Homeowners KPI	Provides a breakdown of households by the likelihood the dwelling is owned or rented: - Definite Owner - Probable Owner - Probable Renter - Definite Renter - Unknown
Average Household Income KPI	Shows the average household income for the selected dataset.
Average Age KPI	Shows the average age of subscribers within the selected dataset.
Age Range (Adult) chart	Shows the distribution of subscribers by age range: 18-24 25-34 35-44 45-54 55-64 65-74 75+

Chart/KPI	Description
Household Size chart	Displays the distribution of subscribers by number of household occupants. • 1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9+
Household Income Range chart	Displays the distribution of subscribers by household income: • \$0k-\$14k • \$15k-\$19k • \$20k-\$29k • \$30k-\$39K • \$40k-\$49k • \$50k-\$74k • \$75k-\$99k • \$100k-\$124k • \$125k-\$174k • \$175k+
Dwelling Type chart	Shows the distribution of subscribers by dwelling type: • Single-family dwelling unit (SFDU) • Multi-family dwelling unit (MFDU) • Mobile Home • Condo • Business • Unknown
Owner/Renter chart	Shows the number of subscribers who either own or rent their home.

Chart/KPI	Description
Duration of Residence chart	Shows the distribution of subscribers by their length of residence: • 0-6 months • 7-10 months • 1-2 years • 3-5 years • 6-10 years • 11-15 years • 16-20 years • 20+ years
Top Lifestyle Profiles table	Lists the most frequent Lifestyle segments for households within the selected dataset.
Top Tech Profiles table	Lists the most frequent Tech segments for households within the selected dataset.
Top Finance Profiles table	Lists the most frequent Finance segments for households within the selected dataset.

View Small and Medium-Sized Business (SMB) Subscriber Insights

The **Dashboards > Subscribers > Business** page presents detailed insights for your business subscribers.

Business Dashboard Filters

Use the following filters to refine Business dashboard data:

- Date
- Segment
- Region
- Location
- Customer Type
- Service Tier
- Campaign Audience

Business Dashboard Charts and KPIs

See the following table for a description of each chart and KPI.

Chart/KPI	Description
Subscribers with Business Data Match	Shows the number of subscribers matched to a business (SMB) data record.
Business Subscribers with Business Data Match	Shows the number of business subscribers (customer type = "business" in billing) matched to a business data record.
SmartBiz Subscribers	Displays the total number of subscribers with SmartBiz.
CommandWorx	Displays the number of subscribers who have onboarded a GigaSpire system in CommandWorx.
Business Subscribers vs SmartBiz Subscribers	Compares the number of business subscribers to the number of SmartBiz subscribers over the selected time frame.
Business Type Details	Shows the number of unique business locations for each business type and business sub-type.
Top Business by Type	Shows up to the top 10 largest business types by number of records.
Business by Annual Revenue per Location	Shows the distribution of businesses by annual revenue per location.
Business by Number of Employees per Location	Shows the distribution of businesses by number of employees.
Age of Business	Shows the distribution of businesses by the age of the business.
Business Details	Provides information for each business in the selected dataset, including business name, contact name, and business address.

Explore Prospect Data

The **Residential Prospects** and **Business Prospects** dashboards provide demographic insights and data filtering for both residential and SMB non-subscribers (prospects).

See the topics in this section for additional information on using the Prospects dashboard.

Considerations

- After a prospect becomes a subscriber, their data move from the Prospects tab to the Subscribers tab.
- You cannot search or view data for individual prospect households.
- You cannot export prospect data from Calix Engagement Cloud.
- The Business Prospects dashboard includes all registered businesses—both online and brick-and-mortar—within your base area.

View Residential Prospect Insights Data

The **Residential Prospects** dashboard presents overall demographic insights for residential prospects within your base area. Engagement Cloud sources prospect data from third-party demographic records. You cannot export prospect data from the Residential Prospects dashboard.

Residential Prospect Filters

Use the following filters to refine residential prospect data:

- Segment
- Household Size
- Household Income
- Age Range
- Technology Usage
- Campaign Audience

Residential Prospect Charts and KPIs

The following table describes each KPI and chart on the Residential Prospects page.

Chart/KPI	Description
Prospects KPIs	- Physical Addresses: Displays the total number of unique physical addresses within the selected dataset. - Households: Displays the total number of unique households within the selected dataset. A household is a unique combination of last name and mailing address. - Records: Displays the total number of prospect records within the selected dataset. A record is a unique combination of name, mailing address, and email address. - Email Addresses: Displays the total number of unique email addresses (emailtype = "executable") within the selected dataset.
Families KPI	Shows how many households, as a percentage of the total number of households within the selected dataset, that likely have children.
Homeowners KPI	Displays the percentage of total households within the selected dataset that are owned.
Average Household Income KPI	Displays the average household income for all households within the selected dataset.
Average Age KPI	Displays the average age of members in a household for all households within the selected dataset.
Age Range (Adult) chart	Shows the distribution of households by age range.
Household Size chart	Provides a breakdown of households by number of household occupants.
Technology Usage chart	Provides a breakdown of households by level of technology usage and adoption: - No Tech - Low Tech - Mid Tech - High Tech
Household Income Range chart	Provides a breakdown of households by household income.

Chart/KPI	Description
Owner/Renter chart	Provides a breakdown of households by the likelihood the dwelling is owned or rented: • Definite Owner • Probable Owner • Probable Renter • Definite Renter • Unknown
Duration of Residence chart	Provides a breakdown of households by duration of residence.
Dwelling Type chart	Provides a breakdown of households by dwelling type: • Single-family dwelling unit (SFDU) • Multi-family dwelling unit (MFDU) • Mobile Home • Condo • Business • Unknown
Top Lifestyle Profiles table	Provides details for up to the top 10 largest Lifestyle segments.
Top Tech Profiles table	Provides details for up to the top 10 largest Tech segments.
Top Finance Profiles table	Provides details for up to the top 10 largest Finance segments.

View Small and Medium-Sized Business (SMB) Prospect Insights Data

This topic describes each chart and KPI on the **Dashboards > Business Prospects** dashboard.

The Business Prospects dashboard presents overall insights for business prospects within your base area. Engagement Cloud sources data from third-party business records. You cannot export business prospect data from Engagement Cloud.

SMB Prospect Filters

Use the following filters to refine SMB prospect data.

- Segment
- Employee Count
- Annual Revenue

- Small/Home Office
- Public Sector Organization
- Campaign Audience

SMB Prospect Charts and KPIs

The following table describes each KPI and chart on Dashboards > Business Prospects.

Chart/KPI	Description
Prospects KPIs	• Businesses: Total number of unique business names within the selected dataset. • Locations: Total number of unique business locations within the selected dataset. • Multi-Location Businesses: Percentage of businesses within the selected dataset with multiple locations.
Potential Channel Coverage KPIs	Shows the percentage of marketable businesses by campaign type: • Direct Mail: Percentage of businesses within the selected dataset with a valid street address. • Phone (Verified): Percentage of businesses within the selected dataset with an active phone number. • Email: Percentage of businesses within the selected dataset with a unique, valid email address.
Top Business Sectors chart	Lists up to the top 10 business sectors with the greatest number of unique locations.
Business Sector Details table	Shows the total number of unique locations for each business sector within the selected dataset.
Business by Annual Revenue chart	Provides a breakdown of businesses by annual revenue. Axes: • Horizontal axis: Revenue range • Vertical axis: Number of business locations
Businesses by Number of Employees per Location chart	Provides a breakdown of businesses by number of employees at each business location. Axes: • Horizontal axis: Employee count range • Vertical axis: Number of business locations

Chart/KPI	Description
Age of Business chart	Provides a breakdown of businesses by number of years in business. Axes: • Horizontal axis: Age range • Vertical axis: Number of business locations
Business Details table	Provides details for businesses within the selected dataset, including: • Business Name • Business Address • Contact • Title • Business Phone • Phone Verification Level • Business Email • Website • Industry Type into the search box to search by any one of the

Work with Residential Demographic Segments

From **Segments**, you can view and create custom subscriber and prospect demographic segments and view pre-built demographic segments. Click on a recommended segment to view additional segment details.

Pre-Built Demographic Subscriber Segments

Pre-built demographic segments combine demographic attributes (for example: age range) with Lifestyle and/or Tech segments to create key subscriber and prospect personas.

Engagement Cloud provides the following pre-built demographic segments for subscribers.

Segment	Description	Demographic Segments Used	
Family	Identifies subscriber households likely to have occupants under the age of 18. Subscribers in this segment may want products and features that ensure online safety for children (for example: parental controls and Bark). See Family Segment Applications in the Reference Information section for a full list.	Lifestyle: • Networked Neighbors • Country Squires • Winner's Circle • Executive Suites • Fast-Track Families • Kids & Cul-de-Sacs • New Homesteaders • Middleburg Managers • Township Travelers • Pickup Patriarchs • Big Sky Families • White Picket Fences • Pools & Patios • Second City Startups • Bright Lights, Lil' City • Kid Country, USA	Technology: • Technovators • Plugged-In Families • Tech Nests • Connected Country • You and I Tunes • High-Tech Society • Calling Circles • Dish Country • Smart Gamers • Bundles Burbs • Kids & Keyboards • Gadgets Galore • Multi-Media Families • Rural Transmissions

Segment	Description	Demographic Segments Used	
Senior	Identifies subscribers aged 65 and older who may prioritize affordability and reliability.	Lifestyle: • Upper Crust • Movers & Shakers • Money & Brains • Gray Power • Big Fish, Small Pond • Cruisin' to Retirement • Urban Elders • Mayberry-ville • Empty Nests • Country Casuals • Traditional Times • Toolbelt Traditionalists • Hometown Retired • Domestic Duos • Heartlanders • American Classics • Simple Pleasures • Lo-Tech Singles • Back Country Folks • Golden Ponds • Crossroad Villagers • Park Bench Seniors	Technology: • Cinemaniacs • Analoggers • Internet Hinterlands • Low Speed Boomers • Rural Transmissions • Broadband Boulevards • Opting Out • Antenna Land • Tech Skeptics • Bucolic Basics • Landline Living • Old-Time Media • Discounts and Deals • Dial-Up Duos • Satellite Seniors • Early-Bird TV • Tech-Free Frontier • The Unconnected • Last to Adopt

Segment	Description	Demographic Segments Used	
Tech Savvy	Identifies subscribers likely to quickly adopt new technology and services. These subscribers may have many devices and a higher service tier.	Lifestyle: • Young Digerati • Upward Bound • The Cosmopolitans • Up-and-Comers • Connected Bohemians • Young and Influential • Urban Achievers	Technology: • Technovators • Plugged-In Families • Tech Nests • Connected Country • You & I Tunes • High-Tech Society • Generation WiFi • Calling Circles • Dish Country • Smart Gamers • WiFi Warriors • Cyber Sophisticates • The Pragmatics • Bundled Burbs • Kids & Keyboards • New Technorati • Gadgets Galore • Multimedia Families • Cyber Strivers • Internet Hinterlands • Video Vistas • IM Nation • Techs and the City • Plug & Play

Segment	Description	Demographic Segments Used	
Work from Home	Identifies subscribers with high data usage from work-related applications (for example: Microsoft Outlook). Subscribers in this segment expect a stable, reliable internet connection. This segment uses a combination of flow data from the subscriber's system and demographics to identify segment members.	Lifestyle: • Upper Crust • Networked Neighbors • Movers & Shakers • Young Digerati • Country Squires • Winner's Circle • Money & Brains • Executive Suites • Fast-Track Families • Upward Bound • Kids & Cul-de-Sacs • New Homesteaders • Beltway Boomers • Middleburg Managers • Township Travelers • Pickup Patriarchs • Up-and-Comers • Home Sweet Home • Pools & Patios • Connected Bohemians • Young & Influential	Technology: • You & I Tunes • Wi-Fi Warriors • Satellites & Silos • Cyber Sophisticates • The Pragmatics • Time Shifters • New Technorati • Cinemaniacs • Analoggers • Internet Hinterlands • Low-Speed Boomers • Rural Transmissions • Plug & Play • Broadband Boulevards • Opting Out

Segment	Description	Demographic Segments Used
Gamer	Identifies subscribers with gaming consoles/devices and high data usage from gaming applications (for example: Twitch). Subscribers in this segment expect a stable connection that delivers a lag-free gaming experience. This segment uses a combination of flow data from the subscriber's system and demographics to identify segment members.	Lifestyle: • Networked Neighbors • Country Squires • Winners Circle • Executive Suites • Fast-Track Families • Kids & Cul-de-Sacs • New Homesteaders • Beltway Boomers • Up-and-Comers • Pools & Patios • Second City Startups • Young & Influential • Bright Lights, Lil' City • Kid Country, USA • Multi-Culti Families • Small-Town Collegiates Technology: • Generation WiFi • Calling Circles • Dish Country • Smart Gamers • Bundled Burbs • Video Vistas • IM Nation • Techs and the City • Techtown Lites
SmartBiz Upsell	Identifies all existing subscribers who 1) match to an SMB data record, and 2) do not have SmartBiz. Use this segment to identify business subscribers for SmartBiz upsell opportunities.	N/A

Pre-Built Demographic Prospect Segments

Engagement Cloud provides the following pre-built demographic segments for prospects.

Segment	Description
Family	Identifies households likely to have occupants under the age of 18. Subscribers in this segment may want products and features (such as parental controls and Bark) that ensure online safety for their children.

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Segment	Description
Senior	Identifies prospects aged 65 and older who may prioritize affordability and reliability.
Tech Savvy	Identifies prospects likely to quickly adopt new technology and services. These subscribers have many devices, several managed services subscriptions, and a higher-tier service package.
Work from Home	Identifies prospects with high data usage from work-related applications (for example: Microsoft Outlook). Subscribers in this segment expect a stable, reliable internet connection.
Gamer	Identifies prospects with gaming consoles/devices and high data usage from gaming applications (for example: Twitch). Subscribers in this segment expect a stable connection that delivers a lag-free gaming experience.
SmartBiz Acquisition	Identifies business prospects within your base area that would likely benefit from SmartBiz.

Create Residential Demographic Segments

This topic describes how to create demographic segments for residential subscribers and prospects in Calix Engagement Cloud.

To create a residential demographic segment

1. Go to **Segments**.
2. Select either **New Segment > Subscribers** or **New Segment > Residential Prospects**.
3. Enter a name for the segment.
4. (Optional) Enter a description of the segment.
5. Apply the desired demographic attribute filters:
 - a. Click **Manage Filters**.
 - b. Click **+ Add Filters**.
 - c. Select a filter from the drop-down menu. See *Demographic Attribute Filters* (on page [163](#)) for filter descriptions.
 - d. Configure the filter options:
 - Select an operator. Options vary depending on the selected filter.
 - Select or enter the filter criteria.
 - e. Repeat steps b through d for each desired filter.
 - f. Click **Save**.
6. Select a region on the map using one of the following methods:

Note: The map supports up to one inclusion boundary (free-form or circle) and one exclusion boundary (free-form or circle).

- Draw a free-form boundary:
 - a. Click **Select Boundaries**.
 - b. Select either **Inclusion Boundary** or **Exclusion Boundary**.
 - c. Click on the map to set the initial point.
 - d. Move the cursor to a different spot on the map and click again to set the next point.
 - e. Continue until you enclose the desired area.
 - f. To close the shape, double-click. Once the shape is closed, all boundary lines turn solid.
- Place a circular boundary:
 - a. Click **Place circle**.
 - b. Select either **Inclusion circle** or **Exclusion circle**.
 - c. Click on the map to set the center point of the circle.
 - d. Enter the desired radius (in miles) in the popup dialog, and then click **Save**.

Note: The dialog may take a few seconds to appear.

7. Click **Apply Selection**. To reset the boundary, click **Clear Boundaries**.

The Preview table lists subscribers in the segment who meet the filter criteria. Click **Download CSV** to export the segment to a CSV file.

8. Click **Save Segment**.

Demographic Attribute Filters

Engagement Cloud Plus provides the following attribute filters for creating residential demographic segments.

Filter Category	Filter	Description	How to Apply the Filter
Account Information	ZIP Code List	Filters businesses by ZIP code. This filter populates ZIP code lists uploaded via Configuration > ZIP Code Filter Lists.	Select an operator, and then select the desired list(s).
Campaigns	Campaign Audience	Filters by groups of households previously used in a campaign.	Select an operator, and then select the desired audience(s).
Demographics	Age Range	Filters households by age range: • 18-24 years old • 25-34 years old • 35-44 years old • 45-54 years old • 55-64 years old • 65-74 years old • 75+ years old	Select an operator, and then select the desired age range(s).
	Business or Home Type	Filters households by type of business or home: • Business • Condo • Multi-family dwelling unit (MFDU) • Mobile Home • Single-family dwelling unit (SFDU) • Unknown	Select an operator, and then select the desired unit type(s).
	Demographic Data Matched	Identifies households matched to a demographic data record.	Select an operator, and then select either Y or N.

Filter Category	Filter	Description	How to Apply the Filter
	Education	Filters households by education level: • College • Graduate School • High School • Some College • Some High School or Less	Select an operator, and then select the desired education level(s).
	Families (Presence of Children)	Filter for subscriber households likely comprised of at least one adult with children present (Yes, No).	
	Finance Profile	Filters households by pre-defined finance segment. See the Finance Audience Segments article for more information.	Select an operator, and then select the desired demographic finance segment(s).
	Home Owner / Renter	Filter for subscribers by the likelihood they either own or rent their residence: • Definite Renter • Probable Renter • Probable Owner • Definite Owner • Unknown	Select and operator, and then select the desired option(s).
	Household Income	Filters households based on income range.	Select an operator, and then select the desired income range(s).

Filter Category	Filter	Description	How to Apply the Filter
	Household Occupation	Filters households by occupation: • Administrative-Managerial • Business Owner • Clerical-White Collar • Craftsman-Blue Collar • Doctors • Farm • Federal Employee • Federal Employee Retired • Foreman • Healthcare • Home Office • Homemaker • Lawyers-Judges • Management • Military • Military Retired • Not Available • Operatives • Professional-Technical • Proprietors • Retired • Sales-Service • Service Workers • Source Occupation Given But Not Interpretable • Student • Supervisors • Teachers-Librarians • Unskilled	Select an operator, and then select the desired occupation(s).
	Household Size	Filters households by number of occupants.	Select an operator, and then select the desired option(s).

Filter Category	Filter	Description	How to Apply the Filter
	Length of Residence	Filters households by length of time they have lived at their address: • 0-6 mos. • 1-2 yrs • 3-5 yrs • 6-10 yrs • 7-10 mos. • 11-15 yrs • 16-20 yrs • 20+ yrs	Select an operator, and then select the desired option(s).
	Lifestyle Profile	Filters households by pre-defined lifestyle segment. See the Lifestyle Audience Segments article for more information.	Select an operator, and then select the desired profile(s).
	Marital Status	Filters households by marital status: • At least 1 married • At least 1 single • Both married and single	Select an operator, and then select the desired option(s).
	Number of Adults	Filters households by number of adults (occupants aged over 18) in the household.	Select an operator, and then select the desired option(s).
	Number of Children	Filters households by number of children in the household.	Select an operator, and then select the desired option(s).
	Tech Profile	Filters households by pre-defined tech segment. See the Tech Audience Segments article for more information.	Select an operator, and then select the desired profile(s).
Prebuilt Prospect Segment	Family	Includes prospect households in the Family recommended segment.	Select an operator, and then select either <i>false</i> or <i>true</i> .

Filter Category	Filter	Description	How to Apply the Filter
	Gamer	Includes prospect households in the Gamer recommended segment.	
	Senior	Includes prospect households in the Senior recommended segment.	
	Tech Savvy	Includes prospect households in the Tech Savvy recommended segment.	
	Work From Home	Includes prospect households in the Work from Home recommended segment.	

Work with Business Segments

From **Segments**, you can view and create custom business subscriber and business prospect segments and view pre-built business segments. Click on a pre-built segment to view additional segment details.

Pre-Built Small and Medium-Sized Business (SMB) Segments

Engagement Cloud provides two pre-built segments for business prospect marketing:

- The **SMB Upsell** segment identifies existing business subscribers for SmartBiz upsell opportunities. This segment includes all your existing subscribers who 1) match to an SMB data record¹, and 2) do not have SmartBiz. View segment insights from Segments > Subscribers and use the segment as a starting point for further data refinement and segmentation.

Note: When creating a new segment, you can identify business subscribers either by applying only the 'SMB Upsell' segment filter or by applying multiple other filters (for example: 'SmartBiz' and 'SMB Matched'). The SMB Upsell segment filter alone may return fewer results than when using a combination of other filters because the SMB Upsell segment only includes SMB data records with specific business types (see note 1 below).

- The **SMB Acquisition** segment identifies business prospects² within your base area. View segment insights from Segments > Prospects and use the segment as a starting point for further data refinement and segmentation.

¹ SMB data records include a data field called 'Business Type' that may contain one of the following values, based on North American Industry Classification System (NAICS) classifications:

- | | |
|--|--|
| • Agriculture, Forestry, Fishing, and Hunting | • Management of Companies and Enterprises |
| • Mining, Quarrying, and Oil and Gas Extraction | • Administrative and Support and Waste Management and Remediation Services |
| • Utilities | • Educational Services |
| • Construction | • Health Care and Social Assistance |
| • Manufacturing | • Arts, Entertainment, and Recreation |
| • Wholesale Trade | • Accommodation and Food Services |
| • Retail Trade | • Other Services (except Public Administration) |
| • Transportation and Warehousing | • Public Administration |
| • Finance and Insurance | |
| • Real Estate and Rental and Leasing | |
| • Professional, Scientific, and Technical Services | |

Recommended segments (both Upsell and Acquisition) exclude SMB data records with a Business Type field that either is empty or contains a different value from those above.

² A business prospect is a non-subscriber within your base area with an SMB data record.

Create Business Segments

This topic describes how to create business segments in Calix Engagement Cloud.

To create a business segment

1. Go to **Segments**.
2. Select either **New Segment > Subscribers** or **New Segment > Business Prospects**.
3. Enter a name for the segment.
4. *(Optional)* Enter a description of the segment.
5. Apply the desired demographic attribute filters:
 - a. Click **Manage Filters**.
 - b. Click **+ Add Filters**. See *Business Segment Attribute Filters* (on page [170](#)) for filter descriptions.
 - c. Select a filter from the drop-down menu.
 - d. Configure the filter options:
 - Select an operator. Options vary depending on the selected filter.
 - Select or enter the filter criteria.
 - e. Repeat steps b through d for each desired filter.
 - f. Click **Save**.
6. Select a region on the map using one of the following methods:

Note: The map supports up to one inclusion boundary (free-form or circle) and one exclusion boundary (free-form or circle).

- Draw a free-form boundary:
 - a. Click **Select Boundaries**.
 - b. Select either **Inclusion Boundary** or **Exclusion Boundary**.
 - c. Click on the map to set the initial point.
 - d. Move the cursor to a different spot on the map and click again to set the next point.
 - e. Continue until you enclose the desired area.
 - f. To close the shape, double-click. Once the shape is closed, all boundary lines turn solid.
- Place a circular boundary:

- a. Click **Place circle**.
- b. Select either **Inclusion circle** or **Exclusion circle**.
- c. Click on the map to set the center point of the circle.
- d. Enter the desired radius (in miles) in the popup dialog, and then click **Save**.

Note: The dialog may take a few seconds to appear.

7. Click **Apply Selection**. To reset the boundary, click **Clear Boundaries**.

The Preview table lists subscribers in the segment who meet the filter criteria. Click **Download CSV** to export the segment to a CSV file.

8. Click **Save Segment**.

Business Segment Attribute Filters

Engagement Cloud Plus provides the following attribute filters for creating business segments.

Filter Category	Filter	Description	How to Apply the Filter
Account Information	ZIP Code List	Filters subscribers by ZIP code. This filter populates with ZIP code lists uploaded from Configuration > ZIP Code File Upload. See <i>Manage ZIP Code Filter Lists</i> (on page 135) for more.	Select an operator, and then select the desired ZIP code(s) from the list: - To select an entire list, select the checkbox next to the list name. - To select ZIP codes within a list, click on the expand arrow for the list, and then select the checkbox for each ZIP code.
Business Information	Age of Business	Filters businesses by business age.	Select an operator, and then select the desired option(s).
	Annual Revenue	Filters businesses by annual revenue.	Select an operator, and then select the desired option(s).
	Business Data Matched* <i>*subscribers only</i>	Filters for subscribers matched to a third-party business data record.	Select an operator, and then select either <i>Y</i> or <i>N</i> .
	Business Operational	Filters for businesses that are still in business/operational.	Select an operator, and then select either <i>Yes</i> or <i>No</i> .

Filter Category	Filter	Description	How to Apply the Filter
	Business Sector	Filters businesses by industry. This filter uses two-digit business type classifications from the North American Industry Code System (NAICS).	Select an operator, and then select the desired sector(s).
	Educational Institution	Filters for businesses that are educational institutions.	Select an operator, and then select either <i>Y</i> or <i>N</i> .
	Employee Count	Filters businesses by number of employees.	Select an operator, and then select the desired option(s).
	Government Entity	Filters for businesses that are government entities.	Select an operator, and then select either <i>Y</i> or <i>N</i> .
	Industry	Filters business subscribers by industry. This filter uses six-digit business type classifications defined by the North American Industry Code System (NAICS).	Select an operator, and then select the desired industry.
	Multi-Location	Filters for whether a business has more than one office/location. (For example, a bank with multiple branches or a restaurant chain with multiple franchise locations.)	Select an operator, and then select either <i>Y</i> or <i>N</i> .
	Non-Profit Entity	Filters for businesses that are non-profit organizations.	Select an operator, and then select either <i>Y</i> or <i>N</i> .

Filter Category	Filter	Description	How to Apply the Filter
	Phone Verification Confidence	Filters businesses by the likelihood that the phone number is still connected and valid: • Not Verified (<i>phone number likely disconnected and/or no longer valid</i>) • Verified, low confidence • Verified, moderate confidence • Verified, high confidence • Verified, highest confidence (<i>highest likelihood phone number still connected and valid</i>)	Select an operator, and then select the desired option(s).
	Small Office/Home Office	Filters for whether the business is a small office/home office.	Select an operator, and then select either Y or N.
	US Postal Service Indicator	Filters businesses by the type of mail delivery address: • Residential • Business • Commercial Mail Receiving Agency	Select an operator, and then select the desired option(s).
Campaigns	Campaign Audience	Filters by groups of subscribers previously used in a campaign.	Select an operator, and then select the desired audience(s).
Pre-built Business Prospect Segment	SmartBiz Acquisition	Includes prospect households in the SmartBiz Acquisition recommended segment.	Select an operator, and then select either <i>false</i> or <i>true</i> .

Create and Deploy Prospect Campaigns

This topic describes how to create and deploy prospect campaigns in Calix Engagement Cloud.

From the **Campaigns** page, you can deploy email and direct mail campaigns to both residential and business prospects. Additionally, you can choose to deploy direct mail campaigns to prospects who have opted out from email marketing communications but still have a mailable address. A prospect's opt out preference is recorded on their third-party data record.

For residential and business prospects, Engagement Cloud sends the following information to marketing channels:

- Residential:
 - First Name
 - Last Name
 - Address (combined street, city, state, ZIP code)
 - Email Address

This information comes from the prospect's third-party demographic data record.

- Business:
 - First Name
 - Last Name
 - Address (combined street, city, state, ZIP code)
 - Email Address
 - Phone

This information comes from the business prospect's third-party SMB data record.

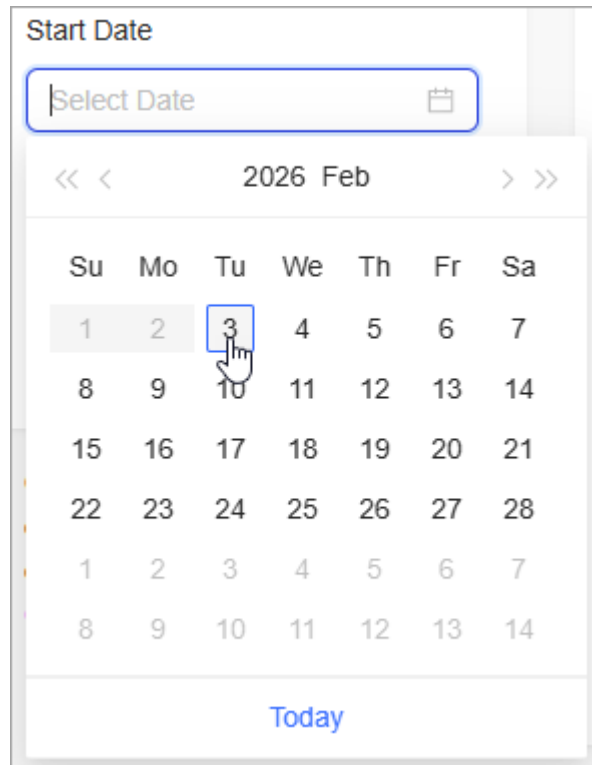
To create a scheduled prospect campaign

1. Go to **Campaigns**.
2. Select either **New Campaign > Residential Prospects** or **New Campaign > Business Prospects**.
3. Enter a helpful name and description (optional) for the campaign.
4. Select channels:
 - Mailchimp (email and direct mail)
 - Constant Contact (email)
 - Facebook (social media)
 - HubSpot (email and direct mail)
 - Postalytics (direct mail)

Note: The Channels section displays only the channels configured in Campaigns > Channels.

5. Select the **Start Date** and the **End Date**.

- Click into the **Select Date** field to open the calendar filter. Use the left and right-facing single arrows to advance by month and the double arrows to advance by year. Click **Today** to select the current date.



6. Select segment(s) to include in and exclude from the campaign.

7. Configure channel-specific settings:

- (Optional - Residential campaigns only) For Mailchimp and HubSpot: Select the **Include additional Direct Mail Only** audience list checkbox to send the campaign to prospects who have opted out from email communications.
- For Postalytics: Select the corresponding Postalytics drip campaign.

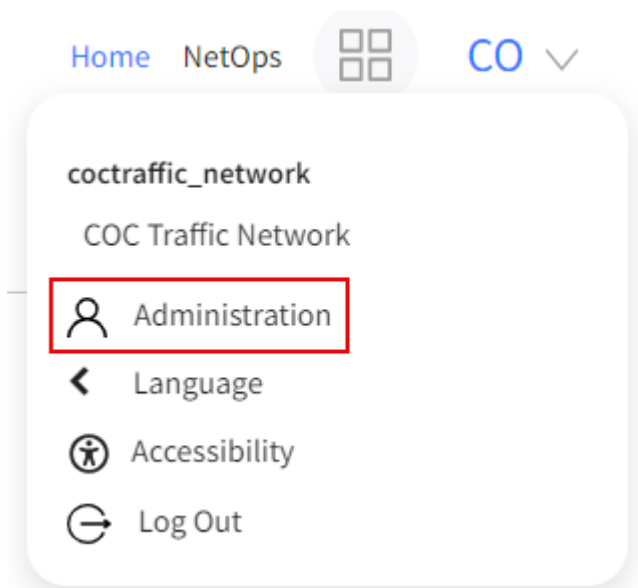
8. Click **Save as Draft** to save your settings and edit later, or click **Save & Deploy** if ready to deploy.

Once deployed, you can monitor campaign progress and results on the campaign *Results* (on page [96](#)) tab.

Chapter 5

Administering Calix Cloud Applications

See topics in this section to administer Calix Cloud applications for your organization. Access to Administration requires the OrgAdmin role.



Set Up Users and Roles

Roles belong to an organization. Permission sets associated to each role provide access to specific functionality in Calix Cloud, and each permission has an allowed action of Read only or Write and Read. Multiple roles can be assigned to a user.

Calix Cloud supports the following role types:

- System roles are pre-defined and always present.
 - **OperationBusinessInsights:** A role with access to KPI and trend charts related to Operations Cloud on a single landing page. This role cannot be modified.
 - **OrgAdmin:** A role with access to all areas of the Administration section of Cloud Cloud. This role cannot be modified.
 - **APIUser:** A role to execute Calix public APIs. Only a user with this role can execute HTTP requests from the *Developer Portal* <https://developers.calix.com/>. Assign this role to an independent Calix user account intended for API execution only. This role cannot be modified.
 - **SiteAdmin:** A role for federated users which may provide access to all areas of the Administration section of Calix Cloud and/or access to the Users, Deactivated Users, Account Management, and/or Roles areas. A user with the OrgAdmin role can modify this role as needed.
 - **ServiceBusinessInsights:** A role with access to KPI and trend charts related to Service Cloud on a single landing page. This role cannot be modified.
 - **MarketingBusinessInsights:** A role with access to KPI and trend charts related to Engagement Cloud on a single landing page. This role cannot be modified.
- Modifiable default roles for each Calix Cloud application based on personas.
- Custom roles that you can define for each Calix Cloud application.

Note: A role cannot be associated to permission sets across different Calix Cloud applications.

Set Up Calix Cloud Application Users

During the Calix Cloud onboarding process, users are added to your organization. To add an additional user, the user must *register* <https://www.calix.com/user-registration.html> for a My Calix account using an organization email address which then syncs with the Cloud. Once processed, the user appears on the Administration > Users page.

Guidelines

- You cannot add new users from Calix Cloud.

- Do not change user passwords from Calix Cloud; Calix Cloud uses the same login credentials as the My Calix portal.

Set Up Roles for Calix Engagement Cloud

From **Administration > Roles**, Organization Administrators can view, add, edit, or delete roles for Calix Engagement Cloud. The Organization Administrator can then associate users with roles to provide them with the permissions needed to perform tasks within Engagement Cloud. Calix Cloud includes pre-defined Engagement Cloud roles that include permissions for viewing and using subscriber data, revenue data, campaigns, and Dashboards. Organization Administrators can assign roles to existing users, create new users, and create custom roles.

Assign and create roles with the following permissions:

- **Campaign:** Provides users with access to the Campaigns section of Engagement Cloud. Creating campaigns requires *Write and Read*.
- **Explored Data:** Provides users with access to the Dashboards section of Engagement Cloud.
- **Revenue:** Provides users with access to all revenue-related data appearing on the campaign Results, Dashboards, and Home pages. (This permission is *Read only*.)
- **Subscriber:** Provides users with access to the Subscriber Dashboard. (This permission is *Read only*.)

Guidelines

- To see the latest role and permission updates, users must log out and then log back in to the application.
- Calix Cloud permissions for Engagement Cloud do not provide access to the Administration panel. Access to the Administration panel requires the *OrgAdmin* role.

To view existing Calix Engagement Cloud roles

1. From **Administration > Roles**, go to **Engagement Cloud Roles**.
2. View existing roles:
 - *MarketingReadOnly* limits users to only viewing data on the Home, Dashboards, Segments, Campaigns, and Configuration pages. Users with this role cannot create campaigns or segments.

- *MarketingAllAccess* allows users to view and use all features within Engagement Cloud, including create campaigns and segments.



To add roles

1. Go to **Administration > Roles > Engagement Cloud Roles**.
2. Click **Add New Role**.
3. Type a **Role Name**.
4. Type a **Description** (optional).
5. Select **Permissions** for the role.
6. For each permission selected, choose **Read only** or **Write and Read**.
7. From **Add Users**, type the user's name or e-mail address or use the drop-down to locate the user.

If the user does not appear, go to **Administration > Users** to create a new user.

Note: You can also associate users with roles from the **Administration > Users** page. See *Set Up Calix Cloud Application Users* (on page [176](#)).

8. Click **Add** to populate the list.
9. Repeat **Steps 7-8** to locate and add users.
10. Select the user name(s) in the list.
11. Click **Save**.

To edit roles

1. Click the *pencil* associated with the role.
2. Update the role as desired.
3. Click **Save**.

To delete roles

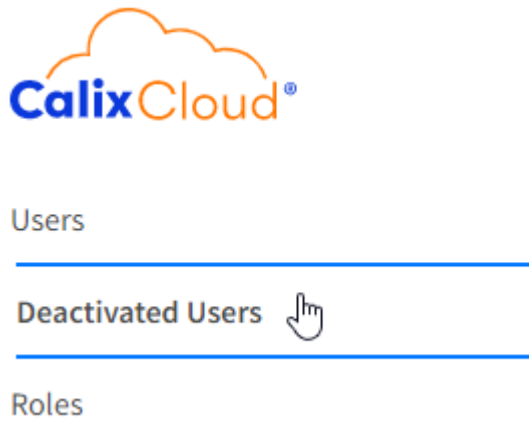
1. Select the role to delete and click the *trashcan*.
2. Confirm to delete.

Deactivate Users

From the **Administration > Deactivated Users** page, Organization Administrators can deactivate Calix Cloud access for users who no longer need access or should not have access to the application.

To view deactivated users

Go to **Administration > Deactivated Users**:



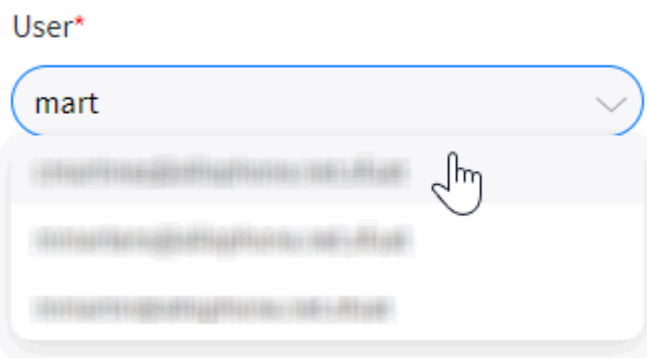
A list of deactivated user names display with the reason and deactivated date.

To deactivate a user

1. Click **Search Users to Deactivate**.

A blue, rounded rectangular button with the text 'Search Users to Deactivate' in white.

2. From the User drop-down list, select a user to deactivate.



3. Select a reason for the deactivation.

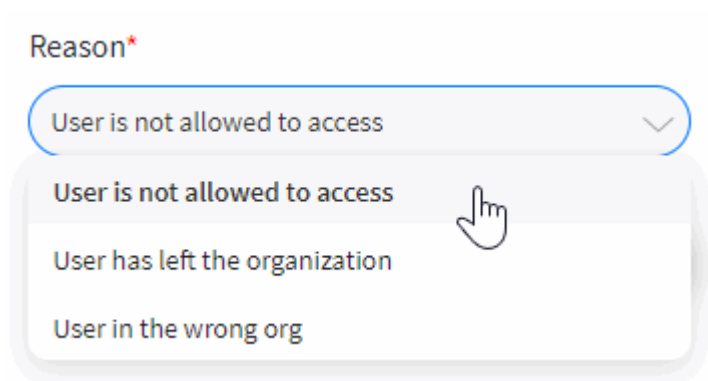
Reason*

User is not allowed to access

User is not allowed to access

User has left the organization

User in the wrong org



4. Click **Save**.

Configure Calix Cloud Settings

View and Configure Calix Cloud ACS Settings

From the **Administration > ACS Configuration** page, Organization Administrators can view and configure the following Calix Cloud auto configuration servers (ACS) Settings:

- **ACS Configuration for CPE (HTTPS, IPv4+IPv6):** This URL is unique per organization and provided by Calix. It allows service providers to establish a bi-directional encrypted communication channel between Calix Cloud and the systems in their network.
 - You can update the username, if desired. Once updated, click **Submit**.
 - View the password by clicking the eye icon.
- **Captive Portal:** Subscribers can self-install systems if service providers have configured a captive portal URL (which is configured on the system on-discovery). The user is then redirected to the captive portal on any browser activity.

Calix Service Cloud ACS Settings

ACS Configuration for CPE (HTTPS, IPv4+IPv6)

HTTPS, IPv4 :

https://gcs-stg.calix.com:8443/12953272/

User Name : *

acs-user-6vRoC

Password : *

.....



HTTPS, IPv4+IPv6 :

https://gcs6-stg.calix.com:8443/12953272/

HTTP IPv4 :

http://gcs-stg.calix.com:8080/12953272/

Captive Portal

URL :

Server Alias :

View Organization Information

From the **Administration > Organization Info** page, Organization Administrators can view the Organization SPID (Service Provider ID), entitlements, and subscription dates:

Users

Deactivated Users

Roles

ACS Configuration

Organization Info

Flow Configuration

Organization ACL

SSO Configuration

Call Outcome

Blocked Page Templates

TechMedia

Service Provider ID (SPID) : a10LVF

EXOS Framework

Start Date

End Date

Status

Active

ExperienceIQ And ProtectIQ

Start Date

End Date

Status

Active

Calix Operations Cloud

Start Date

End Date

Status

Active

Organization SPID

Calix provides each organization a unique SPID, which is a 10 digit alpha-numeric identifier that is essential for network operations involving Calix GigaSpires. The SPID is used to assign GigaSpires to your specific org instance within the multi-tenant Calix Cloud infrastructure. All GigaSpires are required to have a SPID assigned upon turn-up.

Application Subscription Dates

Based on your organization's entitlement subscriptions, you can view start, end, and trial end dates.

Set Up the Flow Configuration

If enabled for your organization, Organization Administrators can use a variety of tools to investigate application and location traffic patterns. The Flow Configuration feature requires a Calix Operations Cloud entitlement.

From the **Administration > Flow Configuration** section of Calix Cloud, Organization Administrators can perform the following actions:

- View and/or set up **Network** definitions, such as **Devices**, **Subnets**, and **Radius Servers**.

Note: IPv6 subnet addressing and validation are supported.

- View and/or set up **Locations**.
- View and/or set up **Applications**.
- View and/or set up **Endpoints**.

Set Up the Network

From **Administration > Flow Configuration > Network**, Organization Administrators can view and set up device mapping, subnets, and Radius servers.

Set Up Device Mapping

Organization Administrators can view, create, update, delete, export, and import flow device definitions for their organization.

Guidelines

-  In the majority of cases, Calix Operations Cloud processes Netflow data for upstream, downstream, and internal communication within the network and ignores remote to remote communication, where both source and destination IPs are outside of your network subnets.
- Device IP, Sampling Rate, and Flow Type data are imported. Default information is imported for the remaining fields.

To view flow device definitions

To view flow device definitions, go to **Administration > Flow Configuration > Network > Devices**.

- To view the Host Name, click the **i** for information.

Note: The **Host Name** is acquired by DNS lookup based on the router IP address.

- To locate definitions based on specific criteria, use the **Search** field. The system returns matches for **Device IP**, **Receiving Port**, **Host Name**, **Flow Type**, and/or **Sampling Rate**.

To create flow device definitions

1. From **Network > Devices**, click **Create**.
2. Specify the flow device definition:
 - **Name:** Specify a name for the device.
 - **Host Name:** Specify the host name.
 - **Device IPv4:** Specify the router IP (IPv4).
 - **Device IPv6:** Specify the router IP (IPv6).
 - **Ignore Upstream:** Select Yes (Y) to ignore flow traffic going out of the network. The default is No (N).
 - **Ignore Downstream:** Select Yes (Y) to ignore flow traffic coming into the network. The default is No (N).
 - **Ignore Local:** Select Yes (Y) to ignore drop flow records from the device where the flow address source and destination match the subscriber subnet (which is indicative of peer to peer connection). The default is No (N).
 - **Observation Domain:** Specify an observation domain.
 - **Sampling:** There is no default sampling rate. When present, Calix Operations Cloud uses the sampling rate as defined in the received Netflow data. If no sampling rate is defined, the configured sampling rate within Calix Cloud is used instead. The sampling rate is configured on the router to reduce the bandwidth used by the exported Netflow data and to reduce the CPU load on the router if it does not have dedicated Netflow hardware. The same sampling rate that you configure on the router needs to be entered in the flow device definition to enable Calix Operations Cloud to calculate actual traffic bytes accurately. Some level of sampling is required. The level of sampling used should be determined by flow device characteristics, flow type (Netflow, IPfix, sFlow, etc.) and device loading or traffic level. For examples, see *Sampling Rates/WAN Bandwidth/Accuracy Loss Examples* (on page [187](#)).
 - **Geo-coordinates:** Specify geo-coordinates.
 - **Address:** Specify the address.
3. Click **Submit**.

The flow device definition appears in the table list view.

To update flow device definitions

1. Click the *pencil* for the device definition.
2. Modify the data as desired and click **Update**.

To delete flow device definitions

To delete a single flow device definition:

1. Click the *trashcan* that corresponds with the device definition.
2. Confirm the deletion.

To delete multiple flow device definitions:

1. Select the device definitions, and click **Delete**.
2. Confirm the deletion.

To export flow device definitions

1. Click **Export**.
2. Open or save the exported file.

To import flow device definitions

1. Click **Full Import**.
2. View the message, and click **Okay** to confirm.
3. Browse to locate the file for import, and click **Open**.
4. View the flow device definitions in the preview table.

Note: All existing flow device definition records that are not in the import file are marked for deletion. Confirm the records in the preview table.

5. Click **Submit**.
The flow device definitions are imported.

To import flow device definitions incrementally

1. Click **Incremental Import**.
2. Browse to locate the file for import, and click **Open**.
3. View the flow device definitions in the preview table.
4. Click **Submit**.

The flow device definitions are added to your list.

Sampling Rates/WAN Bandwidth/Accuracy Loss Examples

For example sampling rates, WAN bandwidth usage, and estimated accuracy loss, refer to the following data as guidelines:

Network Traffic Routing through Router	WAN Bandwidth used by Flow Export	Sample Rate	Estimated Loss of Accuracy Due to Sampling*
1 Gbps	8 Mbps	1:1	0%
1 Gbps	800 Kbps	10:1	0.06 - .012%
1 Gbps	190 Kbps	64:1	0.38 - 0.77%
1 Gbps	80 Kbps	100:1	0.6 - 1.2%
1 Gbps	16 Kbps	500:1	3 - 6%
1 Gbps	8 Kbps	1000:1	6 - 12%

*Measured usage is always under actual usage, never over.

Manage Subnets

Subnets are used to define IP address ranges within your network as included or excluded. Calix Cloud drops all traffic that is not permitted by a subnet. From to **Administration > Flow Configuration > Network > Subnet**, Organization Administrators can view, create, delete, export, and import subnets for the organization.

To view subnets

To view subnets, go to **Administration > Flow Configuration > Network > Subnet**.

To locate subnets based on specific criteria, use the **Search** field.

To create subnets

1. From **Network > Subnet**, click **Create**.
2. Define the subnet:
 - **Subnet:** Select SubnetIPv4 or SubnetIPv6.
 - **Static:** Select Yes to create a assign a static subnet (default = no). When traffic arrives for an IP address within a static subnet, Calix Cloud automatically creates the endpoint with a mapped type of static.

- **SubnetIPv4:** If you selected SubnetIPv4, type a subnet in IPv4 CIDR format. For example, 192.0.2.0/24.
 - **SubnetIPv6:** If you selected SubnetIPv6, type a subnet in IPv6 CIDR format. For example, 2001:DB8::/32.
 - **Excluded:** Select Yes to exclude the subnet from your network (default = No). Use this setting when subnet A belongs to subnet B, and subnet A does not belong to your organization. Calix Cloud does not monitor traffic from excluded subnets.
3. Click **Submit**.
- The subnet appears in the table list view.

To update subnets

1. Click the pencil icon for the subnet.
2. Modify the data as desired and click the **Update**.

To delete subnets

For a single subnet:

1. Click the trashcan that corresponds with the subnet.
2. Confirm the deletion.

For multiple subnets:

1. To select all displayed subnets, click the checkbox in the Selected column header.



OR

To select specific subnets, click the checkbox corresponding to each subnet.

2. Click **Selected**.
3. Confirm the deletion.

To export subnets

1. Click **Export**.

2. Open the unformatted Excel CSV file from your downloads folder, and adjust the column widths as needed.

To import subnets

1. Click **Full Import**.
2. View the message, and click **Okay** to confirm.
3. Browse to locate the file for import, and click **Open**.
4. View the subnets in the preview table.

Note: All existing subnets that are not in the import file are marked for deletion. Confirm the records in the preview table.

5. Click **Submit**.
The subnets are imported.

To import subnets incrementally

1. Click **Incremental Import**.
2. Browse to locate the file for import, and click **Open**.
3. View the subnets in the preview table.
4. Click **Submit**.

The subnets are added to your list.

Set Up Radius Servers

Radius servers enable you to authenticate subscribers on your network. For example, if subscribers connect a DSL modem, they would be required to enter PPPoE credentials (username and password). These credentials are passed to a Radius server that checks against a database of known customers and validates if they are allowed to use the network.

Note:  The Radius server configuration is only required if your organization uses Radius for endpoint mapping in the native Calix Operations Cloud deployment.

To view Radius servers

Organization Administrators can view radius servers from **Administration > Flow Configuration > Network > Radius Servers**.

To locate radius servers based on specific criteria, use the **Search** field.

To create Radius servers

1. From **Network > Radius Servers**, click **Create**.
 2. Define the Radius server:
 - **Server:** Type the IP address for the Radius server.
 - **Port:** This port number is read only and set to 1813.
 - **Secret:** Type the radius secret key. Click the eye to view the information.
 3. Click **Submit**.
- The radius server appears in the table list view.

To update Radius servers

1. Click the pencil icon for the Radius servers.
2. Modify the data as desired and click **Update**.

To delete Radius servers

To delete a single Radius server:

1. Click the trashcan icon that corresponds with the Radius server.
2. Confirm the deletion.

To delete multiple Radius servers:

1. Select the Radius servers, and click **Delete**.
2. Confirm the deletion.

Set Up Locations

From **Administration > Flow Configuration > Locations**, Organization Administrators can view and set up Calix Operations Cloud locations. A location is always a subset of your overall network subnets. A location can be one single IP address, groups of IP addresses, or groups of subnets, which is a convenient method for the service provider to group service offerings and view the traffic for that group when monitoring their network. Locations are defined with an IP Subnet and other attributes and can be a geographic location or a logical location.

Calix Cloud uses a location IP subnet to group flow traffic for each location. Organization Administrators can view, create, update, delete, export, and import locations for their organization.

To view locations

To view locations, go to **Locations**.

To locate locations based on specific criteria, use the **Search** field:

To create locations

1. From **Locations**, click **Create**.
2. Define the location:
 - **Name:** Type the name of the location. (Use a maximum of 64 characters; alphanumeric characters, spaces, hyphens, underscores, and commas are allowed.)
 - **Subnets:** Type subnet(s) and click the **plus sign (+)**.
 - **Region:** Type the region of the location. Specifying a region enables you to group locations into regions and view the Real-time dashboard or report per regions.
 - **Address:** Type an address for the location. (Optional)
 - **Geo:** Type the geographic coordinates of the location. (Optional)
3. Click **Submit**.

The location appears in the table list view.

To update locations

1. Click the pencil icon for the location.
2. Modify the data as desired and click **Update**.

To delete locations

To delete a single location:

1. Click the *trashcan* that corresponds with the location.
2. Confirm the deletion.

To delete multiple locations:

1. Select the locations, and click **Delete**.
2. Confirm the deletion.

To export locations

1. Click **Export**.

Note: The exported file includes location IDs, which serve as internal keys and are useful for modifying location names.

2. Open or Save the exported file.

Import Locations

You can import CSV files to populate locations.

Considerations

- Use the pipe (|) character to separate subnet values in the 'SubnetsV4' and 'SubnetsV6' fields. Separating subnets with a comma (,) results in error and causes Service Cloud to place file contents in the wrong fields in the preview. The 'Geo' field accepts commas to separate latitude and longitude.

To import locations

1. Click **Full Import**.
2. View the message, and click **Okay** to confirm.
3. Browse to locate the file for import, and click **Open**.
4. View the locations in the preview table.

Note: All existing location records that are not in the import file are marked for deletion. Confirm the records in the preview table.

5. Click **Submit**.
The locations are imported.

To import locations incrementally

1. Click **Incremental Import**.
2. Browse to locate the file for import, and click **Open**.
3. View the locations in the preview table.
4. Click **Submit**.

The locations are added to your list.

Set Up Applications

From **Administration > Flow Configuration > Applications**, Organization Administrators can view and/or set up Calix Cloud Application Definitions and Application Groups.

To set up application definitions

To view flow device definitions, go to **Applications > Definitions**.

To locate application definitions based on specific criteria, use the **Search** field.

The system returns matches for **Name**, **Addresses**, **Ports**, and or **Ranged Ports**.

To create application definitions

1. From **Applications > Definitions**, click **Create**.
2. Specify the application definition:
 - **Name:** Type the name of the application definition.
 - **Addresses:** Type the address(es) of the application definition and click the **plus sign (+)**.
 - **Ports:** Type the port(s) of the application definition and click the **plus sign (+)**.
 - **Ranged Ports:** Type the ranged port(s) of the application definition and click the **plus sign (+)**.
3. Click **Submit**.

The application definition appears in the table list view.

To update application definitions (local application definitions only)

1. Click the pencil icon for the application definition.
2. Modify the data as desired and click **Update**.

To delete application definitions (local application definitions only)

To delete a single application definition:

1. Click the trashcan icon that corresponds with the application definition.
2. Confirm the deletion.

To delete multiple application definitions:

1. Select the application definitions, and click **Delete**.
2. Confirm the deletion.

To export local applications

1. Click **Export Local**.
2. Open or save the exported file.

To import local applications

1. Click **Import Local**.
2. Browse to locate the file for import, and click **Open**.
3. View the data in the preview table.
4. Click **Submit**.

The local applications are imported.

To set up application groups

To view application groups, go to **Applications > Application Groups**.

You can filter by **Application Group Name** or **Type** (Global or Local).

To locate application groups based on specific criteria, use the **Search** field.

To create application groups (local application groups)

1. From **Applications > Application Groups**, click **Create**.
2. Specify the application group:
 - **Application Group Name:** Select an existing Application Group Name. If creating a new application group, select the **New** checkbox and type the name of the Application Group Name. The Application Group Name is not required.
 - **Application Name:** Select the Application Name.
3. Click **Submit**.

The application group appears in the table list view.

To update application groups (local application groups only)

1. Click the pencil icon for the application group.
2. Modify the data as desired and click **Update**.

To delete application groups (local application groups only)

To delete a single application group:

1. Click the trashcan icon that corresponds with the application definition.
2. Confirm the deletion.

To delete multiple application groups:

1. Select the application groups, and click **Delete**.

2. Confirm the deletion.

To export local application groups

1. Click **Export Local**.
2. Open or Save the exported file.

To import local application groups

1. Click **Import Local**.
2. Browse to locate the file for import, and click **Open**.
3. View the data in the preview table.
4. Click **Submit**.

The local applications are imported.

Set Up Endpoints

From **Administration > Flow Configuration > Endpoint**, Organization Administrators can configure endpoint mapping sources, view and delete endpoints, manage assigned endpoints, and configure the endpoint age out setting.

View and Delete Endpoints

From the Administration > Flow Configuration > Endpoint > Management page you can view a mapped endpoint list and subscriber endpoints in table format.

From the mapped endpoint list page:

- View all mapped endpoints.
- View aggregated groups, where an endpoint supports multiple IP addresses.

Example: If the subscriber match rule matches three different IP addresses associated with a serial number, you can view the aggregation group.

- Delete any endpoint.

From the subscriber endpoints page:

- View all subscriber endpoints.
- Delete a subscriber endpoint.

To view all mapped endpoints

1. From the persona icon (top right), open the drop-down menu.
2. Go to **Administration > Flow Configuration > Endpoint > Management**.
3. Click **Mapped Endpoint List**.

A table of all mapped endpoints displays.

4. (Optional) To view the IP addresses for an aggregated endpoint:

- a. In the Aggregated Groups column for the endpoint, click the **Show** button.

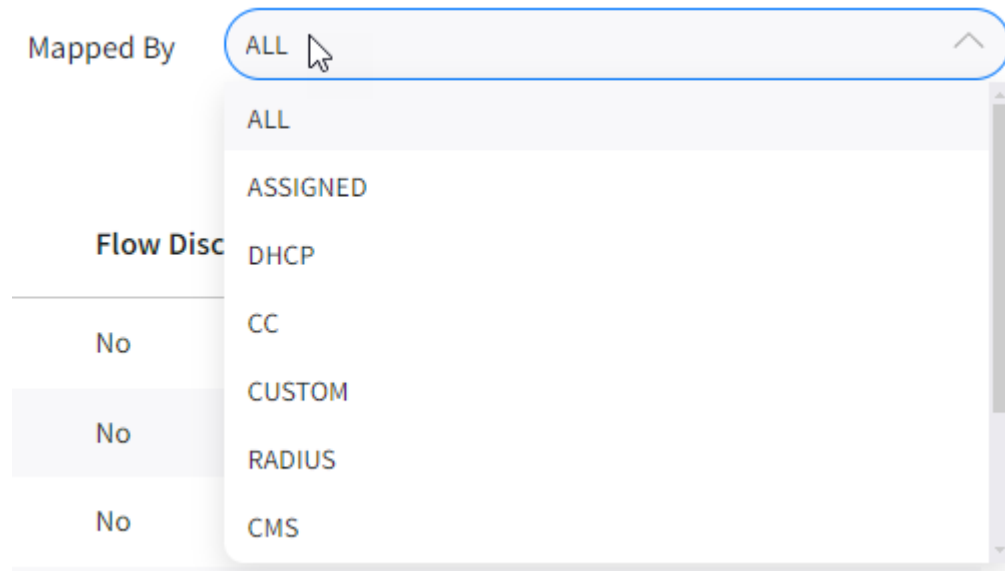
A pop-up window displays the IP addresses associated with the mapped endpoint. This may include IPv4 or IPv6 addresses—including specific IP addresses or prefixes/subnets.

Name	IP	Mapped
CXNK008A74	10.245.236.	CC
CXNK008A74	240e:6a0:37:4	CC
	29a::	
CXNK008A74	240e:6a0:37:4	CC
	::/64	

Note: An aggregation group includes IP addresses mapped by all sources.

- b. Close the window.
5. (Optional) To filter or sort the table:
 - Click in the **Mapped By** field and select a specific mapping source to display.

Note: If an endpoint is mapped by multiple sources, those sources display.



- Click in the **Show** drop-down menu and select 10 (default), 25, 50, or 100 endpoint entries to display per page.
- Click the **Arrow** next to any column heading to sort the table.

To view subscriber endpoints

1. From the persona icon (top right), open the drop-down menu.
2. Go to **Administration > Flow Configuration > Endpoint > Management**.
3. Click **Subscriber Endpoints**.
The endpoint table displays.
4. (Optional) To filter or sort the table:
 - Click in the **Mapped By** field and select a specific mapping source to display.
 - Click in the **Show** drop-down menu and select 10 (default), 25, 50, or 100 endpoint entries to display per page.
 - Click the **Arrow** next to any column heading to sort the table.

To delete endpoints

For a single endpoint:

1. Click the trashcan that corresponds with the endpoint.
2. Confirm the deletion.

For multiple endpoints:

1. To select all displayed endpoints, click the checkbox in the Selected column header.



OR

To select specific endpoints, click the checkbox corresponding to each endpoint.

2. Click **Selected**.
3. Confirm the deletion.

Configure Endpoint Age Out Setting

From the **Administration > Flow Configuration > Endpoint > Settings** page, an Org Admin can configure the endpoint age out setting, which applies to all endpoints except assigned (static) endpoints. If a given endpoint has neither mapping data for flow data for the number of days specified, Calix Cloud purges the endpoint.

To configure the endpoint setting:

1. In the Age Out field, type the number of days (default = 180).
2. Click **Apply**.

Special considerations for RADIUS mapped endpoints

RADIUS mapped endpoints send updates via continuous stream. To reduce system load, a cache mechanism within Calix Cloud drops mapping updates with no tangible change. Updates with no change process every 12 days, while updates with changes process immediately. To prevent unexpected endpoint deletions, Calix recommends that organizations with RADIUS mapped endpoints configure an age-out of more than 12 days.

Other mapping types (e.g., CMS, SMx, Custom, DHCP) have a two-day update cache, meaning Cloud ignores updates with the same IP and endpoint data during that time. For these mapping types, the current minimum age-out setting of five days can be used.

Configure the Organization ACL

From **Administration > Organization ACL**, Organization Administrators can perform the following tasks:

- Add allowed subnets
- Enable/disable Remote Access Protection for GigaSpire EWI access

Configure the Access Control List

From **Administration > Organization ACL > Calix Cloud Access Control List**, Organization Administrators can add allowed subnets from which access is permitted to Calix Service Cloud. The ACL may be configured globally or on a per user basis. Subnets set at the user level take precedence over globally set subnets and override global subnets. If user-level subnets are set, make sure to include the global subnets (if applicable) so they are not overridden.

To add a Subnet (Globally)

1. Go to **Administration > Organization ACL > Calix Cloud Access Control List**.
2. Type the subnet.
3. Click the *plus* icon.

To add a Subnet (User Level)

1. Go to **Administration > Users**.
2. Locate the **Access Control List** section.
3. Type the subnet.
4. Click the *plus* icon.

Configure Remote Access Protection for the GigaSpire EWI

From **Administration > Organization ACL > Remote Access Protection**, Organization Administrators can enable or disable Remote Access Protection for GigaSpire systems (requires EXOS 25.1 or newer). This setting is enabled by default for systems running EXOS 25.2 and newer firmware.

Remote Access Protection uses temporary ACLs to limit exposure of GigaSpire systems to the internet. When a user connects to a system using 'Connect via IPv4' or 'Connect via IPv6', Service Cloud prompts the user to enter an IP address. Service Cloud then creates an ACL for the IP address and enables remote management on the system. After 30 minutes, Service Cloud terminates the session, deletes the ACL entry, and disables remote management.

Considerations

- After a GigaSpire upgrades to EXOS R24.4, Calix Cloud creates ACL entries on the system for Calix ACS IP addresses.
- To ensure seamless interaction between a third party vendor and Calix Cloud, add the vendor's IP space/subnets to the ACL—either globally or at the user level. Otherwise, API calls trigger an error.

To enable Remote Access Protection

1. Go to **Administration > Organization ACL > Remote Access Protection**.
2. Select **Enabled**.
3. Click **Save**.

Once enabled, users are prompted to enter an IP address any time they access the system EWI from Service Cloud.

To disable Remote Access Protection for the system EWI

1. Go to **Administration > Organization ACL > Remote Access Protection**.
2. Select **Disabled**.
3. A confirmation window appears.
4. Click **Yes** to proceed. To revert and keep Remote Access Protection enabled, click **Cancel**.
5. Click **Save**.

Configure Blocked Page Templates

From the **Administration > Blocked Page Templates** page, Organization Administrators can customize the appearance of the web page that displays for subscribers when they attempt to visit a website blocked by one of the following Managed Services:

- ProtectIQ (security)
- ExperienceIQ (parental controls)

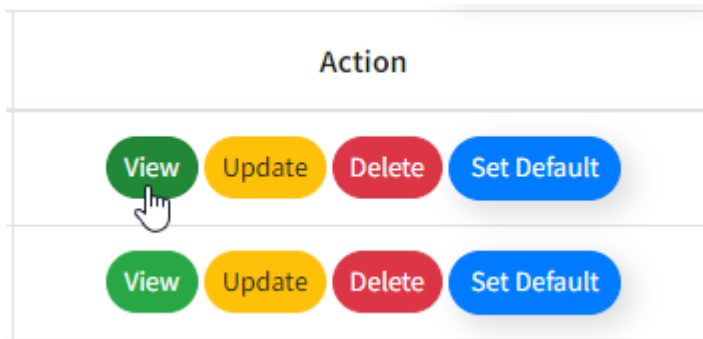
You can specify page colors, logo, and a background image. See the sections below for steps to view, add, update, delete, and set a default block page template.

Limitations

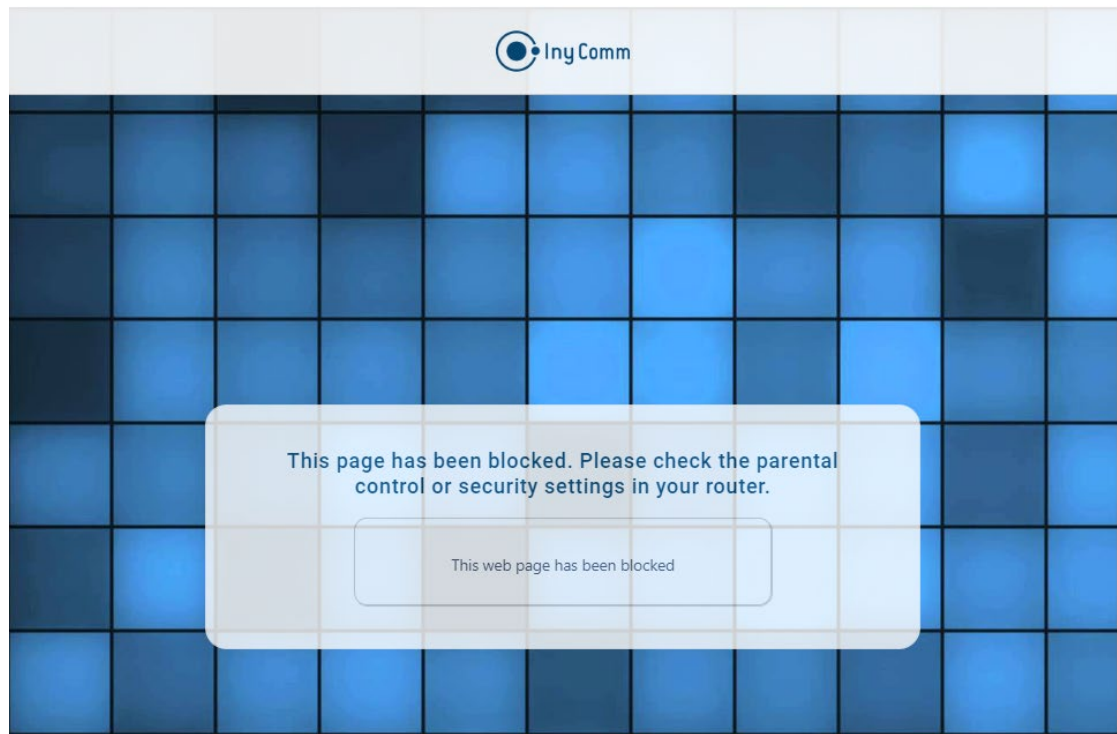
- Blocked page templates do not support HTTPS URLs. Users who visit a blocked HTTPS site see a blank page instead of the blocked page template, while blocked HTTP sites display the template correctly.

To view a Block Page Template

1. Go to **Administration > Blocked Page Templates**.
2. For the desired template, click **View**.



The block page template appears in a new window.



To add a Block Page Template

1. Go to **Administration > Blocked Page Templates**.
2. Click **Add Block Page**.

The Add Block Page Template window displays.

Add Block Template

Preview Blocked Page Templates

Name*

Logo Back Ground Color

Logo Image *

Choose File No file chosen

Body FontColor

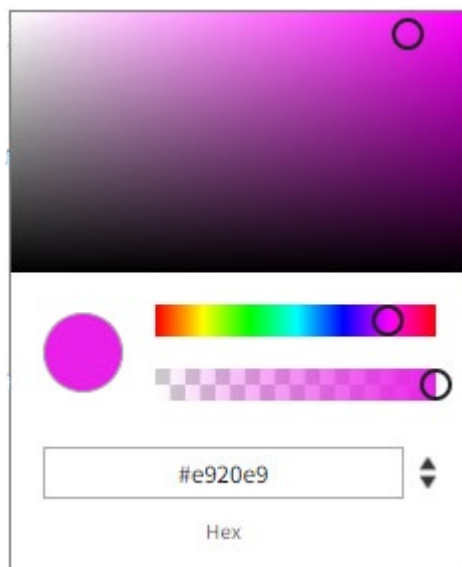
Back Ground Image*

Choose File No file chosen

Submit

3. Complete the following fields:

- Name: Type a descriptive name for the block page template.
- Logo Background Color: Click within the field and select a color from the spectrum. Use the color slider to adjust the color saturation. When ready, click outside the field to confirm the selection. Alternatively, you can type the color value in the field.



- Logo Image: Click **Choose File** to select an image and click **Open**. To avoid image distortion, pre-size/scale your logo to 154 x 62 px image (W x H).
 - Body Font Color: Click within the field and select a color from the spectrum. Use the color slider to adjust the color saturation. When ready, click outside the field to confirm the selection. Alternatively, you can type the color value in the field.
 - Background Image: Click **Choose File** to select an image from your files.
4. Click **Preview** to view the block page template and determine if any formatting adjustments are needed.
 5. Click **Submit** to save the block page template.

To update a Block Page Template

1. Select **Blocked Page Templates**.
2. For the desired template, click **Update**.
3. Update the template as needed. (See the "Add a Block Page Template" section above for guidance.)
4. Click **Preview** to view the block page template and determine if any formatting adjustments are needed.
5. Click **Submit** to save the block page template.

To delete a Block Page Template

1. Select **Blocked Page Templates**.
2. For the desired template, click **Delete**.
3. Click **Yes** to confirm the deletion.

To set a Block Page Template as default

1. Select **Blocked Page Templates**.
2. For the desired template, click **Set Default**.
3. Click **Yes** to confirm the default template.

Provision Single Sign-on

This section describes how to provision Single Sign-on (SSO), which enables users to sign on to Calix Cloud via any external identity provider (IDP) using Security Assertion Markup Language (SAML).

Guidelines

- SSO is optional. The Cloud login page for all users provides two options: Log in with your Calix ID (email/password) or log in with SAML-based SSO.
- Implementing the Calix Cloud SSO feature requires configuration in both the SAML-based external IDP (for example, Google Identity or Azure AD) and Calix Cloud. The OpsAdmin enters the IDP configuration and the Calix Technical Assistance Center (TAC) enters the Calix Cloud configuration.
- You can assign the OrgAdmin role to an SSO user, but the role will only be applied at SSO login if that user already exists as a Calix Cloud user (that is, the same user ID was created via sync from My Calix contacts).
- Be careful with your SSO configuration. If it is incorrect or the SSO provider is unavailable, you will not be able to log in to Calix Cloud. You may open a ticket with Calix TAC to have your SSO configuration removed so that you can log in again.

Create GeoMap Overlays (Layers)

This topic describes how to upload GeoMap files for the Calix Engagement Cloud Map Filter.

From **Administration > GeoMap Overlay**, Organization Administrators can upload files to create layers for the Map Filter in Engagement Cloud. Layers let you display additional information, such as fiber runs and service areas, on the Map Filter to facilitate geographic segmentation.

For more on using Map Filter layers in Engagement Cloud, see the "Using the Map Filter" section in *Create Subscriber Segments* (on page [62](#)).

Considerations

- The GeoMap supports .kml/.kmz files (up to 10 MB) and .json/.geojson files.
- Ensure the files do not contain personally identifiable or sensitive data prior to upload.

To upload GeoMap files

1. Go to **Administration > GeoMap Overlay**.
2. Click **Upload**.
3. Select the file from your computer.

To delete GeoMap files

1. Go to **Administration > GeoMap Overlay**.
2. Click on the *trashcan* icon for the desired file.

Appendix A

Reference Information

Glossary of Terms

This page defines key acronyms and terms used throughout this Help document.

Terms

Active subscriber: A billing subscriber who used the internet (had flow data) within the last 30 days.

Audience: A group of subscribers or prospects created when a campaign is deployed from Engagement Cloud. An audience is associated with a specific campaign. Audiences are static (membership does not change).

Base area: The collection of ZIP codes in which you offer your services. Engagement Cloud serves prospect data (both residential and business) only for prospects within this area.

Business prospect: A non-subscriber business within your base area with an SMB data record.

Business subscriber: An existing subscriber with a customer type of "business" in billing.

Channel (aka marketing channel): A third-party marketing platform (for example: Mailchimp) that integrates with Engagement Cloud for deploying campaigns.

Churn: The measure of how many subscribers discontinued their service.

Churned subscriber: Any subscriber removed from the billing file over the previous six months (excluding the current month).

Demographics: Information that allows marketers to target specific audiences based on shared traits (for example: age, income, location). Engagement Cloud provides third-party demographic data for residential subscribers and prospects.

Device: Internet-enabled subscriber equipment (for example: a phone, a computer, a gaming console) connected to the subscriber's system.

Household: A unique combination of last name and mailing address that represents a physical serviceable location. Multiple subscribers or prospects may inhabit the same household.

Lens: A chart, table, or KPI that displays insights data.

Prospect: A non-subscriber who does not use your services.

Record: A unique combination of a name, mailing address, and email address that represents an individual person or business.

Scheduled campaign: A one-time campaign that deploys an audience to selected channels on a specified date.

Segment: A subset of subscribers or prospects that shares common characteristics, needs, or behaviors. Segments

Subscriber: A customer who uses your service.

System (aka router, aka RG): The subscriber's router and/or mesh/SAT. The RG connects the subscriber's local network (LAN) to the internet (WAN). Typically, RGs also serve as a Wi-Fi "access point" (see definition above) for Wi-Fi client devices in the home, providing both gateway router and Wi-Fi AP functions. RGs may also be integrated into an ONT or modem to provide both the access endpoint/bridge and gateway functions from a single device. This document most commonly refers to RGs as systems.

Triggered campaign: An ongoing campaign that deploys subscribers who meet a specified threshold (e.g., 50 service limit hits within 30 days) on a daily basis. Currently, only the mobile notification and Mailchimp channels support triggered campaigns.

Acronyms

ARPU: Average revenue per user.

KPI: Key performance indicator.

QoE: Quality of Experience.

SMB: Small and medium-sized business.

USOC: Universal Service Order Code.

How ProtectIQ Identifies Threats

ProtectIQ identifies viruses, threats, and intrusions as follows:

- **Viruses** include traditional software viruses such as those embedded in files attached to malicious emails. A virus can do many things, such as:
 - Replicating itself by sending out copies based on emails in your inbox or sent items.
 - Running in the background of the operating system, collecting sensitive information such as logins to different applications, and then periodically transmitting that data back to the hackers.
 - Opening a port on the network to become a relay for other threats to pass through the system as a proxy.
- **Web threats** are like viruses but involve a user clicking on a malicious link that opens an infected file or site.
- **Intrusions** include someone from outside the network trying to enter the network or an unauthorized application on the network trying to communicate externally. ProtectIQ blocks both types of intrusions: it prevents unauthorized clients attempting to get into the network from outside and shuts down bad actors inside the network doing port scans, looking for vulnerabilities, etc.

VPN and Business Applications

The following table lists the VPN and business applications used to identify Work from Home (WFH) subscribers:

Application Type	Application
Business	• INOC • Cisco[®] MGCP • Cisco WebEx[®] • MS FS • MySQL[®] • CVS • MSGC • RMI • RPC • VNC • WBEM • Whois • Xgrid[®] • Citrix[®] • ClearcaseDStorage • File Maker • IBM[®] Informix[®] SQL • HCL Notes • Microsoft Azure[®] • Microsoft Exchange[®] • Microsoft OLAP • Microsoft SharePoint[®] • Microsoft SQL Server[®] • Microsoft System Center • Microsoft Terminal • Microsoft Web • MS-RPC • MS-SSDP • MS Cluster • Microtec • MS SMB • MS MessageQ • NetBIOS • Northern Computers • PC over IP (PCoIP) • PostgreSQL • Printer • Salesforce • SybaseSQL • Symantec[®] • Tanker Int • Telnet • US Signal • VMWare[®] • VMware vMotion[®] • Web Objects • XO Communications • XSR App • X Window System • GoTo Meeting • Zoom
VPN	• ESP • GRE • IKE • SOCKS • OpenVPN™ • Apple VPN • Tunneling

Camera Brands

From Dashboards, you can apply the Camera Brand filter to identify subscribers with specific makes of camera (*Arlo, Nest, Ring, Other*). The 'Other' filter option includes the following brands:

- Amazon[®]
- Blink[®]
- Binatone
- CANARY
- D-Link
- Eufy[®]
- Google
- Kasa Smart
- NETGEAR[®]
- Owlet[®]
- TP-LINK[®]
- Vivotek[®]
- Wyze
- YI Technology
- Unknown

Subscriber to Endpoint Mapping in Calix Engagement Cloud

This topic explains how subscriber to endpoint mapping works in Calix Engagement Cloud.

Any activity—such as online gaming, streaming movies and TV shows, and web browsing—that consumes internet bandwidth generates data traffic on the subscriber's RG system. Service providers can view this traffic data (flow data) in Calix Cloud for troubleshooting subscriber issues and for creating targeted marketing campaigns. To measure flow-based data points (for example: streaming usage, service limit hits) in Engagement Cloud, each subscriber must be correctly associated with a flow endpoint. A flow endpoint is an identifier for a specific set of traffic data. The process of associating subscribers with flow endpoints is called endpoint mapping. Correct endpoint mapping ensures that Engagement Cloud shows complete data for your subscribers.

Endpoint Mapping Behaviors

Subscriber engagement insights lenses populate a subscriber's data 30 days after the subscriber becomes active (is mapped to an endpoint and has flow data).

Unmapped Subscribers

Subscribers that are not matched to endpoint do not show as an active subscriber in Engagement Cloud. Although the Total Subscribers KPI includes unmapped subscribers, unmapped subscribers have incomplete data (they have no flow-related data such as total data usage or service limit hits).

Managing Unmapped Subscribers

Organizations using the billing pipeline (flat billing file) can download a CSV of unmapped subscribers from **Configuration > Billing Data Quality**. Provide this file to your internal team to evaluate and adjust subscriber data mappings in your billing file or billing API.

Triggered Campaign Lifecycle Event Triggers

This topic provides information on Calix Engagement Cloud Lifecycle event triggers. Lifecycle event triggers allow you to deploy Mailchimp email and mobile notification campaigns based on subscriber milestone dates. For example, you can send a welcome email after a subscriber system first checks in to Calix Cloud or send a customer satisfaction survey after a system swap.

Pre-Conditions

The following pre-conditions apply to all Lifecycle Event triggers except 'Support Ticket was Closed':

- Valid billing data for the subscriber.
- Systems associated with subscribers in Calix Cloud. See *Associate Systems to the Subscriber and Add Services* in the *Calix Service Cloud Help* for instructions.

Lifecycle Event Triggers

The following table describes each event trigger and provides the minimum and maximum number of days after each event to deploy the campaign.

Event Trigger	Description	Days after event to send campaign	
		Minimum	Maximum
System is Discovered	Based on the date a system (Calix or third-party) first checks in to Calix Cloud. Converting a GigaSpire to SmartBiz use does not count toward this trigger.	2	1825 (5 years)
System is Swapped	Based on the date a system swap is initiated from either Calix Deployment Cloud or Calix Service Cloud.	2	180
Onboarded CommandIQ	Based on the date a subscriber completes the onboarding process for their system in CommandIQ.	2	1825
Created a Person in CommandIQ	Based on the date a subscriber creates a new user profile (person) in CommandIQ.	2	180
Onboarded an Arlo Camera ¹	Based on the date an Arlo camera checks in to Calix Cloud.	2	365
Support Ticket was Closed ²	Based on the date a Call Outcome record is saved with a 'Resolved' status in Calix Service Cloud.	2	90
¹ Requires subscribers with an active Arlo plan.			
² Requires Calix Service Cloud entitlement.			

Trigger Timing

Engagement Cloud syncs subscribers based on the 'Days after' value set in the campaign workflow. Every night, beginning the night before the scheduled start date, Engagement Cloud syncs subscribers who meet the trigger criteria and deploys the campaign the following day. Engagement Cloud then syncs subscribers nightly and deploys daily until the campaign ends or is paused.

Example

You create a triggered campaign with the following properties:

- Trigger Type: Lifecycle Event
- Event: System is Discovered
- Days after: 5
- Start date: March 15
- End date: May 15

On the evening of March 14, Engagement Cloud syncs all subscribers with systems discovered on March 10 and deploys the campaign on March 15. The evening of March 15, Engagement Cloud syncs subscribers with systems discovered on March 11 and deploys the campaign on March 16. The final subscriber sync occurs the evening of May 14 for systems discovered May 10, and the final campaign deployment occurs on May 15.

Calix Engagement Cloud Plus: Data Types, Usage, and Accessibility

This Help topic describes how Calix Engagement Cloud uses demographic data.

Demographic Data

Demographic data provide insight into the lifestyle, financial, technology usage habits of subscribers and prospects within your base area. Engagement Cloud uses demographic data to power insights lenses on Dashboards and to create pre-built acquisition segments.

Demographic data include:

- Email address
- Household income
- Presence of children in a household
- Education level
- Number of household occupants

From Engagement Cloud, it is not possible to view or download personal identifiable information (PII)—including name, email address, and mailing address—for individual subscribers or prospects. See the *Calix Cloud and EXOS Privacy and Security Policy* for more information.

Base Area

After you define your base area and complete the Engagement Cloud onboarding process, Engagement Cloud populates with demographic data for all households within your base area. Engagement Cloud then attempts to match demographic data to existing subscriber records for filtering upsell and retention segments. Households that do not match existing subscriber records are considered prospects, and their data appears on Dashboards > Residential Prospects. Note that if information on the subscriber record (e.g., address) in Engagement Cloud differs from the demographic data, there is no match, and the subscriber is considered a prospect.

Subscriber Matching Process

Engagement Cloud attempts to match subscriber records to demographic data using the following information:

- ZIP code
- Billing ZIP code
- House or unit number
- Street name
- Postal address

The matching process runs when:

1. The Engagement Cloud instance first populates with demographic data.
2. Demographic data refreshes quarterly.
3. Engagement Cloud deploys a segment to a marketing channel.
4. Engagement Cloud adds subscribers via billing pipeline ingestion or the Cloud API.

Updating the subscriber record does not trigger a match.

When a match is found, Engagement Cloud links demographic data to the subscriber record. Demographic data is residential-only and is based on residential mailing addresses capable of receiving USPS mail (the USPS Delivery Point Validation Address).

Note that subscribers who move to a new address will not have demographic data at that new address.

Look-alike Acquisition Segments

Look-alike acquisition segments leverage demographic (age, gender, income, etc.) and psychographic (financial, technology-based) traits to find similarities between a broadband provider's subscribers and their non-subscribers. Machine learning—a branch of artificial intelligence—is used to continually optimize models. Look-alike acquisition segments can help prioritize where and how you launch your marketing campaigns in a specific service area.

Facebook Look-alike Segments

To create a look-alike segment, Facebook analyzes the data usage habits and behaviors of subscribers in a segment (e.g., Gamers) and builds a profile of similar users. When Engagement Cloud sends a segment to Facebook, it sends two lines of data for each subscriber: one that contains an email address and one that does not contain an email address. These two lines of data allow Facebook to match on primary email address, secondary email address, or direct mail address. Because Facebook can match on two lines of data, it creates larger segments than email marketing platforms such as Mailchimp.

If Engagement Cloud contains multiple email addresses and multiple subscribers for a single household, each email address and subscriber is sent to Facebook separately to increase the match rate.

Social and Email Marketing Campaigns

Email marketing platforms (HubSpot, Mailchimp, Constant Contact) and social marketing platforms (Facebook) match subscriber data sent from Engagement Cloud to user data based on email address, direct mail address, or both. Email and social marketing platforms differ in how they match Engagement Cloud subscribers to users to create segments, which may result in a discrepancy between the number of records sent from Engagement Cloud and the number of records that appear in a marketing platform.

Engagement Cloud only sends email campaigns to subscribers who have opted-in to marketing communications. Direct mail and Facebook campaigns do not require subscribers to opt-in to marketing communications.

Social Marketing Campaigns

Engagement Cloud syncs one record for each matching prospect to Facebook. For each record, Engagement Cloud sends the following information:

- First/Last name
- Email
- Phone
- Address

Facebook uses those fields to match Facebook users and create a list to run marketing campaigns against. The sync from Engagement Cloud to Facebook does not include household grouping.

HubSpot, Mailchimp, and Constant Contact Segments

Engagement Cloud syncs one record for each matching prospect with email type Executable to these marketing channels. For each record, Engagement Cloud sends the following information:

- First/Last name
- Email
- Phone
- Address

Service providers may create email or postal mail campaigns in the marketing channels using these contact lists. The sync from Engagement Cloud to these marketing channels does not include household grouping.

HubSpot seeks an exact match based on the email address provided from Engagement Cloud. Segments created by HubSpot may be smaller than those created by Facebook, but larger than those created by Mailchimp and Constant Contact.

Mailchimp and Constant Contact match on email address. Because Engagement Cloud Plus only sends Mailchimp and Constant Contact email addresses for prospects (non-subscribers) who have opted-in to marketing communications through Claritas and subscribers who have not opted out from marketing communications, segments sent to these marketing channels may be smaller than those sent to other marketing platforms like Facebook or HubSpot.

Calix Engagement Cloud Plus: Email Types for Prospect Marketing

Third party marketing records for prospects may have one of the following email types:

- **Null** indicates the record has no provided email address.
- **Executable** indicates the email address may be used for marketing campaigns.
- **Linkage** indicates the email address may be used to link prospect data to social media accounts. Linkage email addresses may also be used for direct mail marketing campaigns, but may not be used for email campaigns.

By default, Calix Engagement Cloud only deploys executable email addresses to marketing channels for acquisition segments. When you deploy an acquisition campaign from Engagement Cloud, you can select whether to include linkage email addresses for direct mail campaigns.

Calix Engagement Cloud Plus: Subscriber Matching to Demographic Data

This topic explains how Calix Engagement Cloud matches subscribers with demographic data.

Subscriber Matching to Residential Data

Engagement Cloud attempts to match demographic data to subscriber records in Calix Cloud so you can use demographic data to create segments and deploy campaigns. Demographic data are only available for residential addresses within your defined base area.

The matching process compares the following address fields between the subscriber record and the demographic data:

- ZIP code
- House number
- Street
- City
- Unit number (Address Line 2)
- PO box
 - Engagement Cloud supports the following formats for PO box: PO box, PO Box, P O Box, PO BOX, Po Box, po box

The matching process first compares the subscriber's service address with the demographic data. If no match occurs, the process then checks the subscriber's billing address. A successful match occurs when either the subscriber service or billing address matches the demographic data address. Any difference between address fields results in a failed match. For example, if the demographic data includes a unit number but the subscriber record does not, the match fails.

Matching Process Triggers

Subscriber matching occurs when:

1. The Engagement Cloud instance first populates with demographic data.
2. Engagement Cloud adds subscribers via billing pipeline ingestion or the Cloud API.
3. The demographic data refreshes quarterly.
4. A TAC ticket is submitted to rerun the match.

Updating the subscriber record, whether in Calix Cloud or via the billing file, does not trigger a match. Rematching for updated subscriber records occurs in the next quarterly marketing data refresh.

When a match is found, Engagement Cloud links marketing data to the subscriber record and removes the linked record from the prospect data.

Subscriber to Business Record Matching

Engagement Cloud matches subscribers to SMB data records based on address. Engagement Cloud first checks the service address, and if no match is found, checks the billing address.